

# Advanced Modulation and Coding Techniques CU 7101

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# Advanced Modulation and Coding Techniques

## Course Objectives

- To enable the student to understand the **role of the communication medium** in the design approaches for coding and modulation techniques.
- To enable the student to understand the **trade-offs involved** in the design of basic and advanced coding and modulation techniques.
- To expose the student to the advanced baseband **signal conditioning** methods evolved for exploiting the channel and user application characteristics and to familiarize them on the system design approaches.

## Course Outcomes

- The student would be able to demonstrate an **understanding of the trade-offs involved** in the design of basic and advanced coding and modulation techniques and the advanced baseband **signal conditioning** methods evolved for exploiting the channel and user application characteristics.
- Given the **user requirements** and the type of **channel** over which the system has to function the student would be in a position to **apply** his knowledge for designing the **baseband signaling waveforms** that would address the channel impairments.

# Advanced Modulation and Coding Techniques

- **REVIEW OF DIGITAL MODULATION TECHNIQUES**

- Base band and band pass communication; Signal space representation, Linear and nonlinear modulation techniques, M-ary modulation techniques; Spectral characteristics of digital modulation, Spread spectrum modulation techniques

- **RECEIVERS FOR AWGN AND FADING CHANNELS**

- Optimum receivers for AWGN channel -Correlation demodulator, matched filter, maximum likelihood sequence detector, envelope detectors for M-ary signals; Characterization of fading multipath channels, RAKE demodulator, Multiuser detection techniques

- **MULTICARRIER SYSTEMS**

- OFDM- Generation of sub-carriers using the IFFT; Guard Time and Cyclic Extension; Windowing; Peak to Average Power reduction schemes; Multicarrier CDMA- System design, Performance parameters.

# Advanced Modulation and Coding Techniques

- **TRELLIS CODED MODULATION**

- Coded modulation for bandwidth-constrained channels-Trellis coded modulation; Set Partitioning, Four –state Trellis-coded modulation with 8-PSK signal constellation, Eight-state Trellis code for coded 8-PSK modulation, Eight-state Trellis for rectangular QAM signal constellations, Decoding methods and implementation issues.

- **TURBO CODING**

- Introduction-Turbo Encoder, Turbo Decoder, Iterative Turbo Decoding Principles; Modifications of the MAP Algorithm-The Soft-Output Viterbi Algorithm(SOVA); Turbo Coding for AWGN channels, Turbo Coding for Rayleigh Channels, LDPC Codes.

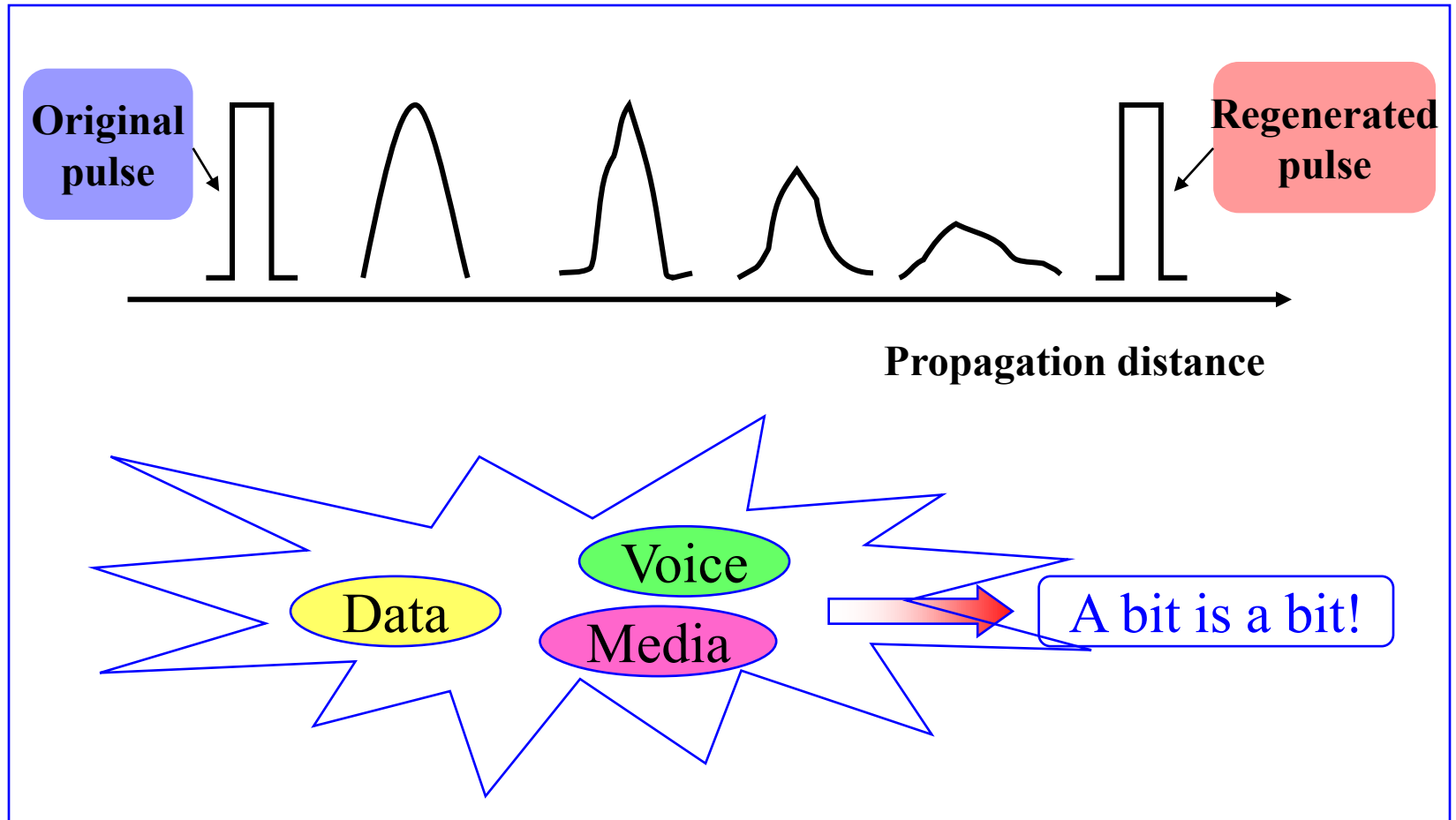
# Digital Communication Fundamentals

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# Analog versus Digital



# Performance Metrics

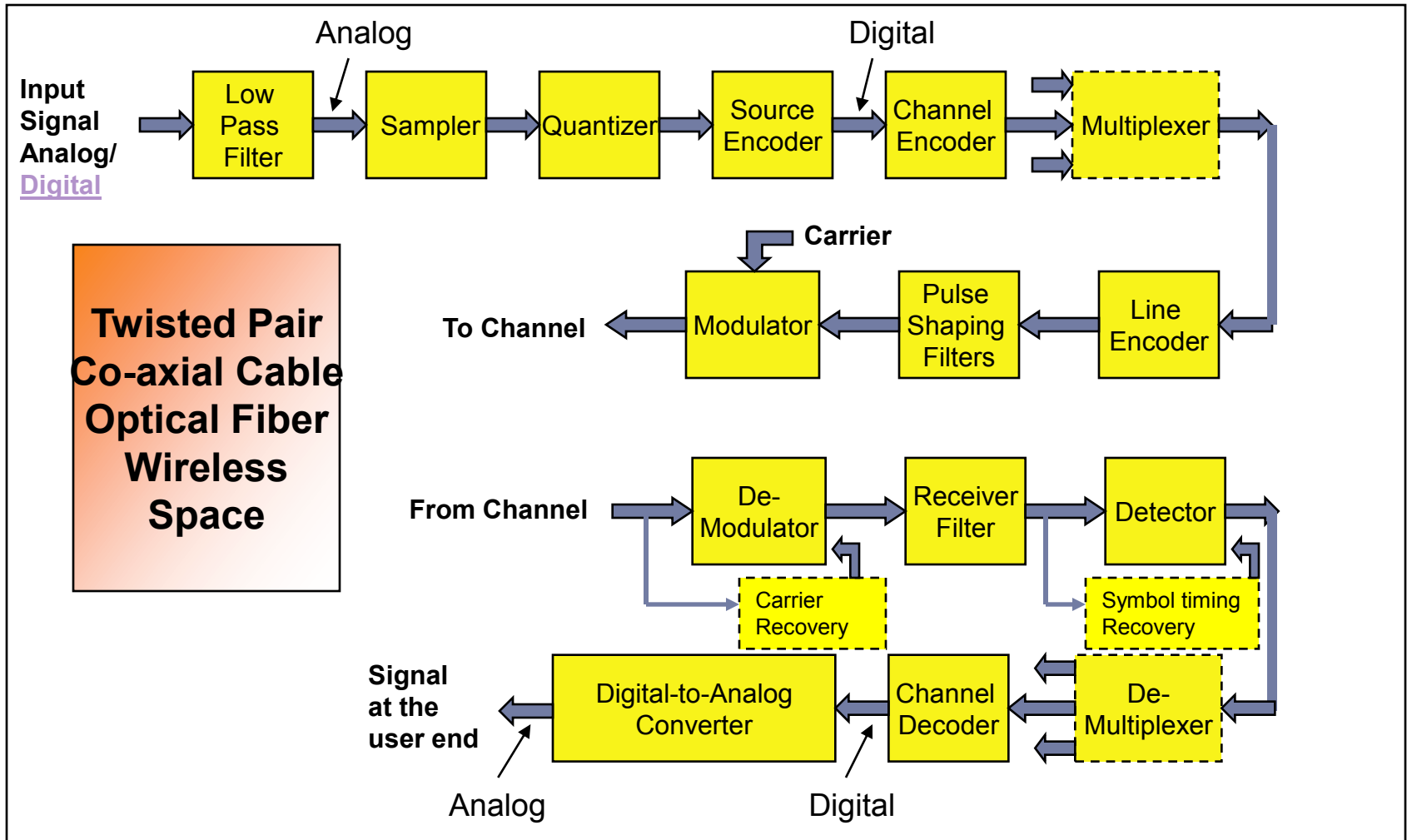
- In Analog communications we want,

$$\hat{m}(t) \cong m(t)$$

- In Digital communication systems,

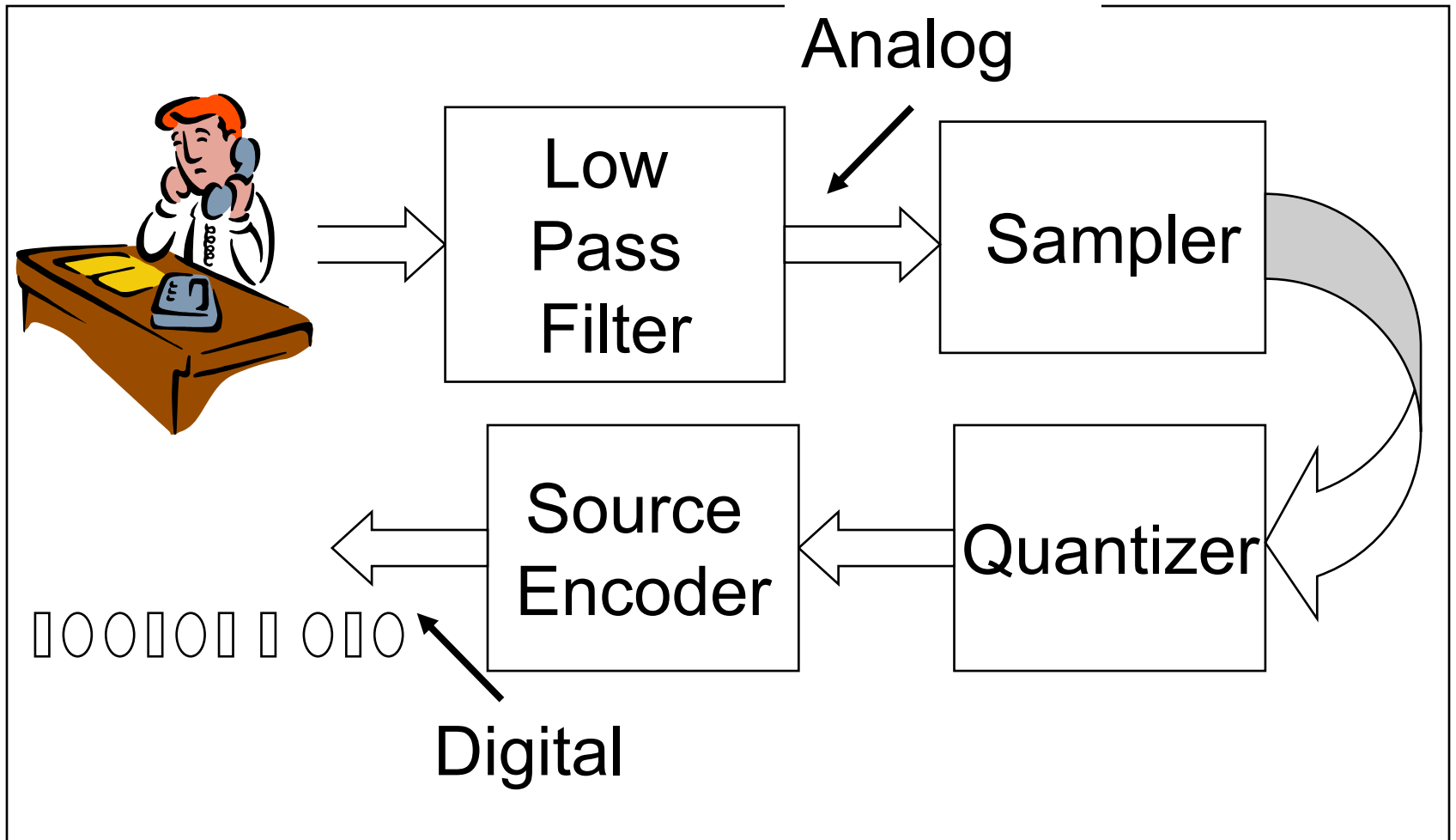
- Data rate (R bps) → Limited by Channel Capacity
- Probability of error (BER)

# Digital communication system



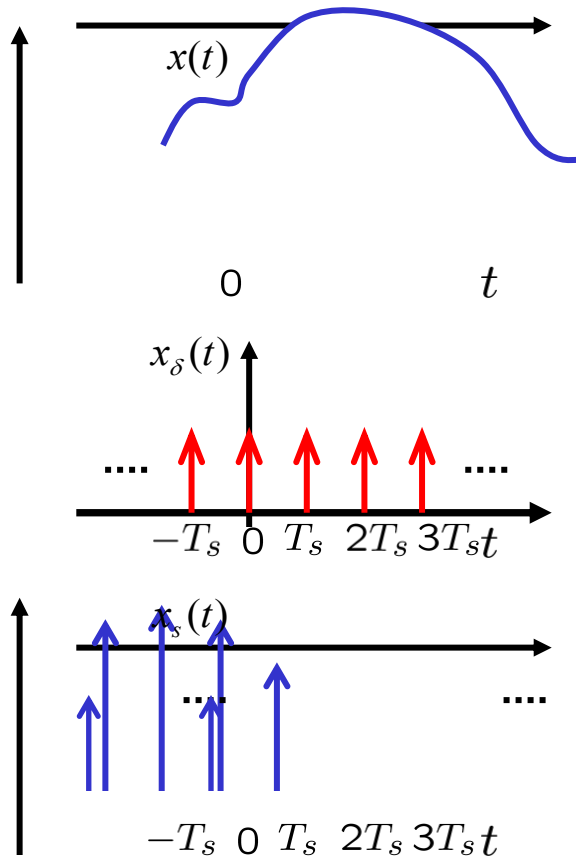


# Analog to Digital Conversion

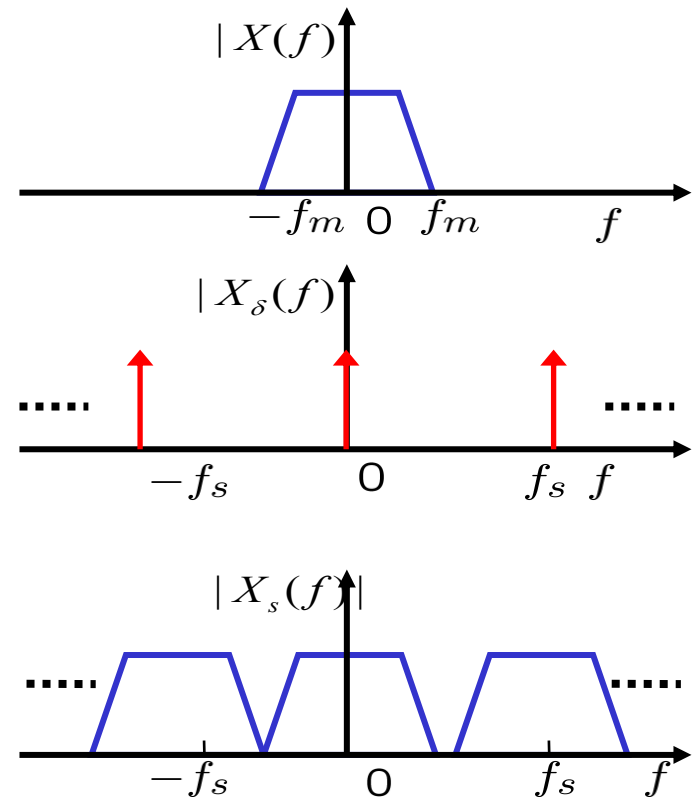


# Sampling Process

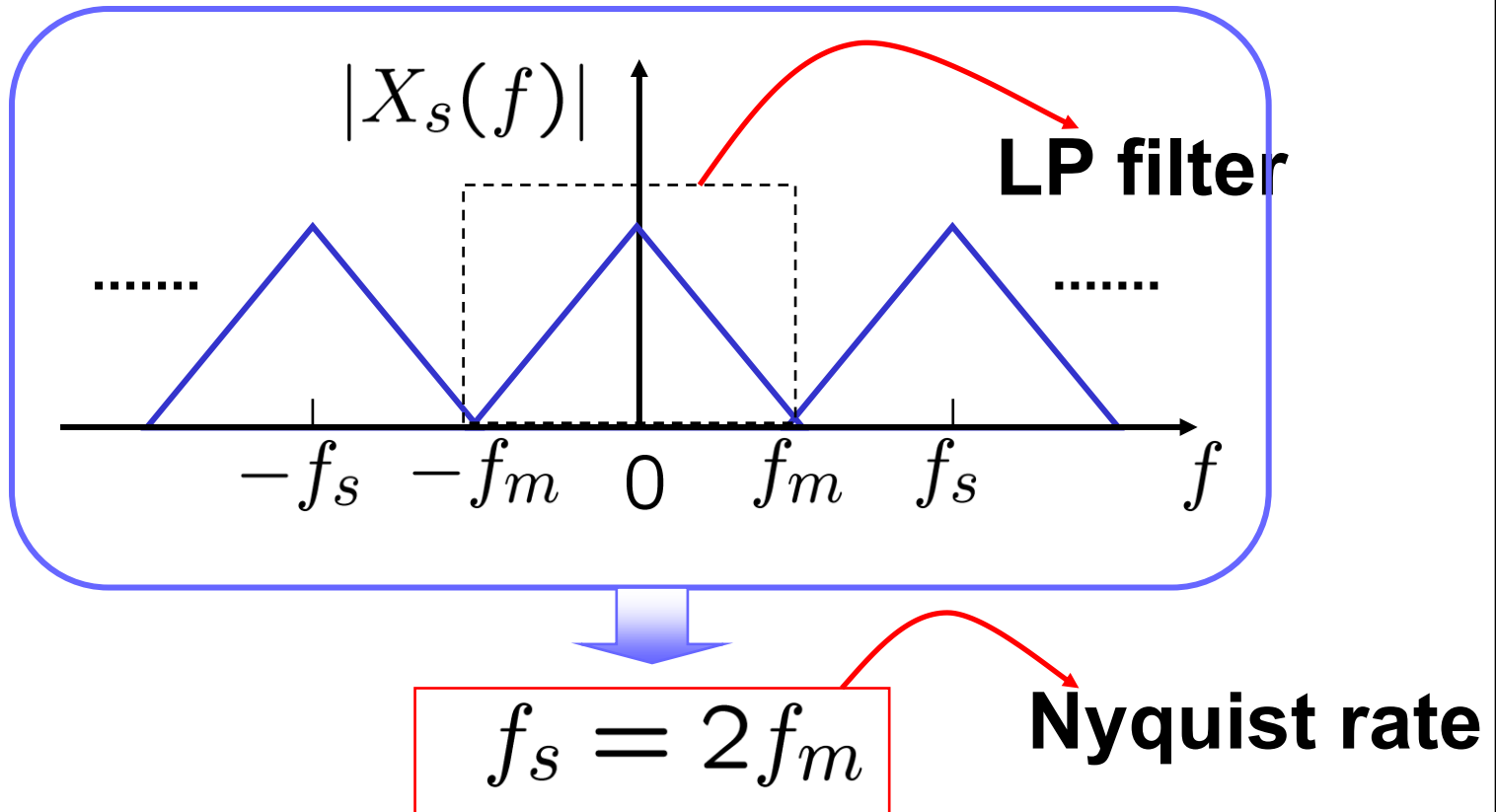
$$x_s(t) = x_\delta(t) \times x(t)$$



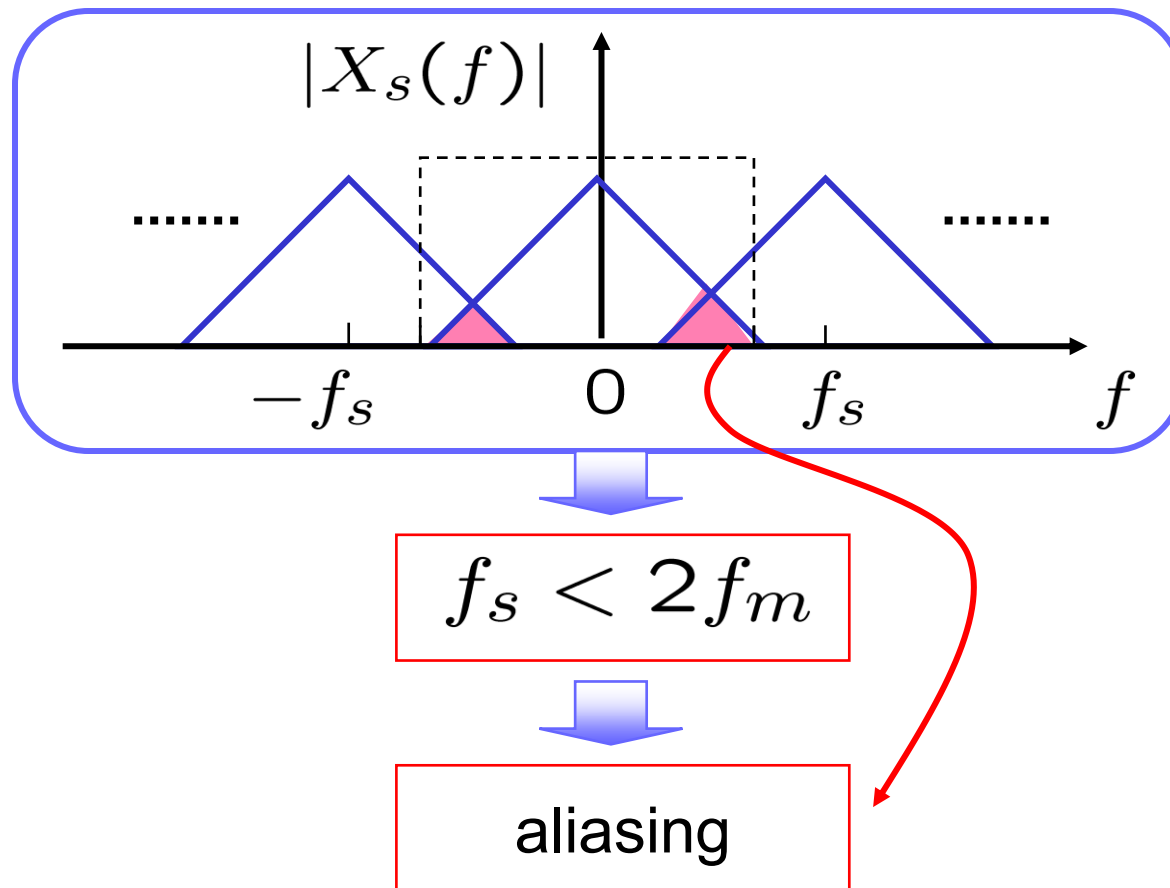
$$X_s(f) = X_\delta(f) * X(f)$$



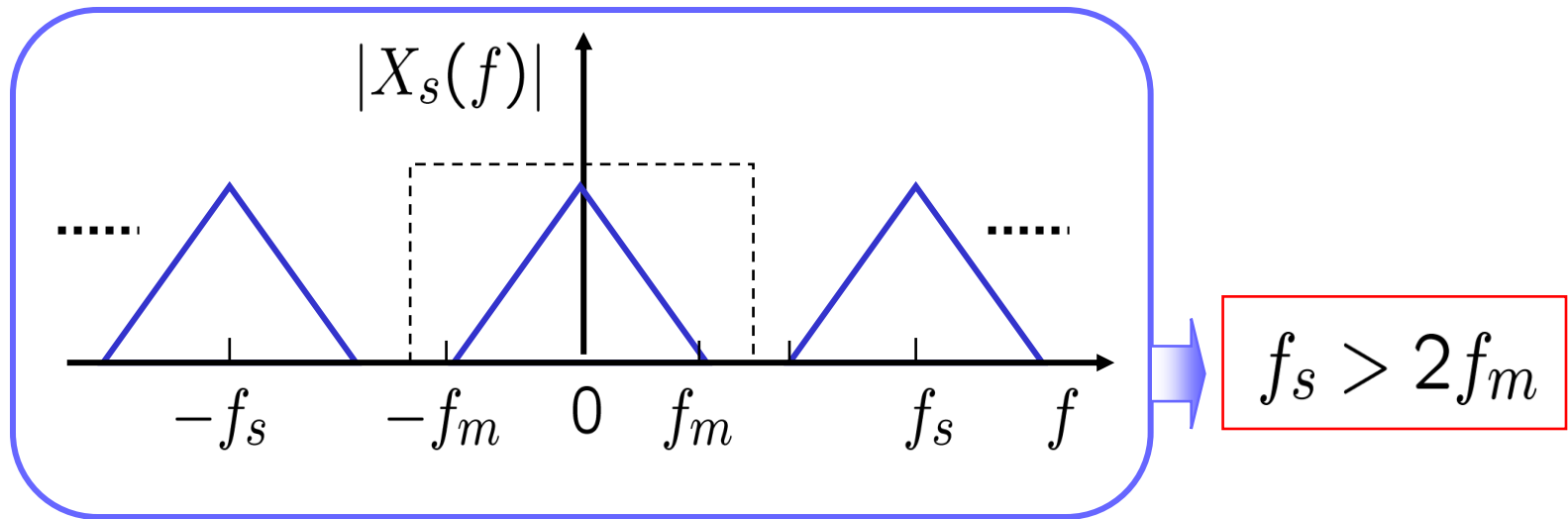
# Aliasing effect



# Aliasing effect

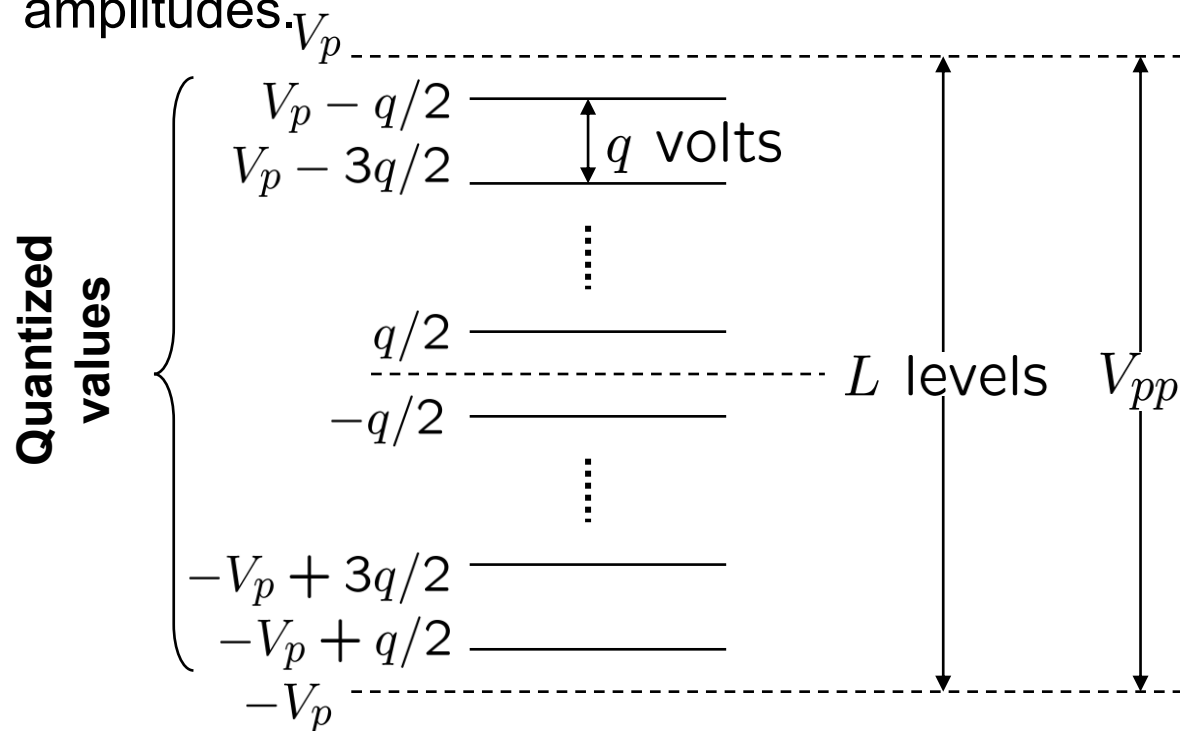


# Aliasing effect



# Quantization

- Amplitude quantizing: Mapping samples of a continuous amplitude waveform to a finite set of amplitudes.



# Quantization

- Average quantization noise power

$$\sigma^2 = \frac{q^2}{12}$$

- Signal peak power

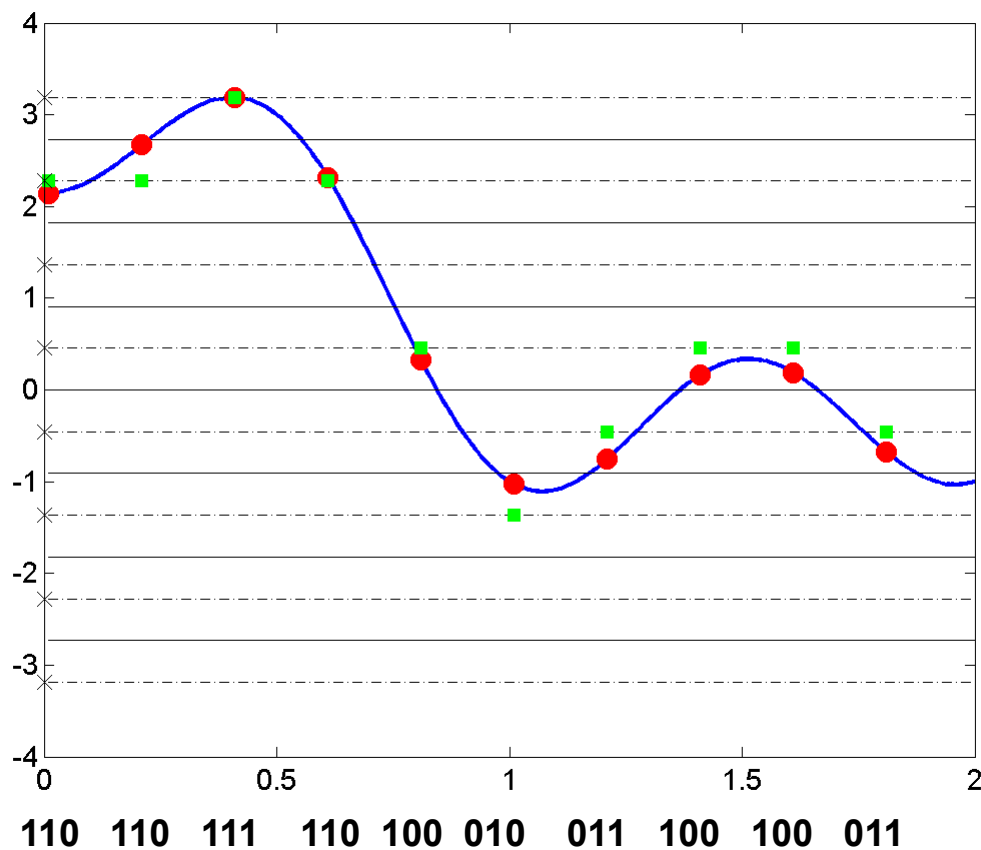
$$V_p^2 = \frac{L^2 q^2}{4}$$

- Signal power to average quantization noise power

$$\left(\frac{S}{N}\right)_q = \frac{V_p^2}{\sigma^2} = 3L^2$$

# Quantization example

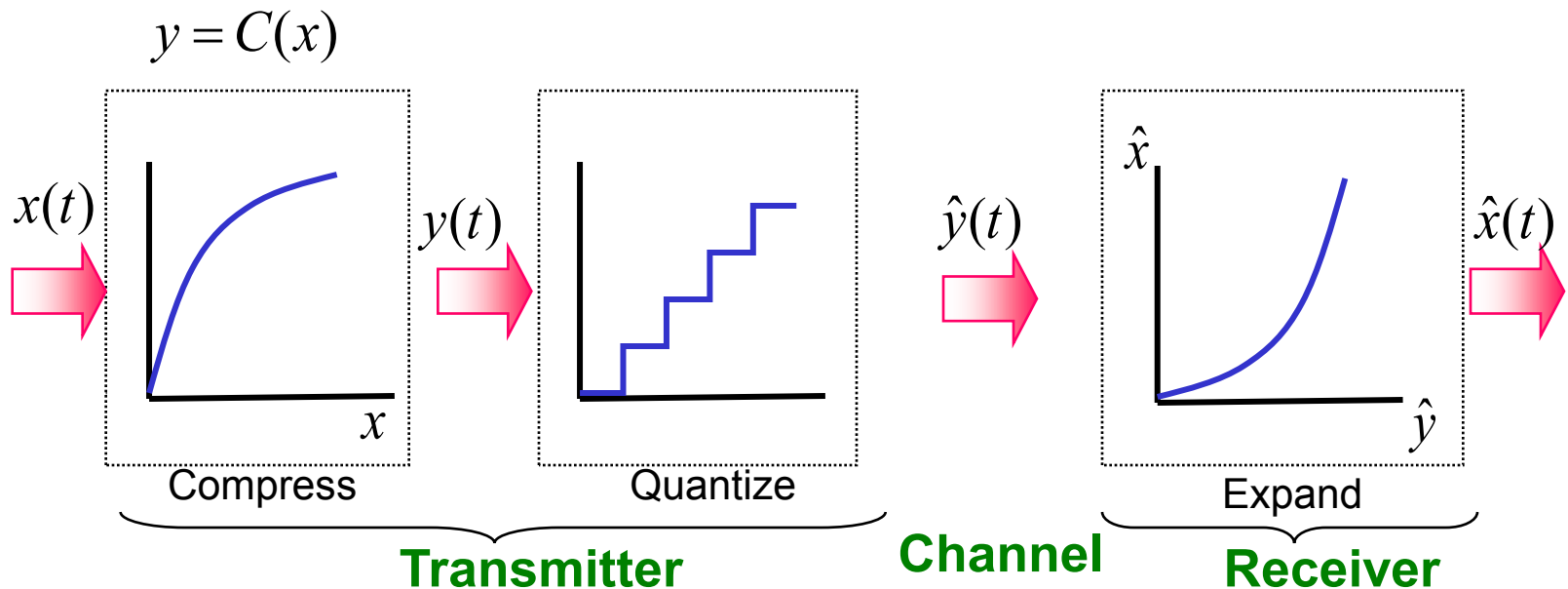
|     | <i>amplitude</i> |
|-----|------------------|
|     | $x(t)$           |
| 111 | 3.1867           |
| 110 | 2.2762           |
| 101 | 1.3657           |
| 100 | 0.4552           |
| 011 | -0.4552          |
| 010 | -1.3657          |
| 001 | -2.2762          |
| 000 | -3.1867          |





# Non-uniform quantization

compression+expansion  $\Rightarrow$  companding



# Source Coding

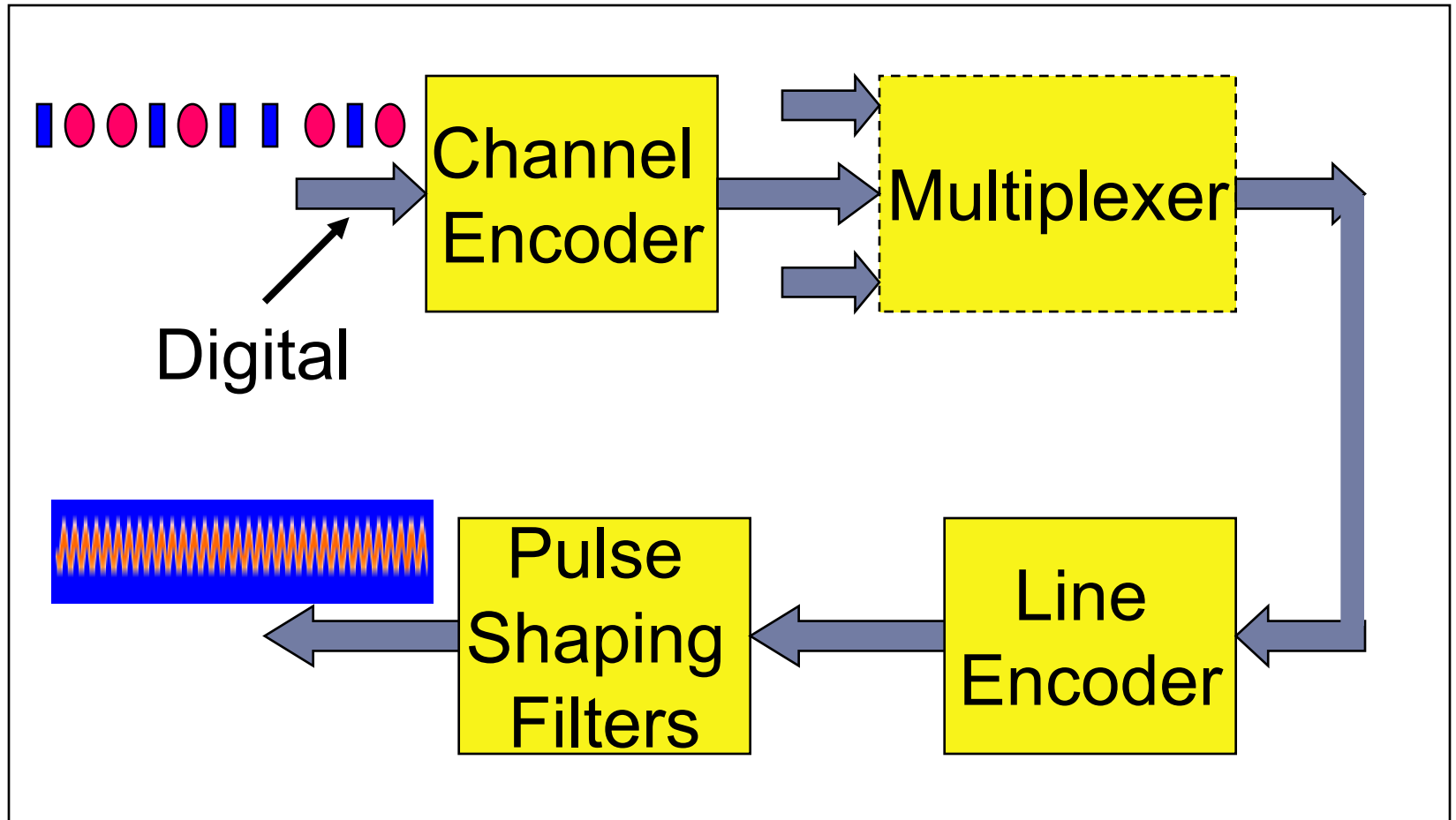
- **Pulse Code Modulation**
- **Delta Modulation**
- **Eliminate redundancy**
  - **Lossless coding**
    - Huffman Coding, Shannon-Fano Coding, etc.
  - **Lossy compression coding**
    - LPC, JPEG, MPEG, etc.

# Source Coding

- **Redundancy elimination maximizes information rate in a transmission**
- **What is Information Rate ?**

**Entropy → Average Information per bit**

# Baseband Signal Processing



# Channel Coding

(Error Control Coding)

- **To reduce the impact of channel errors by controlled introduction of redundancy**
  - **Decrease in effective data rate**
  - **Increased coding gain**

# Channel Coding

(Error Control Coding)

## ➤ Forward Error Correcting Codes

- Linear Block Codes
- Convolutional Codes
- Turbo Codes, LDPC Codes

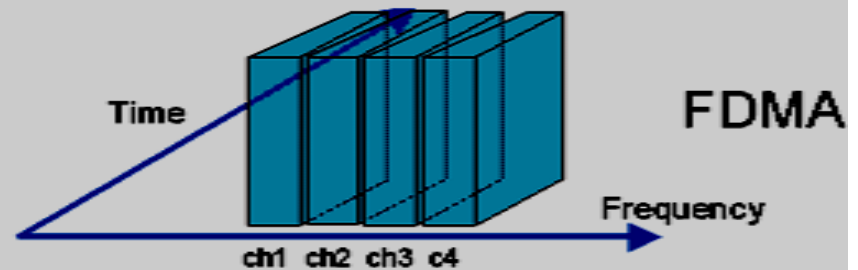
## ➤ Feedback based Error Correcting Codes

- ARQ methods: Stop and wait,  
Go-back N, Selective repeat

# Multiplexing

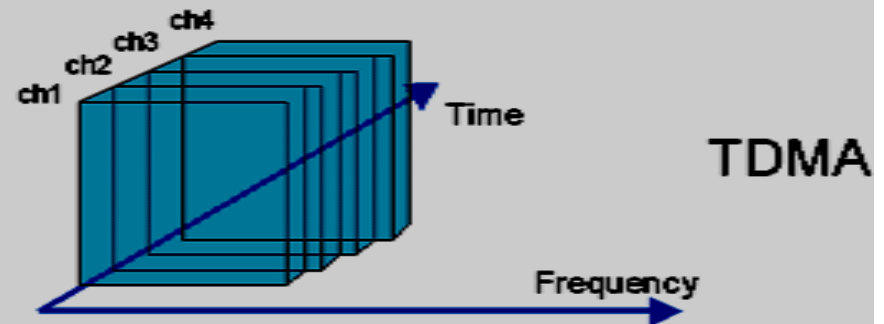
## ■ **FDMA:**

- Oldest, first in PSTN
- Used in GSM uplink/downlink channel separation
- ADSL uplink/downlink & in OFDM



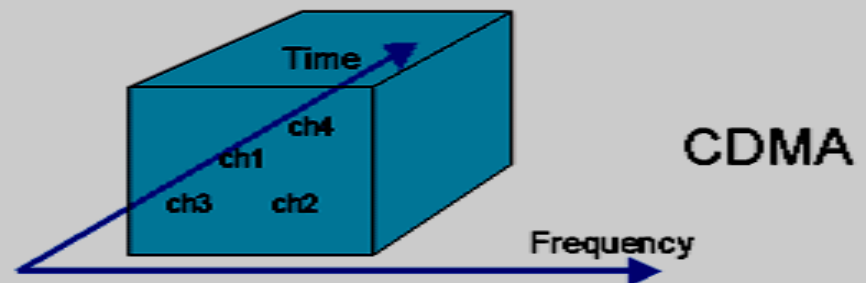
## ■ **TDMA:**

- ISDN-exchanges, GSM calls
- Other applications: DECT



## ■ **CDMA:**

- Supports statistical multiplexing
- Suits well to wireless cellular channel (channel adaptation)



# Line Coding Formats

- **Converting Data into time varying Signals**
- **Must consume least bandwidth**
- **Reduce dc content**
- **Clock extraction for symbol timing**
- **Error correction capability if possible**



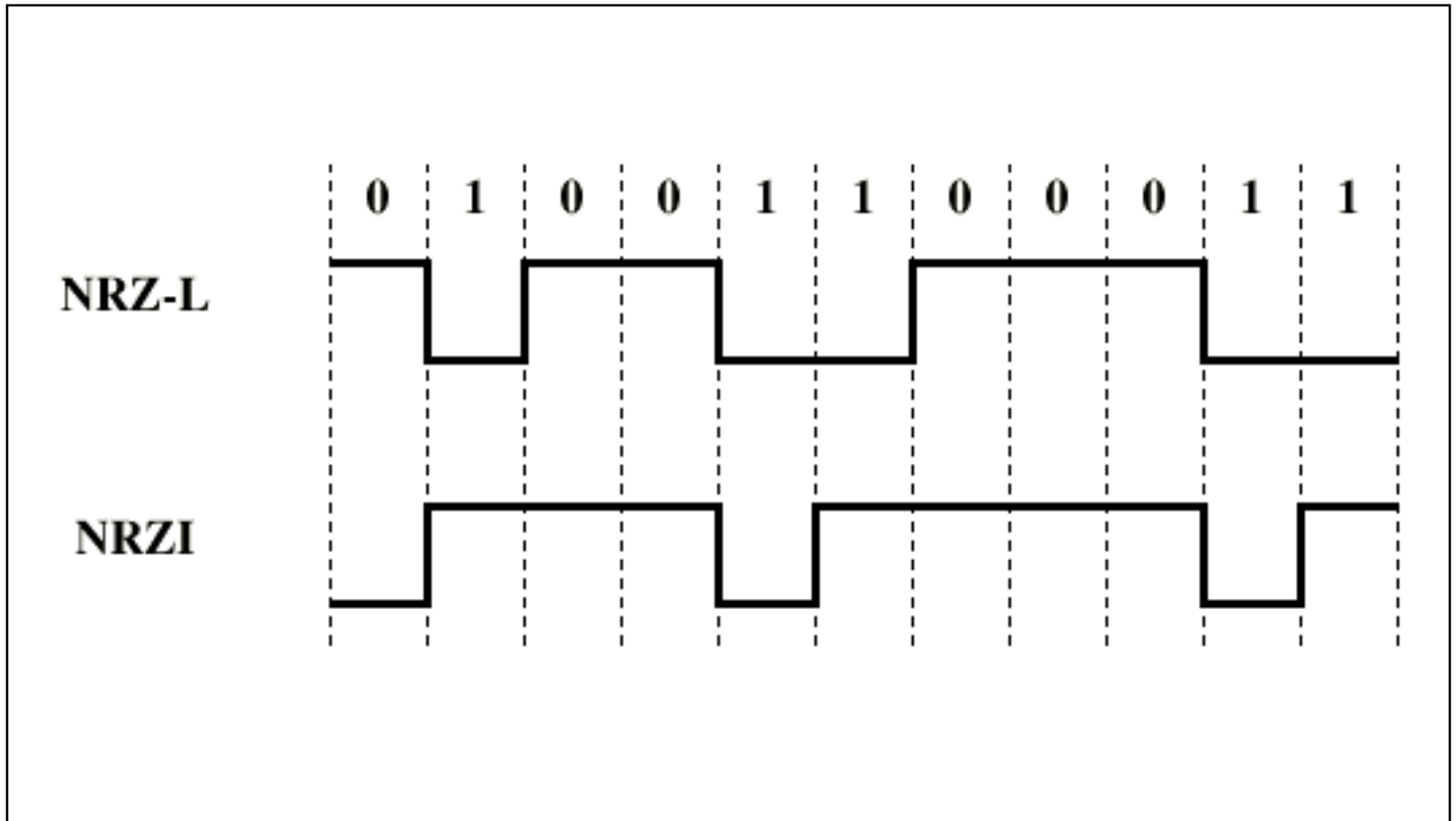
# Line Coding Formats

(Converting Data into Signals)

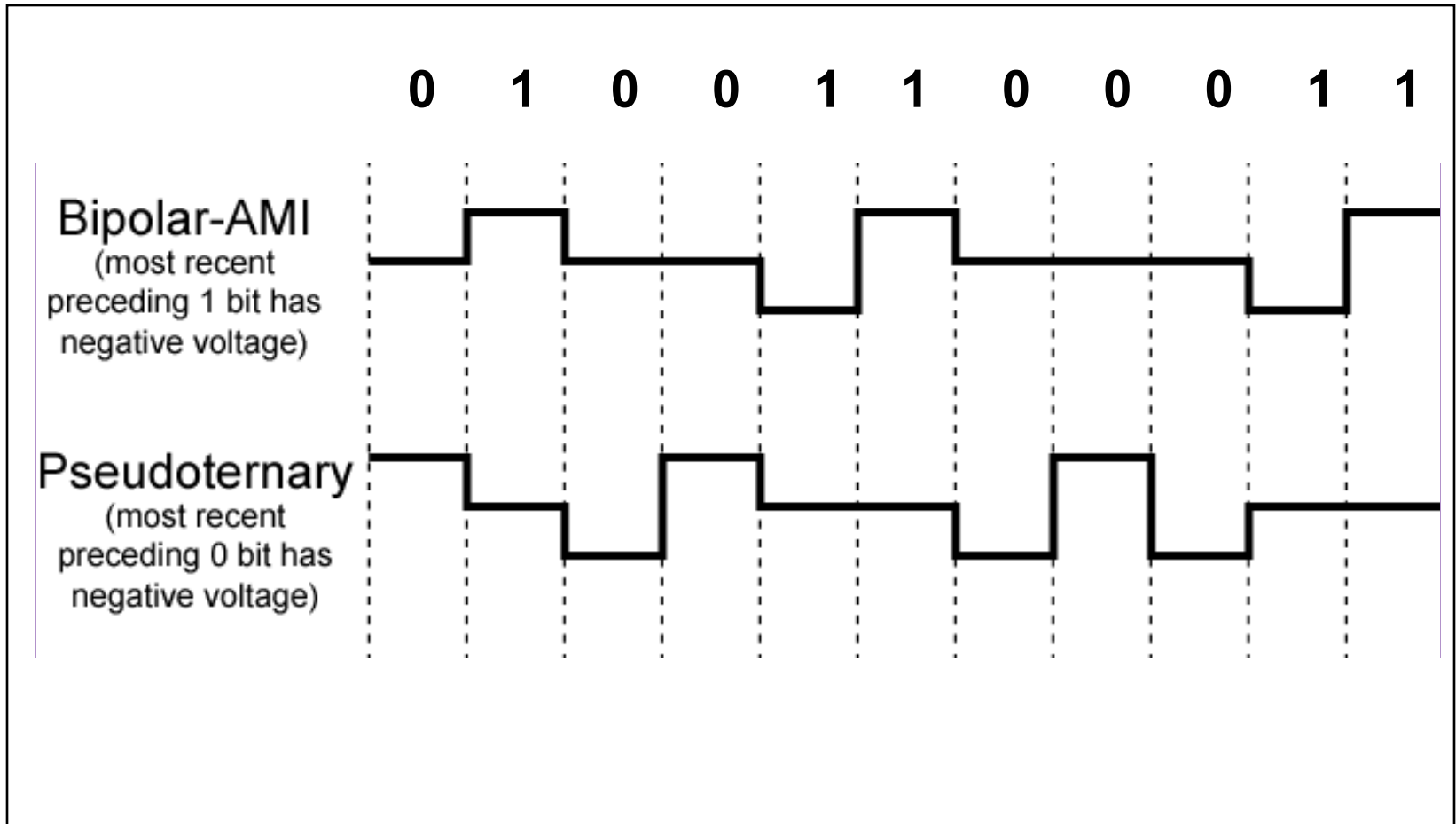
**Numerous techniques –**

- **NRZ-L**
- **NRZ-I**
- **Manchester**
- **Differential Manchester**
- **Bipolar AMI**

# Line Coding Formats

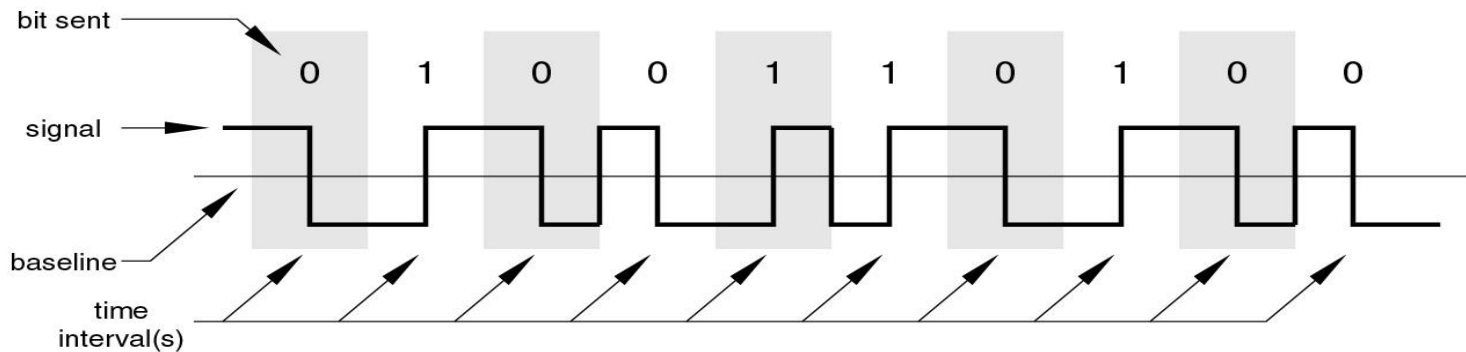


# Line Coding Formats

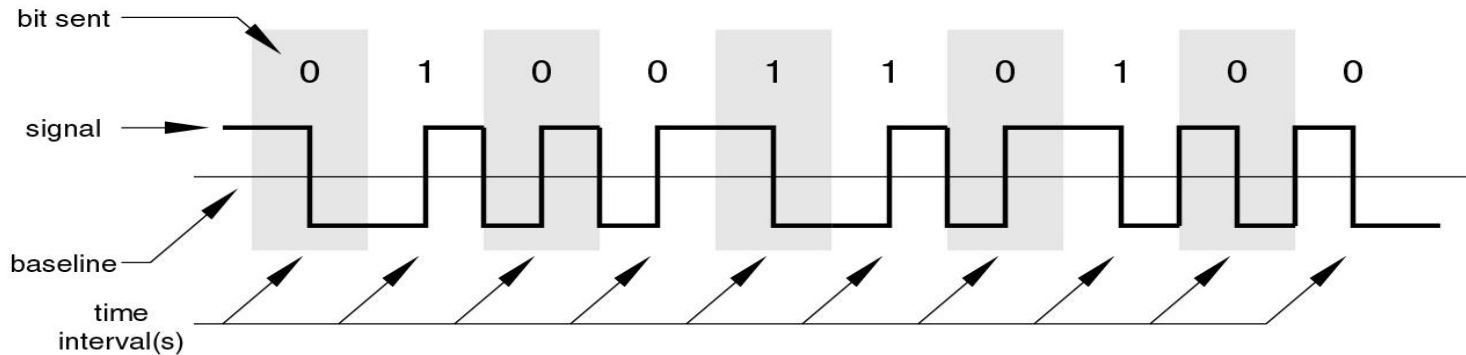


# Line Coding Formats

## Manchester Encoding



## Differential Manchester Encoding

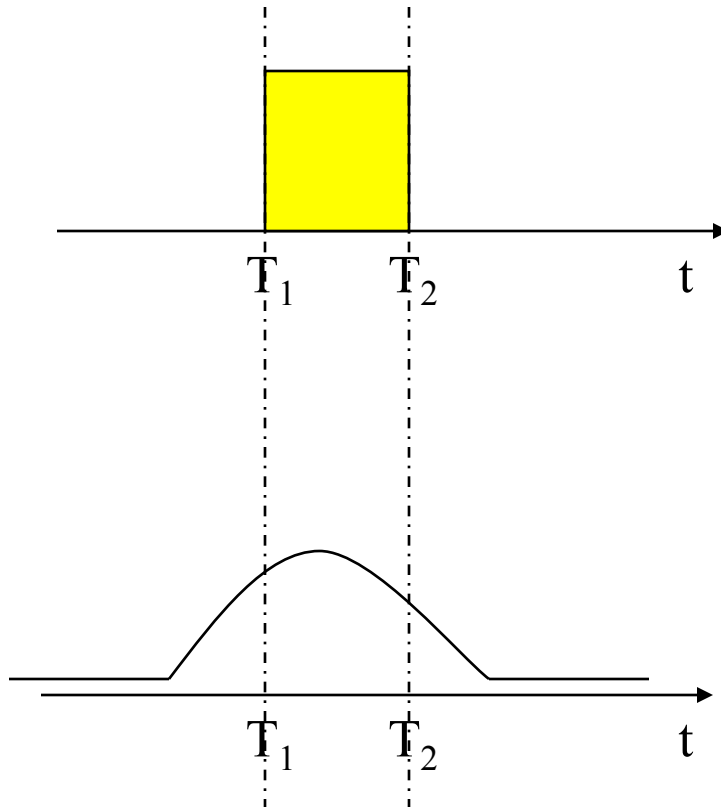


# Pulse Shaping Filters

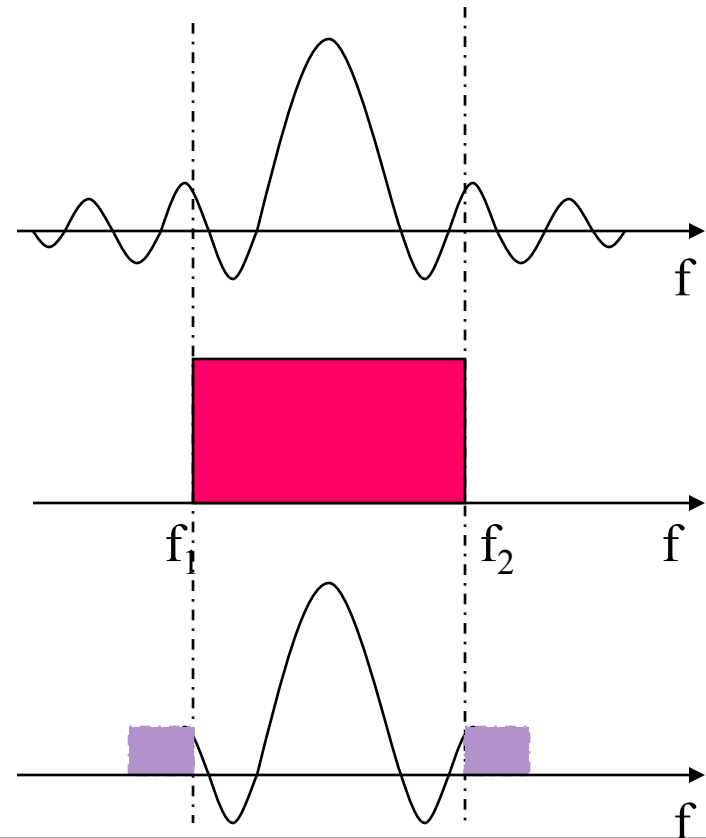
- Time Domain  $\Leftrightarrow$  Frequency Domain
- Bandlimiting signals in frequency domain
  - ➔ spreads signal in time domain
- Inter-Symbol Interference ➔ ISI

# Inter Symbol Interference

- Time Domain

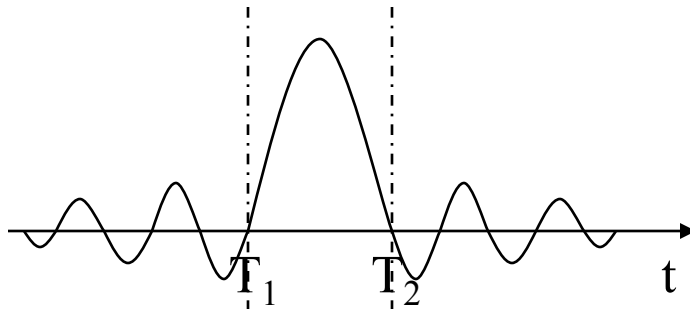


- Frequency Domain

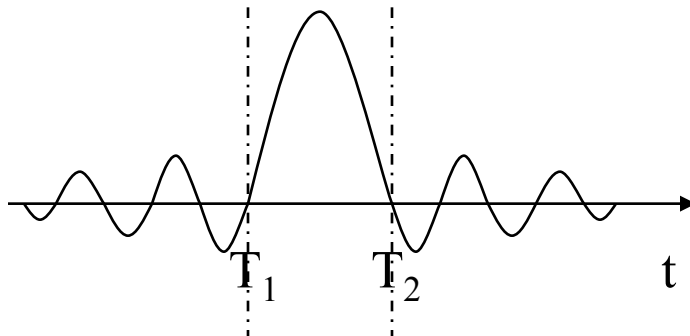
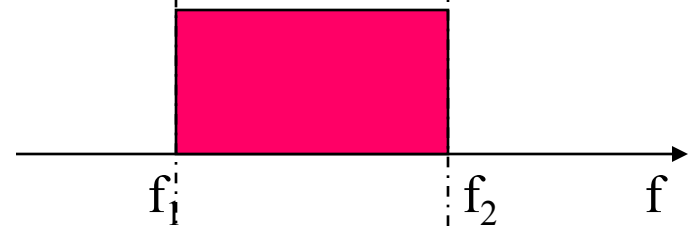
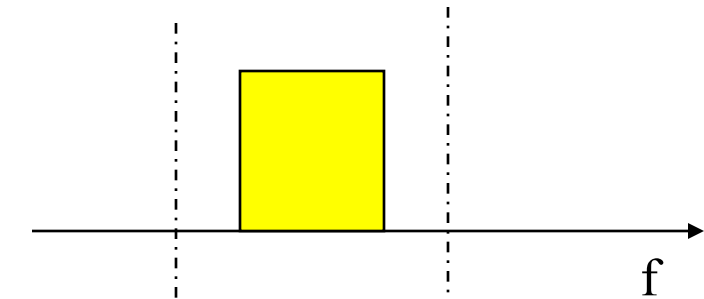


# Nyquist Criterion

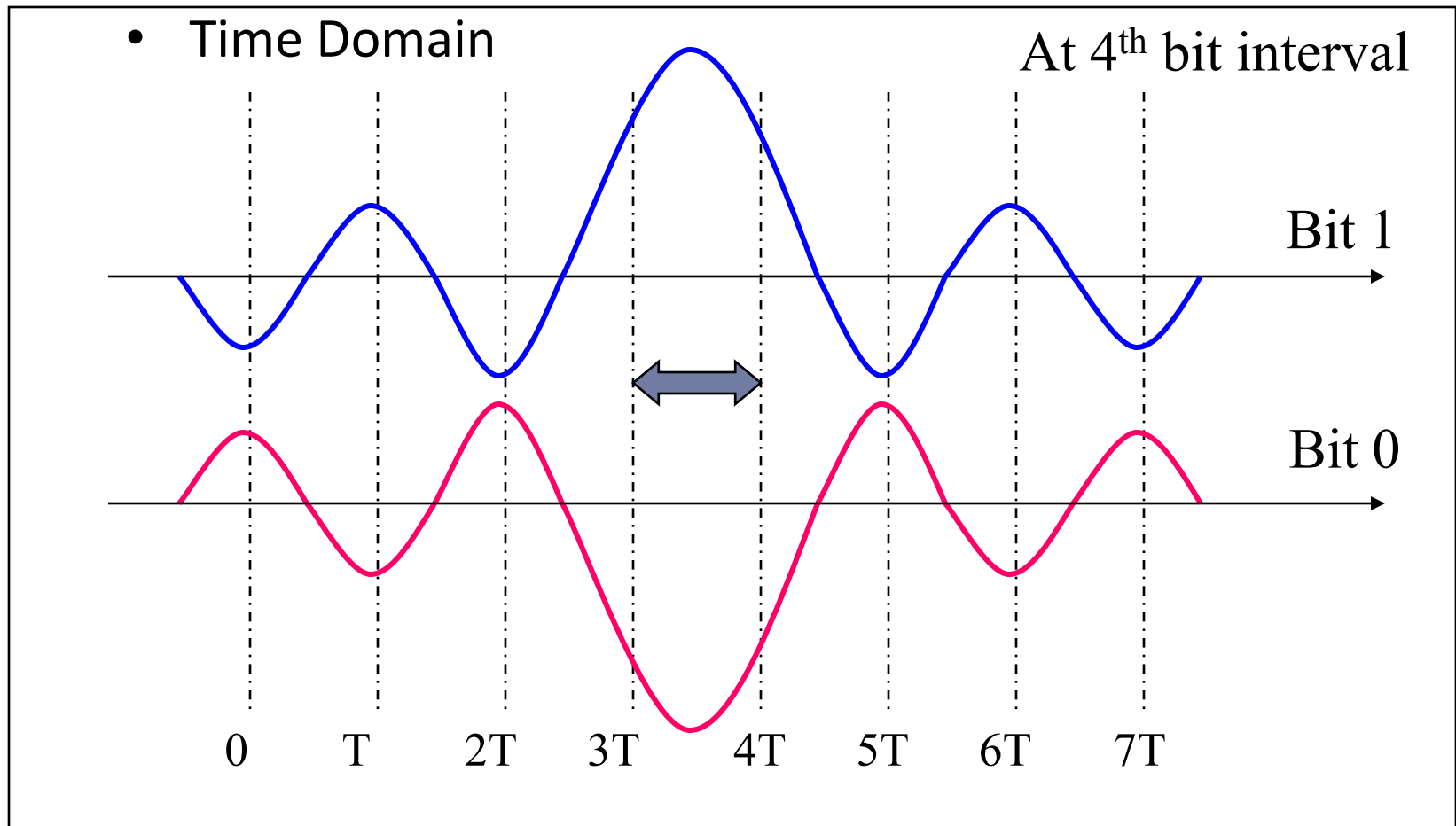
- Time Domain



- Frequency Domain



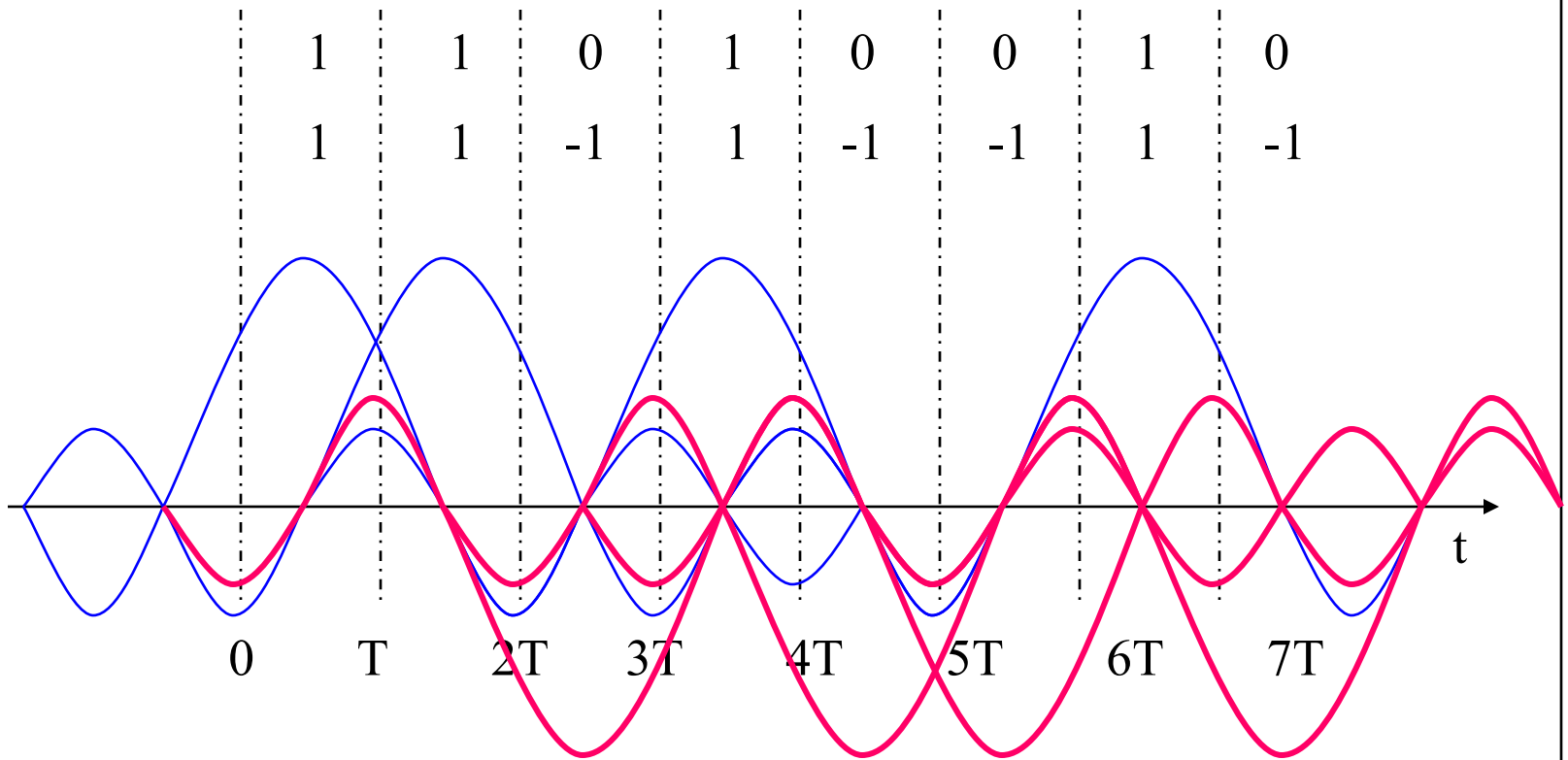
# Application of Nyquist Criterion





# Application of Nyquist Criterion

- Time Domain

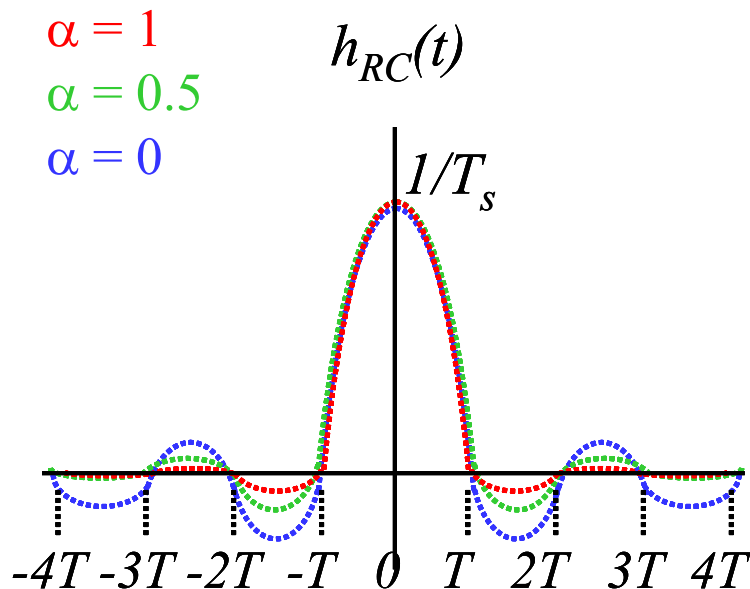


# Pulse Shaping Filters

- **Nyquist Criterion to overcome ISI**
- **Pulse Shaping Filters →**
  - **Raised Cosine Filters**
  - **Gaussian shaping filters**

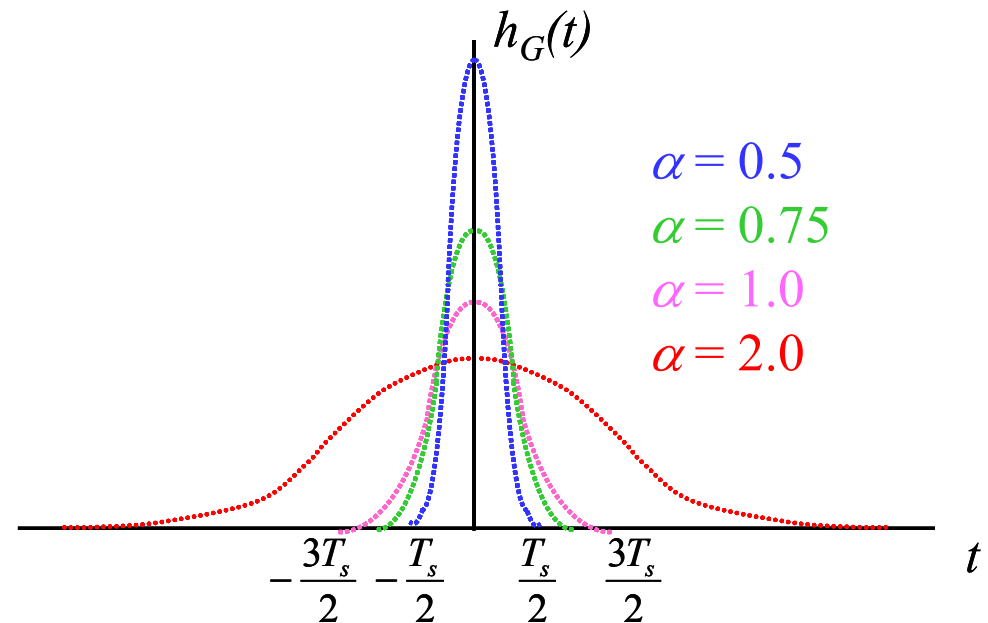
## Baseband RC Filter

- impulse response plotted for  $0 \leq \alpha \leq 1$



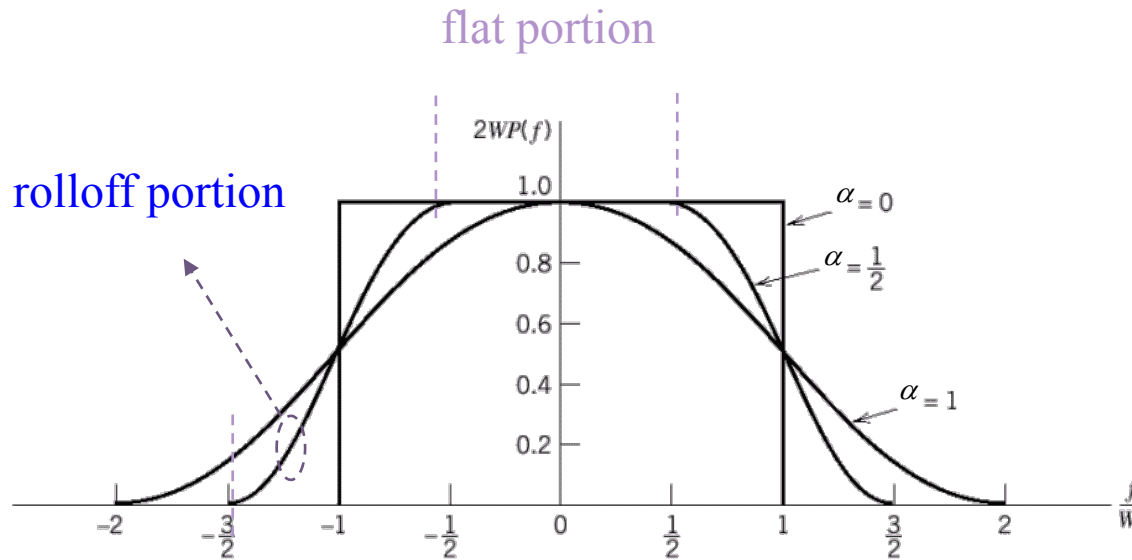
## Baseband Gaussian Filter

- impulse response plotted for different  $B_{3dB}T_s$



# Raised cosine spectrum

## -Frequency response

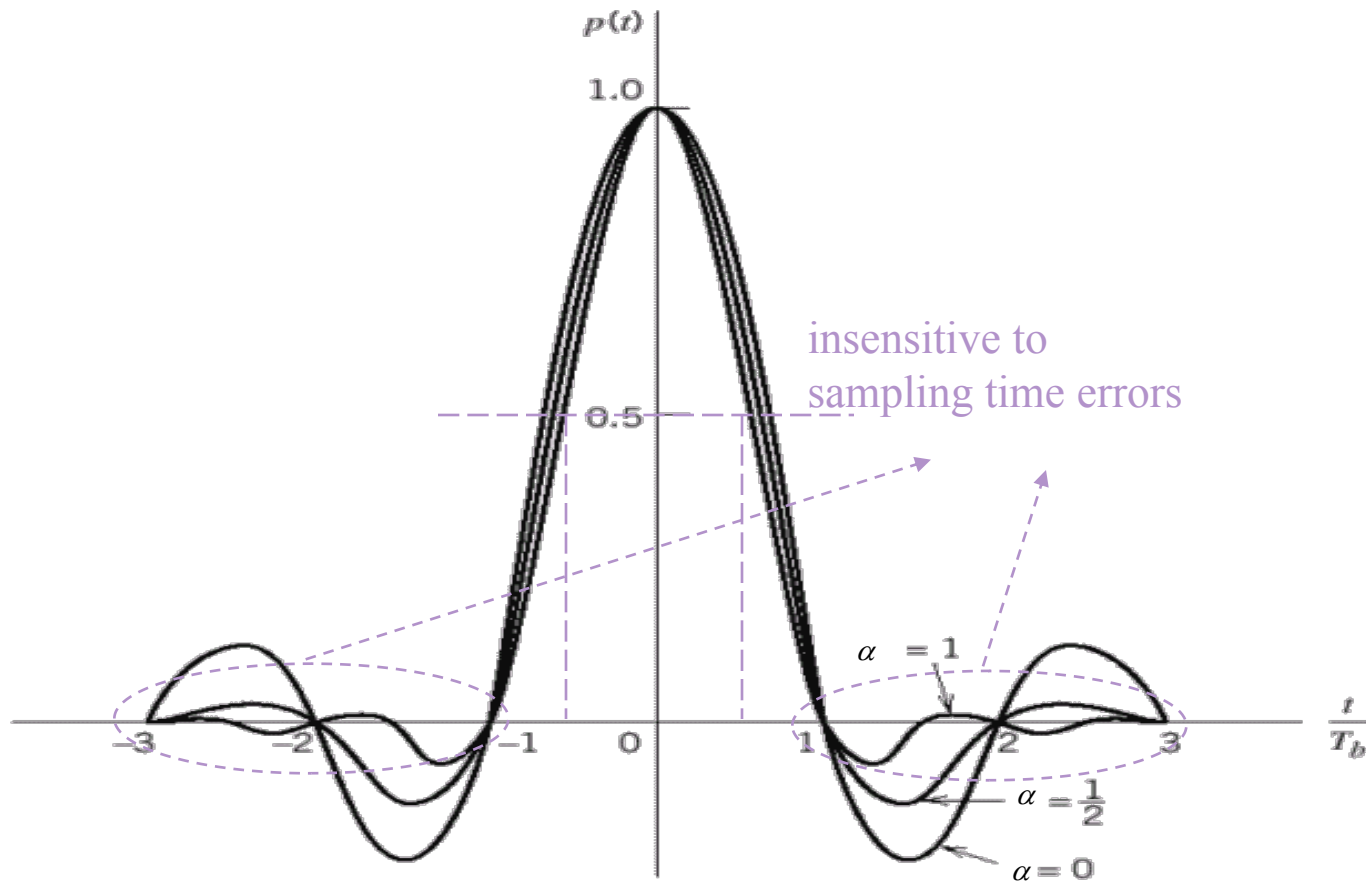


$\alpha = 1 - \frac{f_1}{W} \rightarrow \alpha$  is called the rolloff factor; it indicates excess bandwidth over the ideal solution,  $W$ .

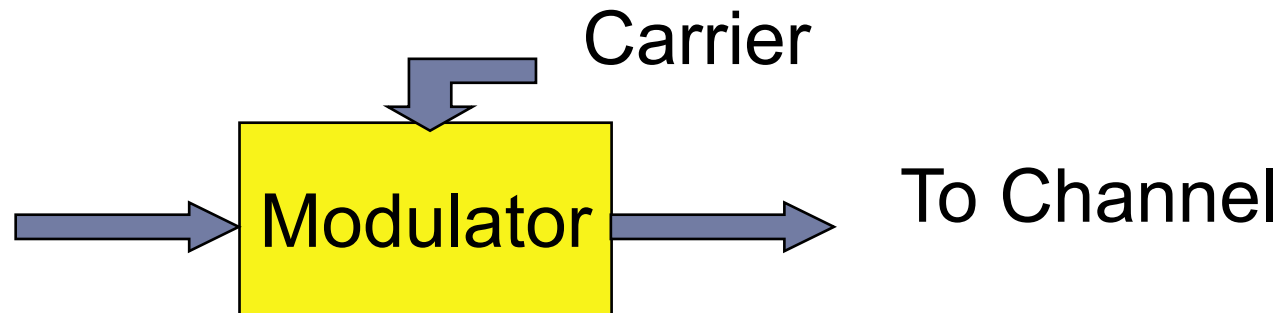
$B_T = 2W - f_1 \rightarrow$  Transmission bandwidth  
 $= W(1 + \alpha)$

# Raised cosine spectrum

## - Time response



# Bandpass Processing



- ✓ **Modulation** is the process of encoding information from a message source in a manner suitable for transmission through the channel.

# Passband Modulation

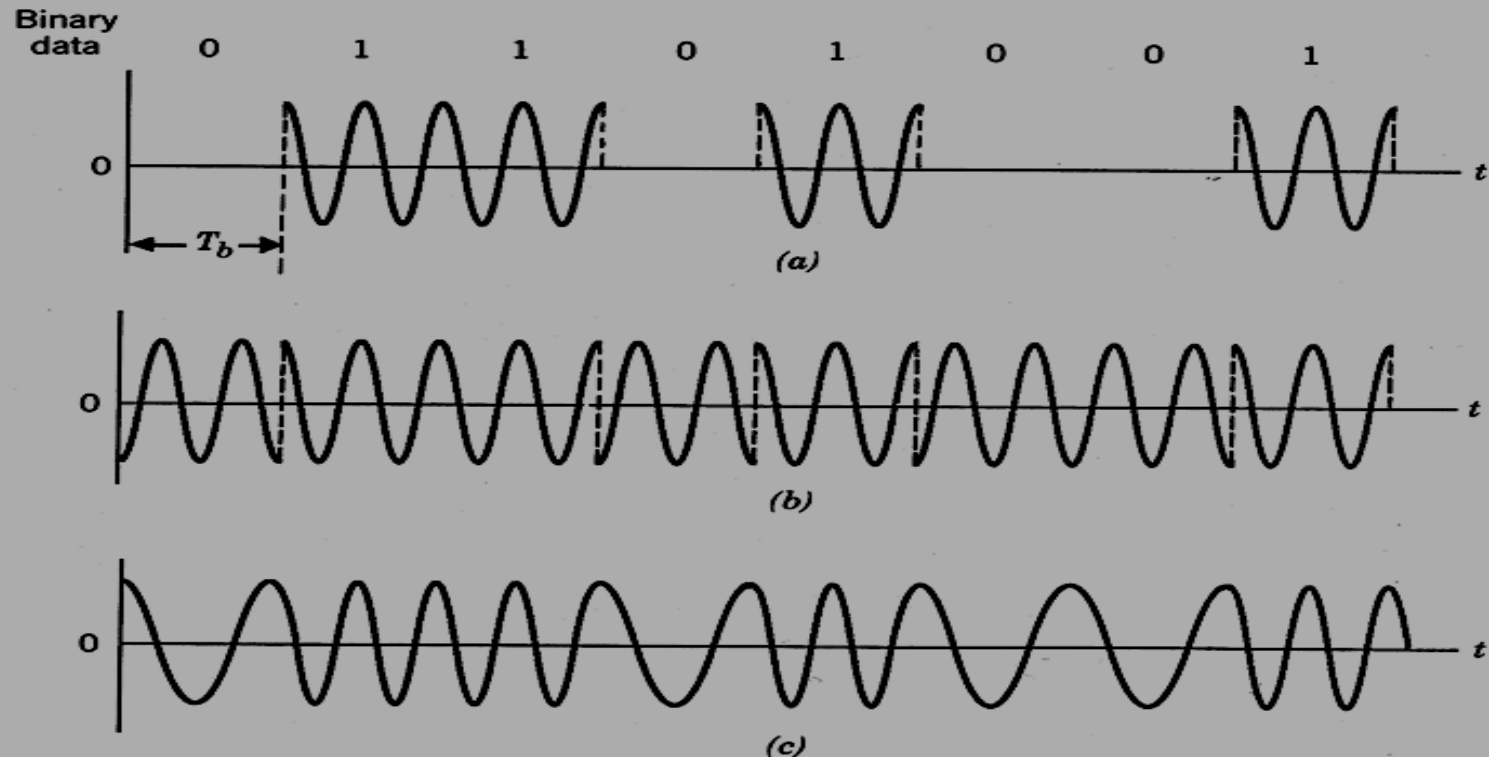
## □ Analog

- AM → Amplitude Modulation
- FM → Frequency Modulation
- PM → Phase Modulation

## □ Digital

- ASK → Amplitude Shift Keying
- FSK → Frequency Shift Keying
- PSK → Phase Shift Keying

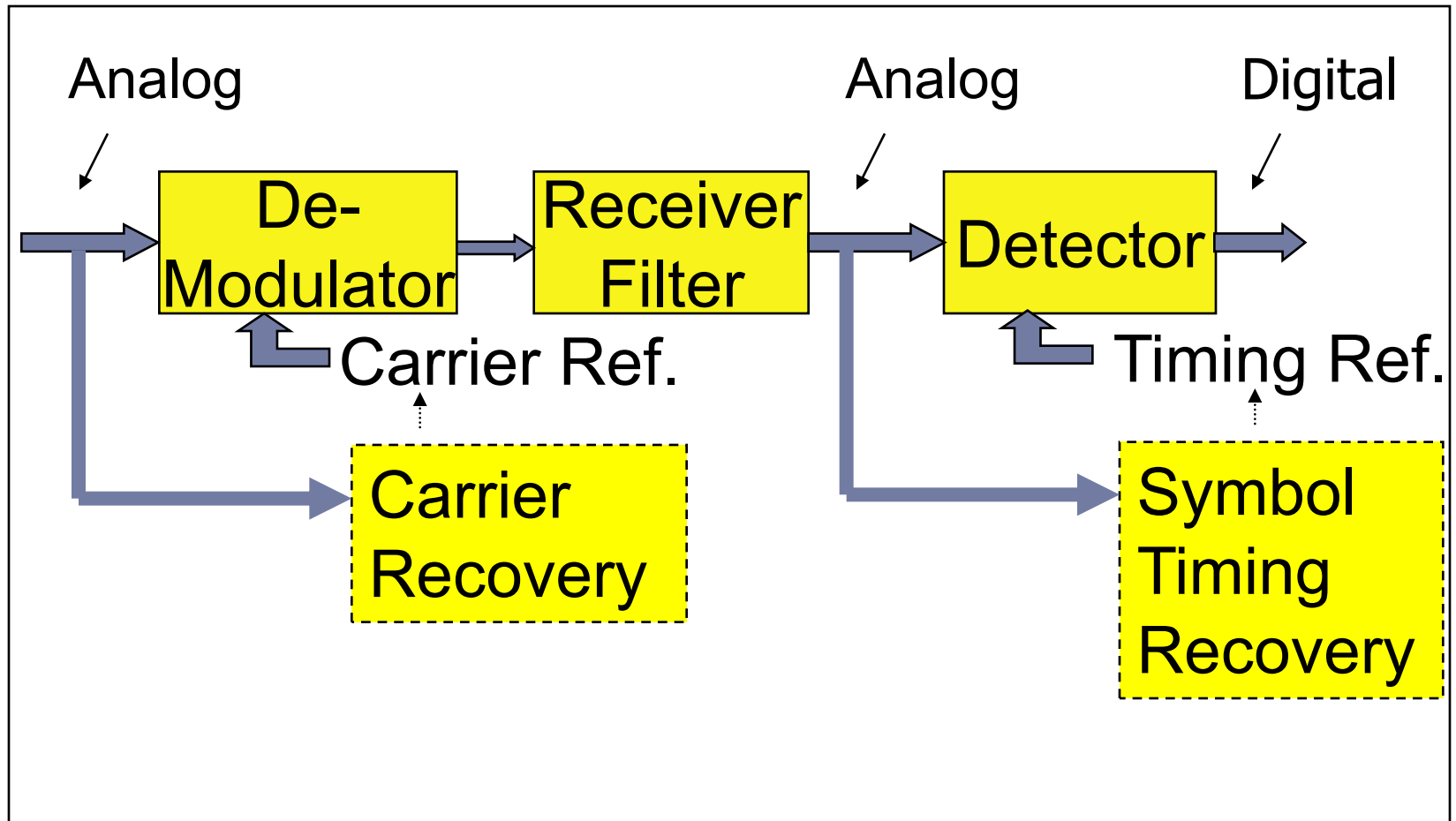
# ASK, FSK, PSK



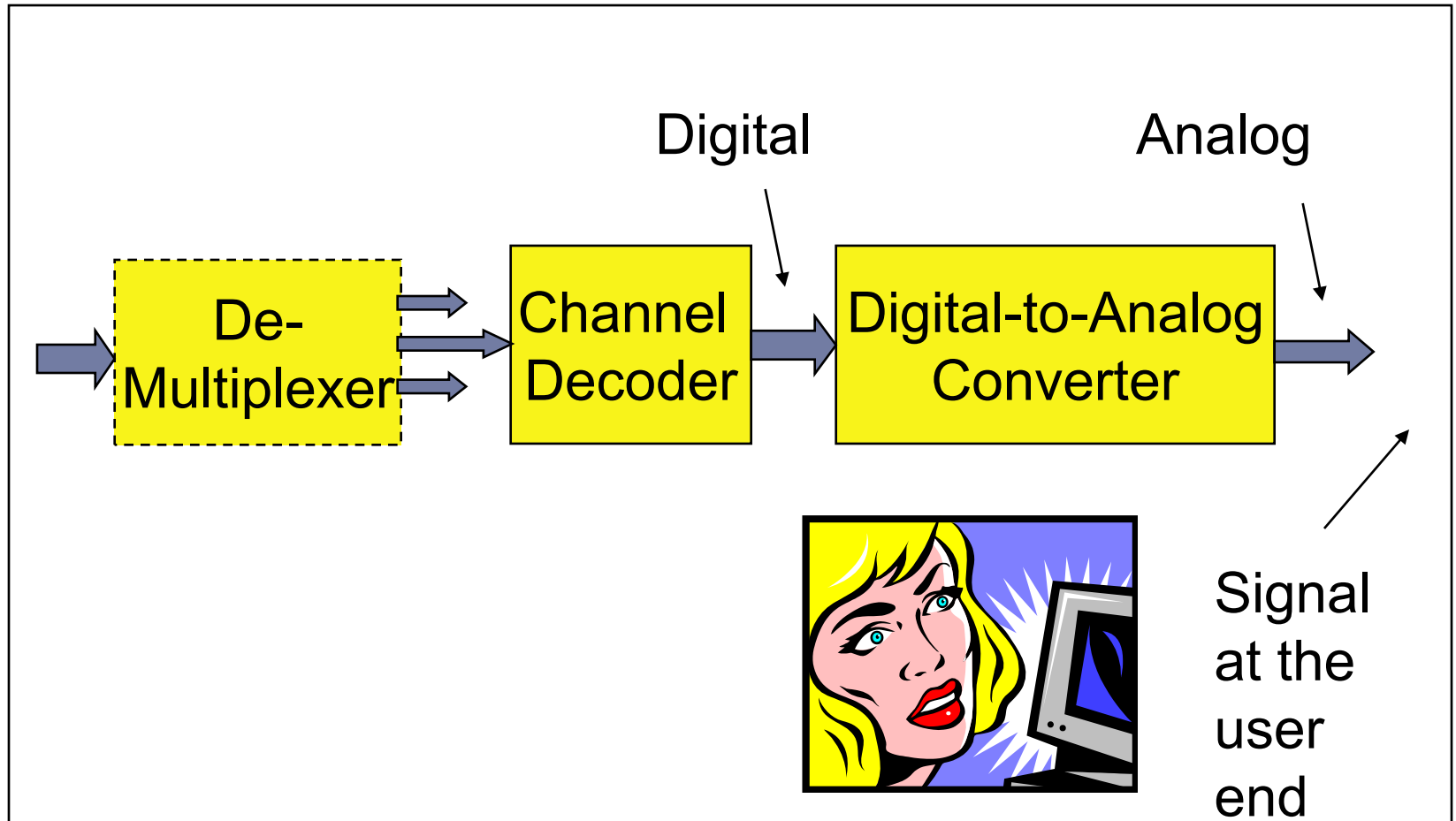
**Figure 8.2** The three basic forms of signaling binary information. (a) Amplitude-shift keying. (b) Phase-shift keying. (c) Frequency-shift keying with continuous phase.



# Bandpass Signal Reception



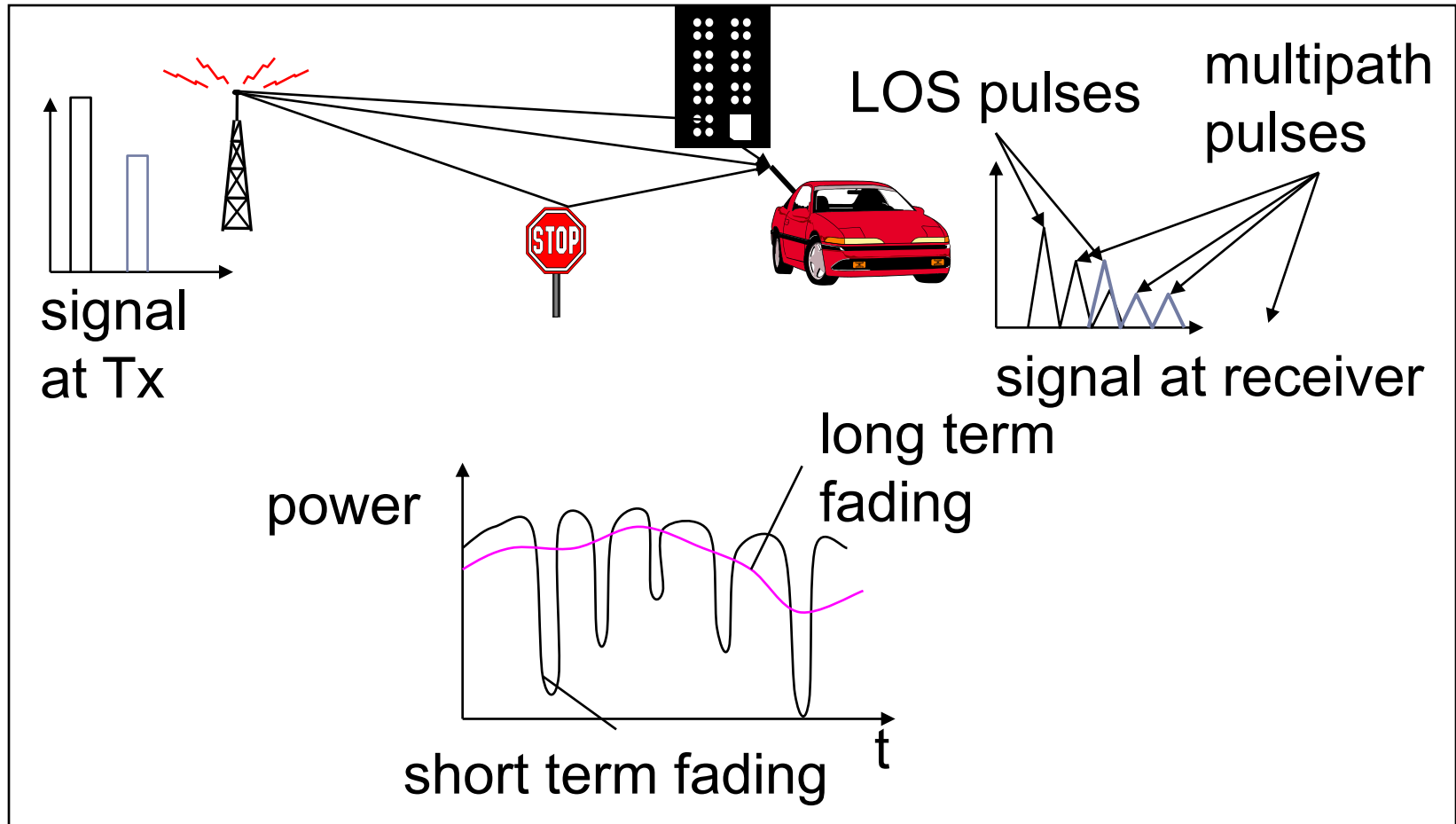
# Signal Decoding and Reception



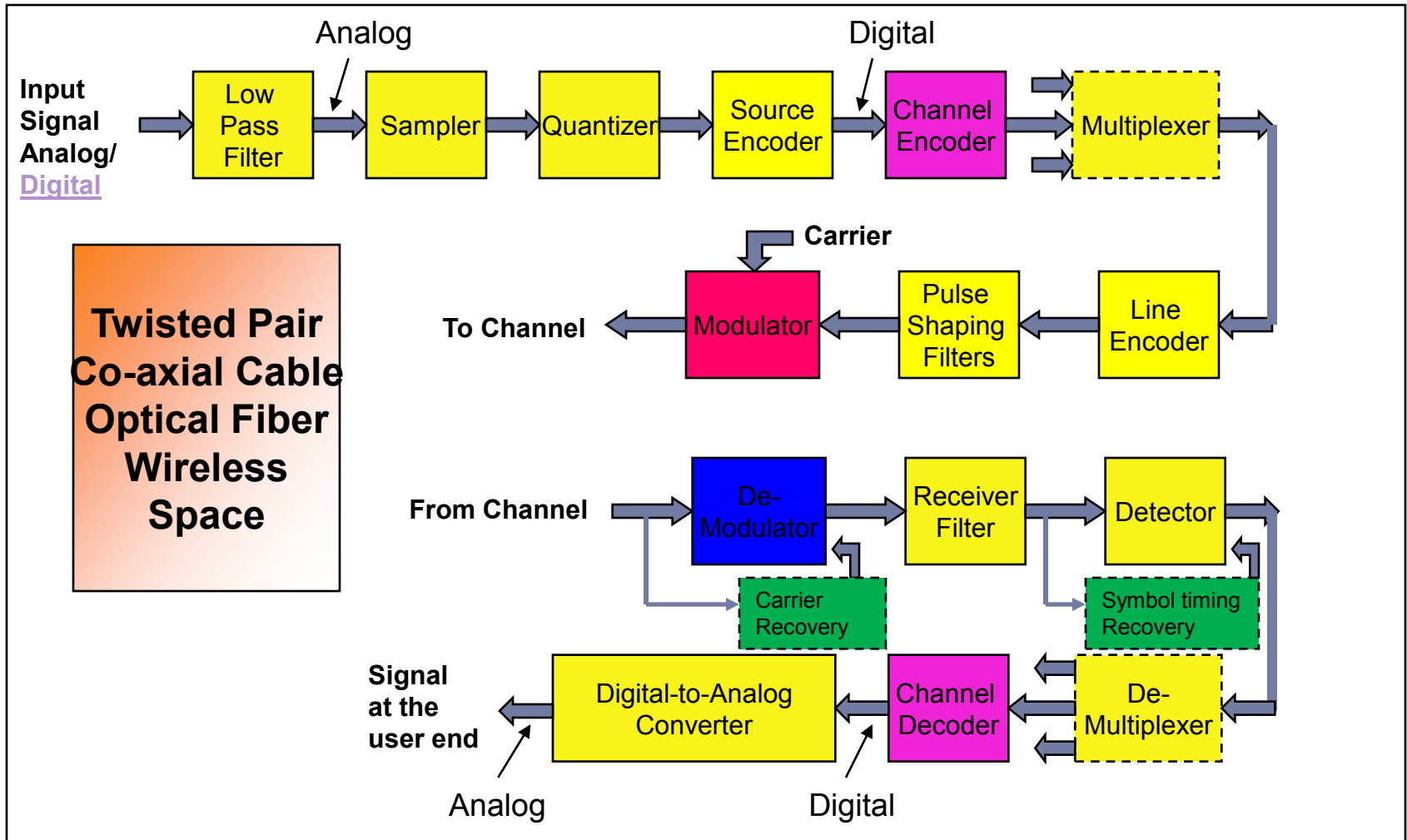
# Goal of Advanced Modulation & Coding Techniques

- **Modulation / coding is difficult task in hostile channels like the mobile radio channels**
  - **Small-scale fading and multipath conditions.**
- **The goal of a modulation / coding scheme is:**
  - **To transport the message signal through the radio channel with best possible quality.**
  - **To occupy least amount of (eg. radio ,RF) spectrum.**

# Multipath Propagation



# Digital communication system



# REVIEW OF DIGITAL MODULATION TECHNIQUES

Base band and band pass communication

Signal space representation

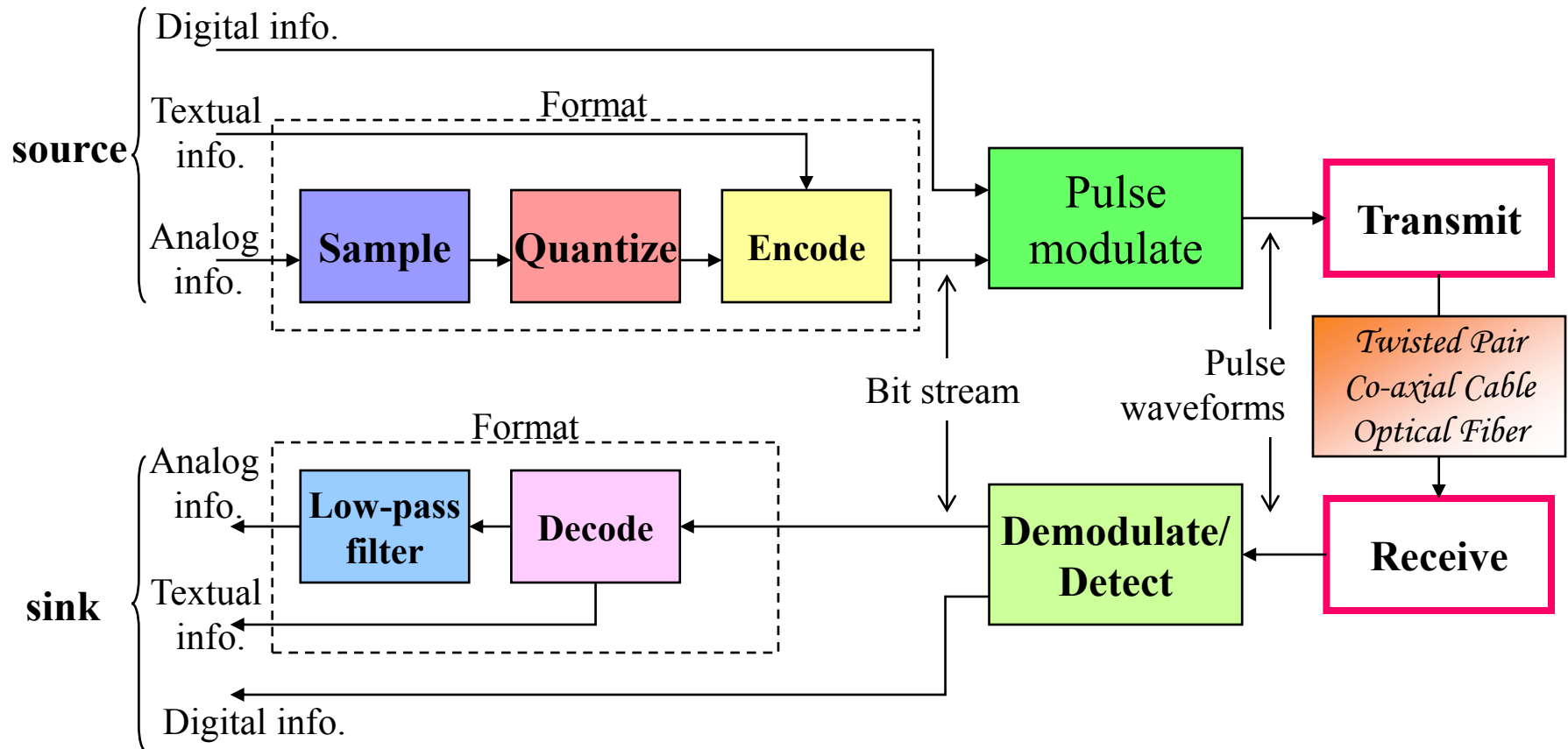
Linear and nonlinear modulation techniques

M-ary modulation techniques

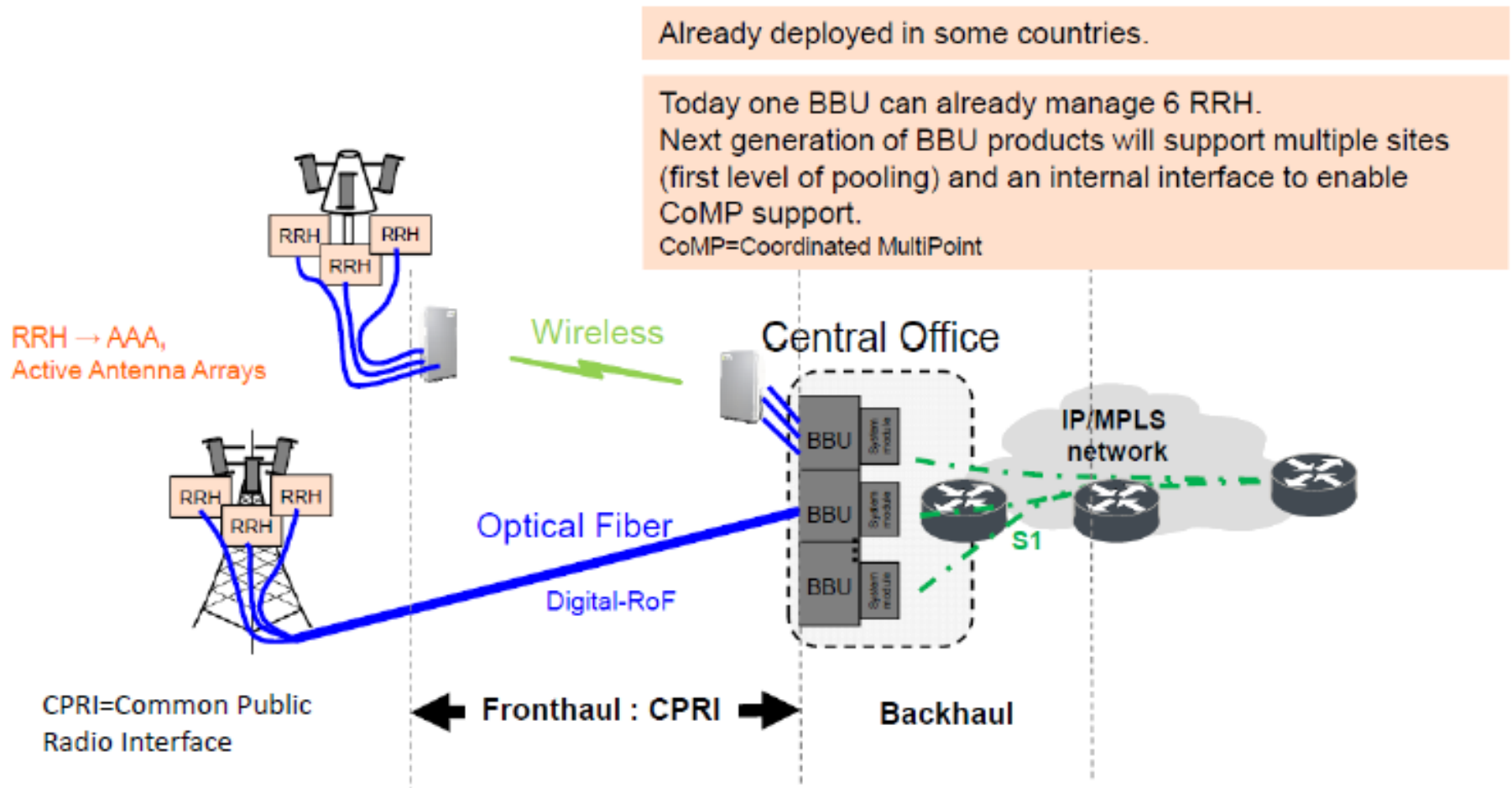
Spectral characteristics of digital modulation

Spread spectrum modulation techniques

# Baseband Communication System



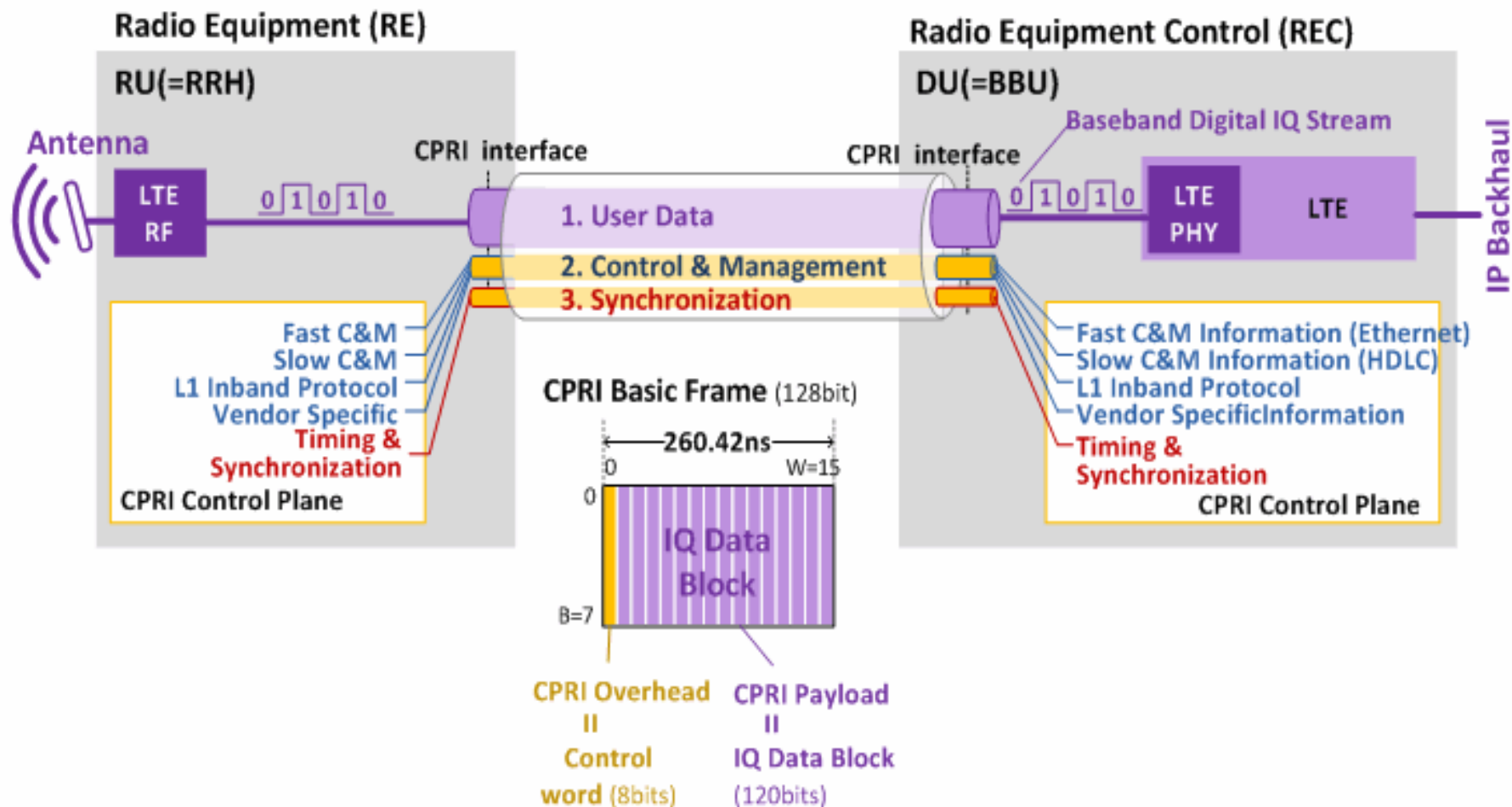
# Example : Cellular Backhaul Transport



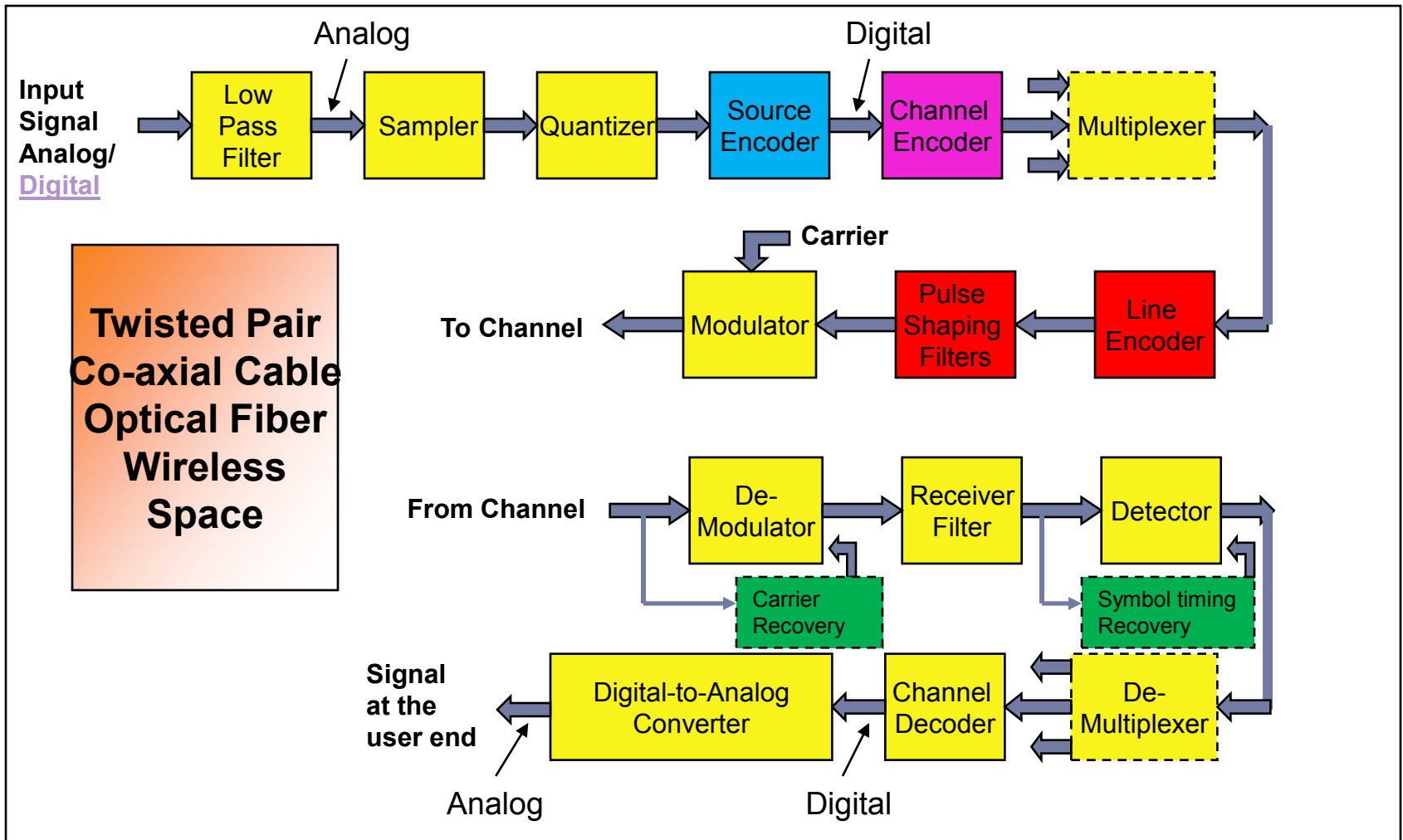
Fiber between BBU and remote radio head (RRH) known as Cloud RAN (CRAN).  
Figure was provided by Anna Pizzinat and Philippe Chanclou from Orange Labs Networks



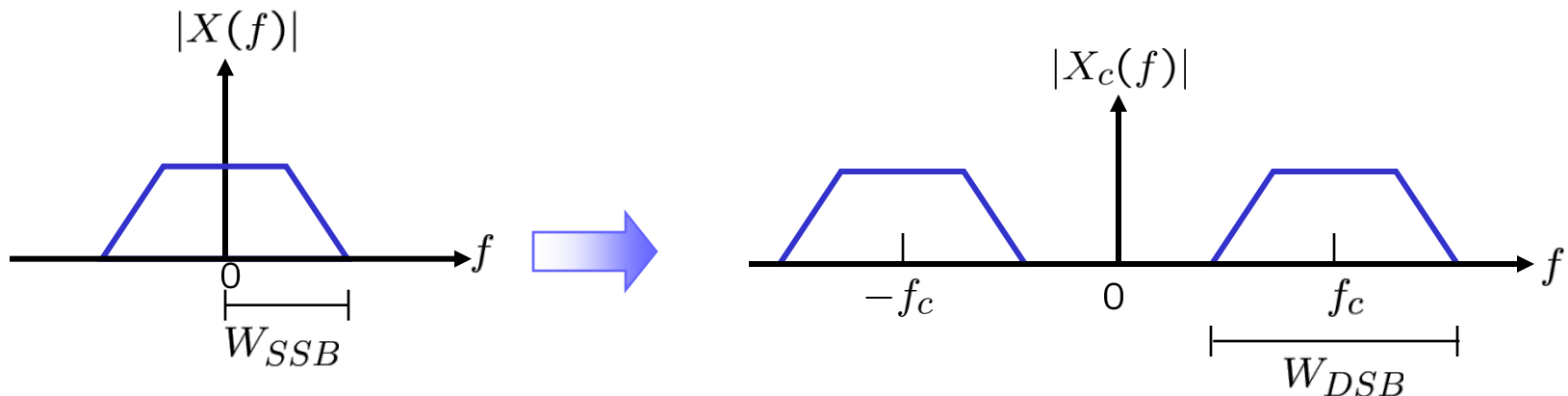
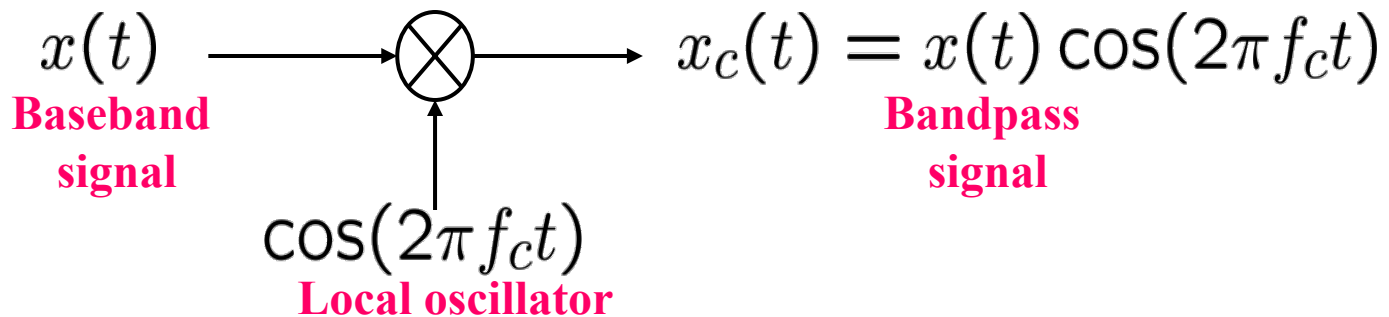
## ■ CPRI Transport Concept



# Bandpass Communication System



# Baseband Versus Bandpass

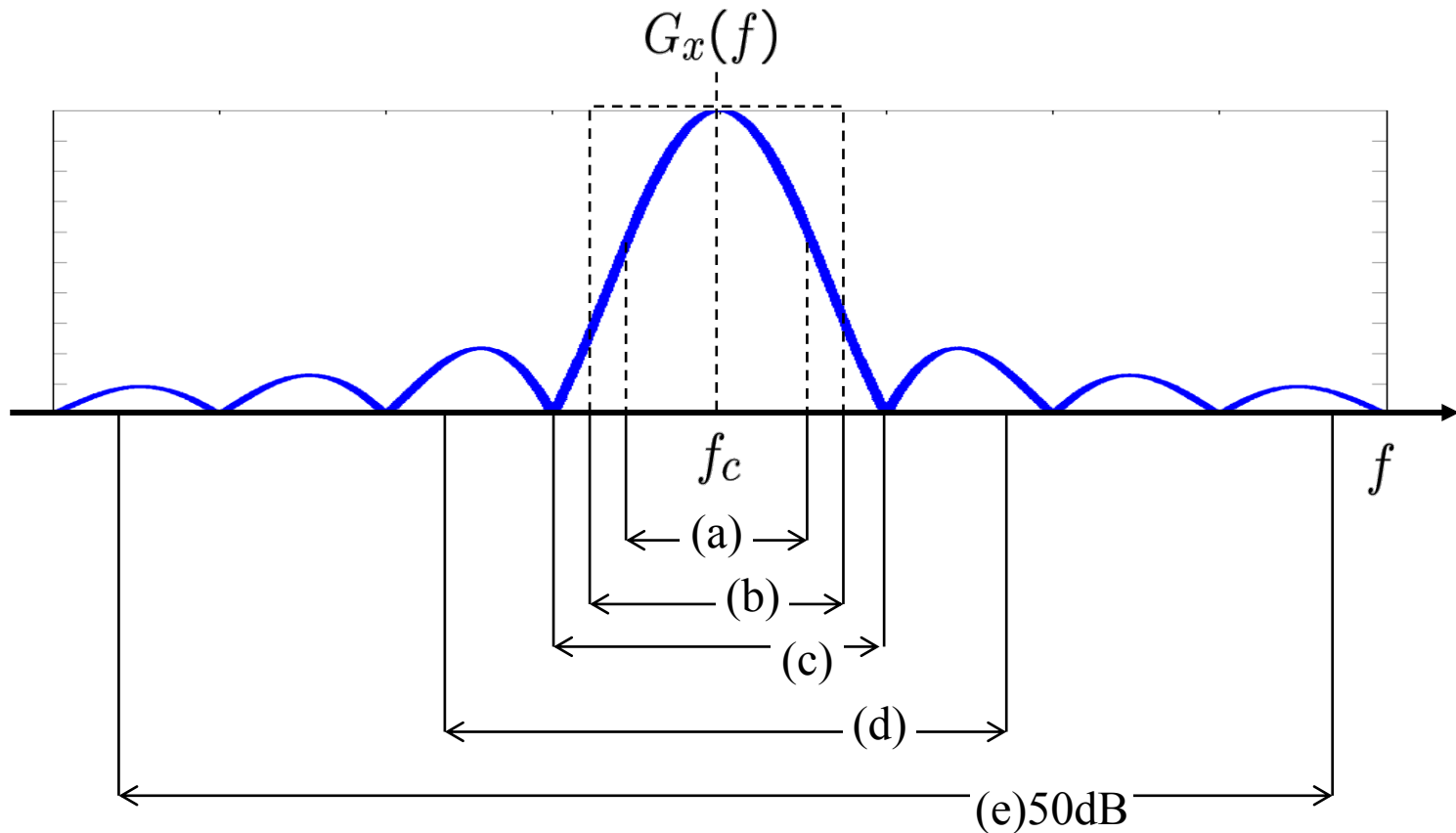


Bandwidth dilemma:

- Bandlimited signals are not realizable!
- Realizable signals have infinite bandwidth!

# Different definitions of bandwidth

- a) Half-power bandwidth
- b) Noise equivalent bandwidth
- c) Null-to-null bandwidth
- d) Fractional power containment bandwidth
- e) Bounded power spectral density
- f) Absolute bandwidth



# Factors that Influence Choice of Digital Modulation/Coding Techniques

- A desired modulation scheme
  - Should provide low bit-error rates at low SNR
    - Power efficiency
  - Should occupy minimum RF channel bandwidth
    - Bandwidth efficiency
  - Should perform well in multi-path and fading conditions
  - Should be easy and cost-effective to implement
- Depending on the demands of a particular system or application, tradeoffs are made when selecting a digital modulation/coding scheme.

# Noiseless Channels and Nyquist Theorem

For a noiseless channel, Nyquist theorem gives the relationship between the channel bandwidth and maximum data rate that can be transmitted over this channel.

Nyquist Theorem

$$C = 2W \log_2 m$$

$C$ : channel capacity (bps)

$W$ : RF bandwidth

$m$ : number of finite states in a symbol of transmitted signal

**Example: A noiseless channel with 3kHz bandwidth can transmit a maximum of 6kbps if the symbols are binary symbols.**

# Shannon's Bound for noisy channels

There is a fundamental upper bound on achievable bandwidth efficiency.

Shannon's theorem gives the relationship between the channel bandwidth and the maximum data rate that can be transmitted over a noisy channel .

## Shannon's Theorem

$$\eta_{B\max} = \frac{C}{W} = \log_2 \left( 1 + \frac{S}{N} \right)$$

C: channel capacity (maximum data-rate) (bps)

B or W : RF bandwidth

S/N: signal-to-noise ratio (no unit)

# Shannon's Theorem

- Shannon theorem puts a limit on transmission data rate, not on error probability:
  - Theoretically possible to transmit information at any rate  $R_b$ , where  $R_b \leq C$  with an arbitrary small error probability by using a sufficiently complicated coding scheme.
  - For an information rate  $R_b > C$ , it is not possible to find a code that can achieve an arbitrary small error probability.



# Power Efficiency of Modulation

- Power efficiency is the ability of the modulation technique to preserve fidelity of the message at low power levels.
- Usually in order to obtain good fidelity, the signal power needs to be increased.
  - Tradeoff between fidelity and signal power
  - Power efficiency describes how efficiently this tradeoff is made

$$\text{Power Efficiency: } \eta_p = \left[ \frac{E_b}{N_0} \text{ required at the receiver input for certain } PER \right]$$

$E_b$ : signal energy per bit

$N_0$ : noise power spectral density

$PER$ : probability of error

# Bandwidth Efficiency of Modulation

- Ability of a modulation scheme to accommodate data within a limited bandwidth.
- Bandwidth efficiency reflect how efficiently the allocated bandwidth is utilized

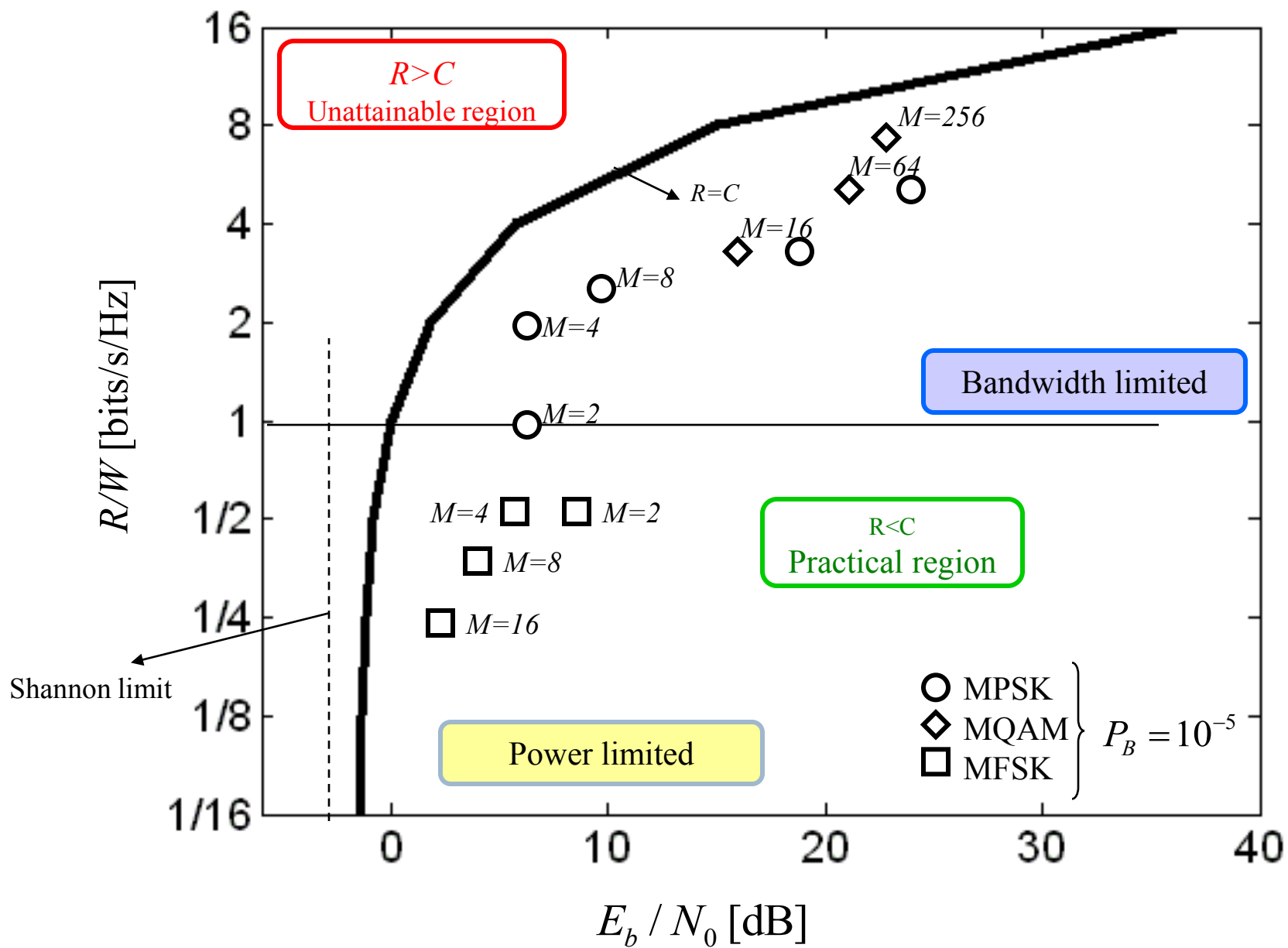
$$\text{Bandwidth Efficiency : } \eta_B = \frac{R}{W} \text{ bps/Hz}$$

R: the data rate (bps)

W: bandwidth occupied by the modulated RF signal

# Tradeoff between BW Efficiency and Power Efficiency

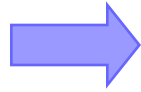
- Adding error control codes
  - Improves the power efficiency
    - » Reduces the required received power for a particular bit error rate
  - Decreases the bandwidth efficiency
    - » Increases the bandwidth occupancy
- M-ary keying modulation
  - Increases the bandwidth efficiency
  - Decreases the power efficiency
    - » More power is required at the receiver



# Shannon limit ...

$$C = W \log_2 \left( 1 + \frac{S}{N} \right)$$

$$\begin{cases} S = E_b C \\ N = N_0 W \end{cases}$$



$$\frac{C}{W} = \log_2 \left( 1 + \frac{E_b}{N_0} \frac{C}{W} \right)$$

As  $W \rightarrow \infty$  or  $\frac{C}{W} \rightarrow 0$ , we get :

$$\frac{E_b}{N_0} \rightarrow \frac{1}{\log_2 e} = 0.693 \approx -1.6 \text{ [dB]}$$

**Shannon limit**

- There exists a limiting value of  $E_b / N_0$  below which there can be no error-free communication at any information rate.
- By increasing the bandwidth alone, the capacity cannot be increased to any desired value.

# Goals in designing a DCS

- Maximizing the transmission bit rate
- Minimizing probability of bit error
- Minimizing the required power
- Minimizing required system bandwidth
- Maximizing system utilization
- Minimize system complexity

# Geometric Representation of Modulation Signals

□ Digital Modulation involves

- Choosing a particular signal waveform for transmission for a particular symbol
- For M possible symbols, the set of all signal waveforms are:

$$S = \{s_1(t), s_2(t), \dots, s_M(t)\}$$

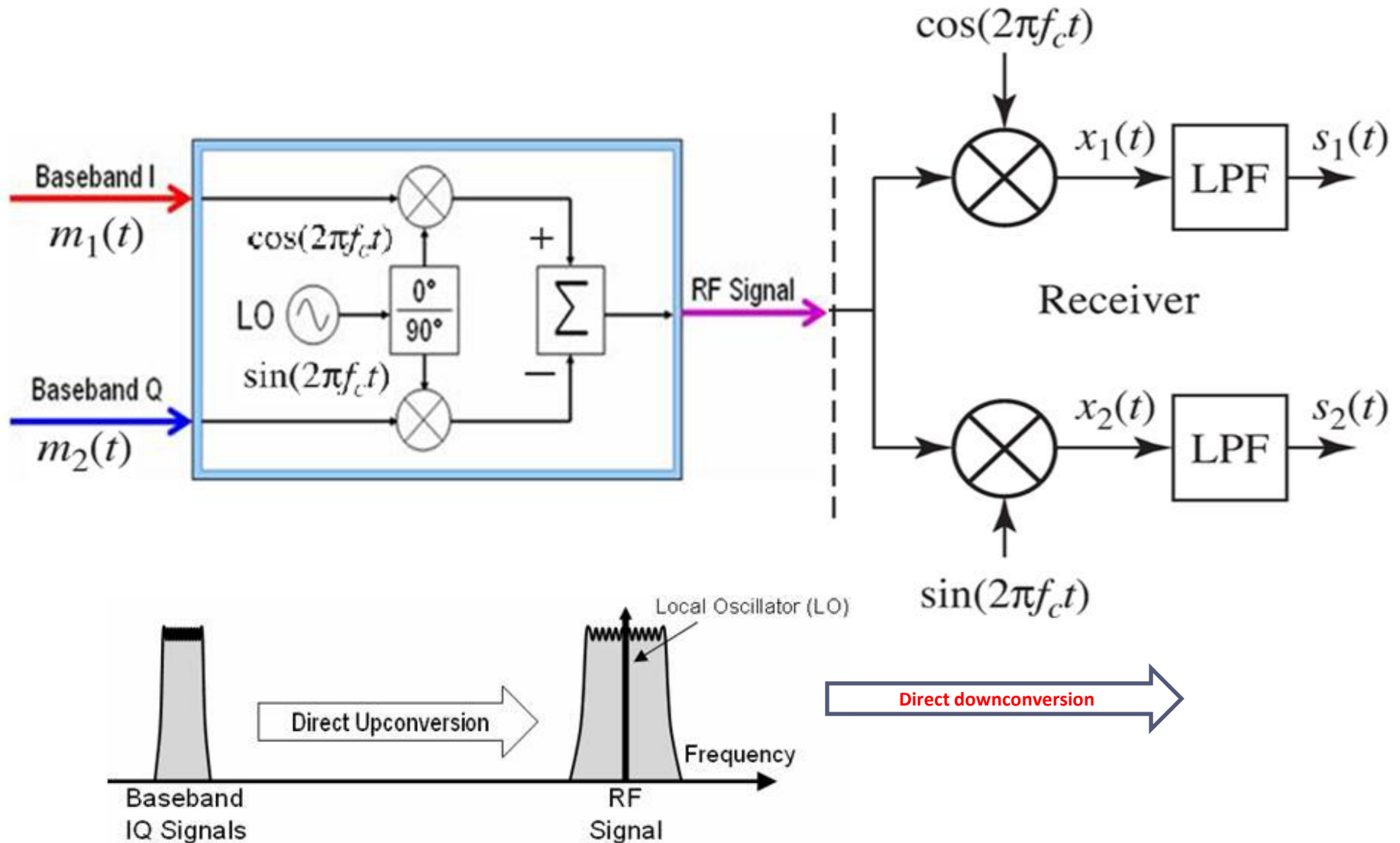
# Symbol to Signal Mapping

## QPSK

| Symbol | Energy Signal                    |
|--------|----------------------------------|
| 00     | $A \cos ( 2\pi f_c t + \phi_1 )$ |
| 01     | $A \cos ( 2\pi f_c t + \phi_2 )$ |
| 11     | $A \cos ( 2\pi f_c t + \phi_3 )$ |
| 10     | $A \cos ( 2\pi f_c t + \phi_4 )$ |



# Quadrature Modulation and Demodulation



# Symbol to Signal Mapping

## QPSK

| Symbol | Txn. Encoder Output<br>(Baseband mod. Signal) | Passband Energy Signal            |
|--------|-----------------------------------------------|-----------------------------------|
| 00     | $A/(2)^{1/2} + jA/(2)^{1/2}$                  | $A \cos (2\pi f_c t + 45^\circ)$  |
| 01     | $-A/(2)^{1/2} + jA/(2)^{1/2}$                 | $A \cos (2\pi f_c t + 135^\circ)$ |
| 11     | $-A/(2)^{1/2} - jA/(2)^{1/2}$                 | $A \cos (2\pi f_c t + 225^\circ)$ |
| 10     | $A/(2)^{1/2} - jA/(2)^{1/2}$                  | $A \cos (2\pi f_c t + 315^\circ)$ |

# Geometric Representation of Modulation Signals

- Eg. For any modulation, each bit / group of bits is mapped to a signal from a signal set  $S$  that has real energy signals.
- Elements of  $S$  can be viewed as points in a **vector space / geometric space**.

# Geometric Representation of Modulation Signals

## ➤ Vector space

- ✓ We can represent any set of  $M$  energy signals  $\{s_i(t)\}$  as a linear combination of  $N$  orthogonal basis signals  $\phi_i(t)$ , where  $N \leq M$ .

$$S = \{s_1(t), s_2(t), \dots, s_M(t)\}$$

$$\phi_1(t), \phi_2(t), \dots, \phi_N(t)$$

# Geometric Representation of Modulation Signals

Let  $\{\phi_j(t) | j = 1, 2, \dots, N\}$  represent

a basis of  $\mathcal{S}$ , such that,

**Any symbol,  $s_i(t) \rightarrow$**

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t)$$

# Geometric Representation of Modulation Signals

- Basis signals  $\approx$   
Coordinate system for **S**
- Number of basis signals  $\rightarrow$   
the **dimension** of the  
vector space.

# Geometric Representation of Modulation Signals

- ✓ Basis signals should be orthogonal to each-other.

$$\int_{-\infty}^{\infty} \phi_i(t) \phi_j(t) dt = 0 \quad i \neq j$$

- ✓ Eg. **Sin (2π f t ) , Cos (2π f t )**

# Geometric Representation of Modulation Signals

- ✓ Each basis is normalized to have unit energy.

$$E_i = \int_{-\infty}^{\infty} \phi_i^2(t) dt = 1$$

$$i = 1, 2, \dots, N$$



# Geometric Representation of signals

$$S = \{s_1(t), s_2(t), \dots, s_M(t)\}$$

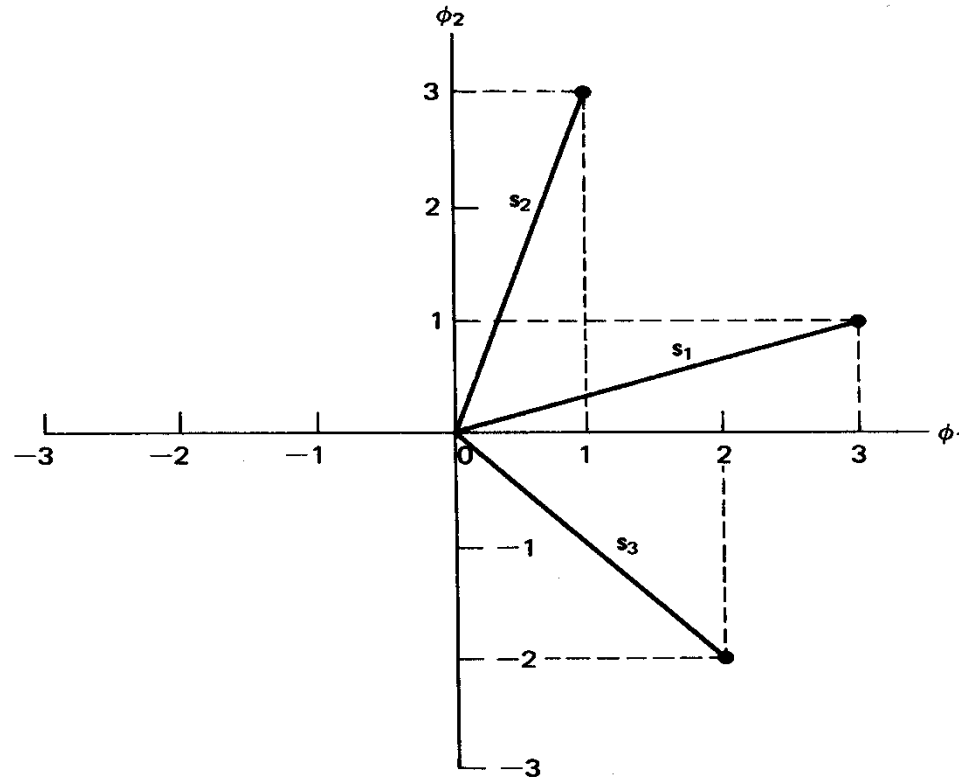
$$\{ \phi_j(t) \mid j = 1, 2, \dots, N \}$$

$$S_i = \begin{bmatrix} s_{i1} \\ s_{i2} \\ \bullet \\ \bullet \\ s_{iN} \end{bmatrix} \quad i = 1, 2, \dots, M$$

# Geometric Representation of Modulation Signals

$$S_i = \begin{bmatrix} s_{i1} \\ s_{i2} \\ \vdots \\ s_{iN} \end{bmatrix}$$

$$i = 1, 2, \dots, M$$

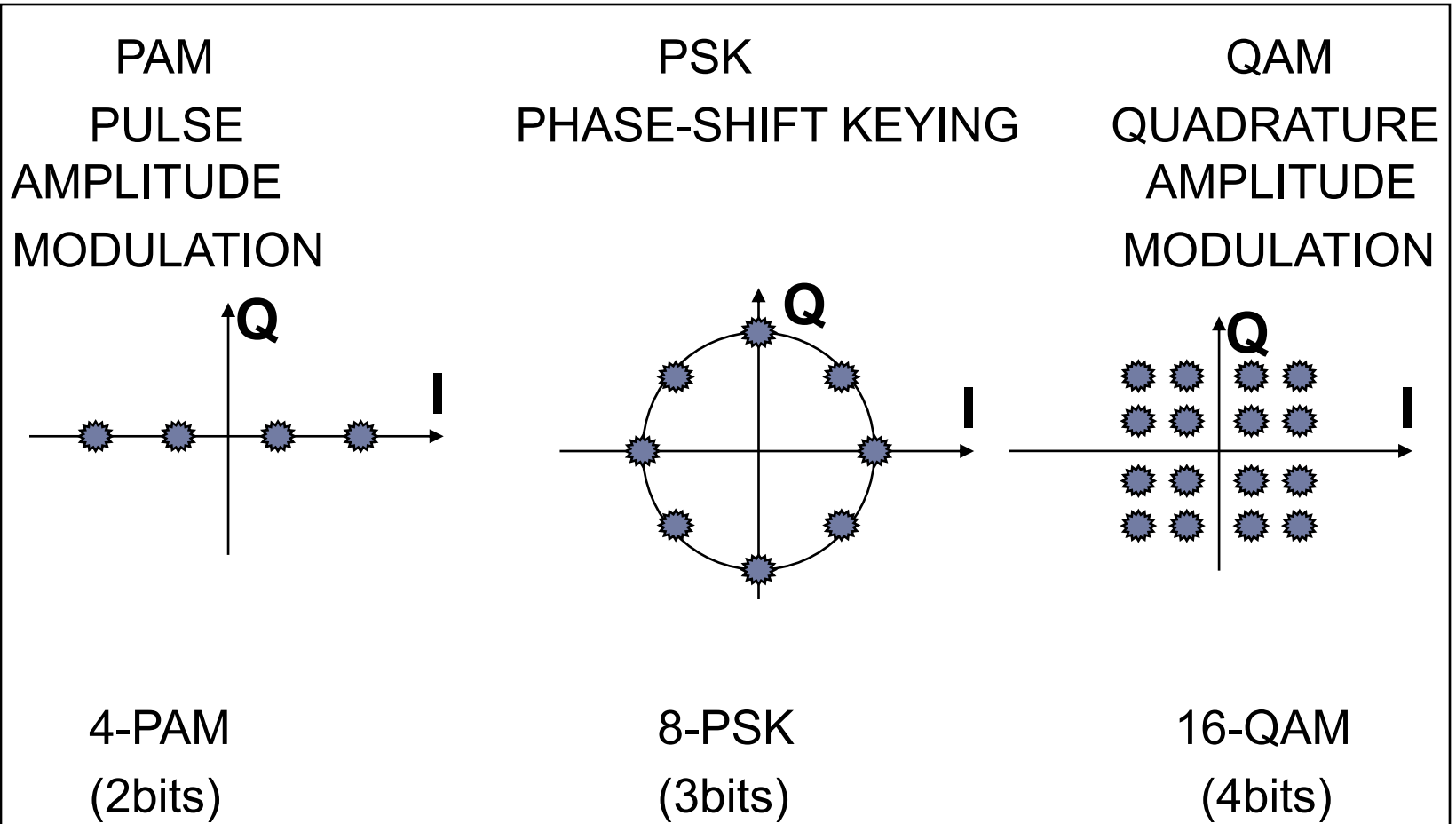


**Figure 8.6** Illustrating the geometric representation of signals for the case when  $N = 2$  and  $M = 3$ .

# Geometric Representation of Modulation Signals

**Gram-Schmidt procedure →  
systematic way to obtain basis  
for any real signal set  $S$**

# Example Constellations



# Constellation Diagram

- Properties of Modulation Scheme that can be inferred from the Constellation Diagram:
  - ✓ Bandwidth occupied by the modulation increases as the dimension of the modulated signal space increases.
  - ✓ Bandwidth occupied by the modulation decreases as the signal-points per dimension increases (getting more dense).

# Constellation Diagram

- Properties of Modulation Scheme that can be inferred from the Constellation Diagram:
  - ✓ Probability of bit error is proportional to the distance between the closest points in the constellation.
    - » Euclidean Distance
    - » BER decreases as the distance increases (sparse constellation).

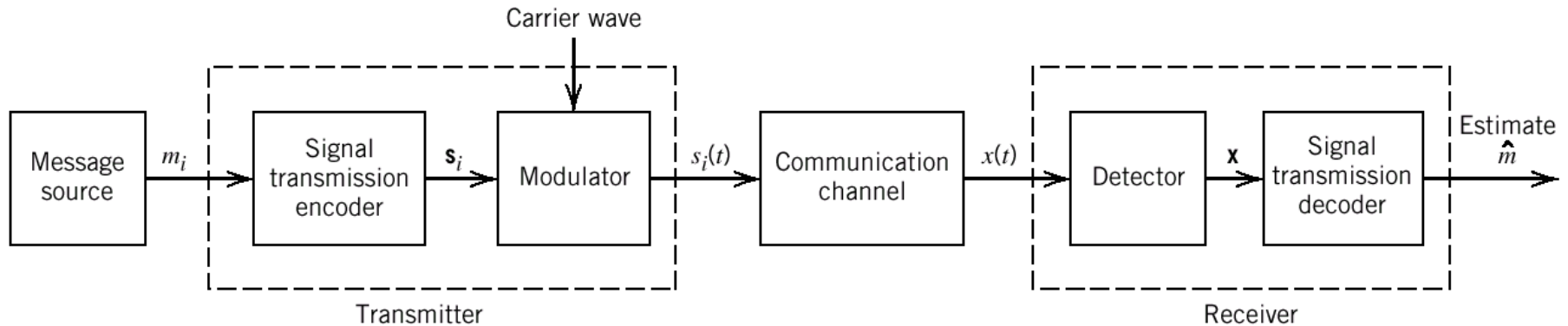
# Euclidean distance

$$d_{ik} = \|\mathbf{s}_i - \mathbf{s}_k\|$$

$$d_{ik}^2 = \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \sum_{j=1}^N (s_{ij} - s_{kj})^2$$

$$= \int_0^T [s_i(t) - s_k(t)]^2 dt$$

# Implementation Model of Digital Modulation

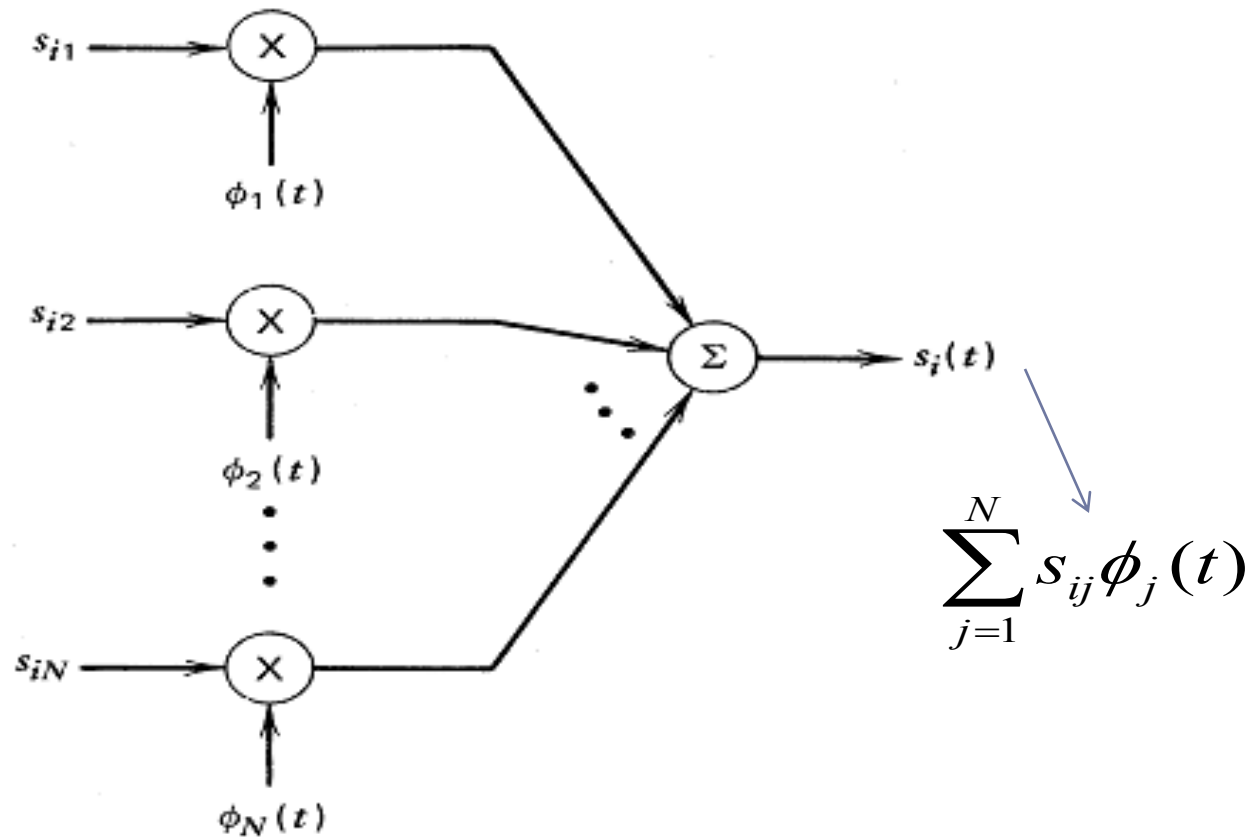


|    |            |                                     |
|----|------------|-------------------------------------|
| 00 | $A + j 0$  | $A \cos ( 2\pi f_c t + 0^\circ )$   |
| 01 | $0 + j A$  | $A \cos ( 2\pi f_c t + 90^\circ )$  |
| 11 | $-A - j 0$ | $A \cos ( 2\pi f_c t + 180^\circ )$ |
| 10 | $0 - j A$  | $A \cos ( 2\pi f_c t + 270^\circ )$ |

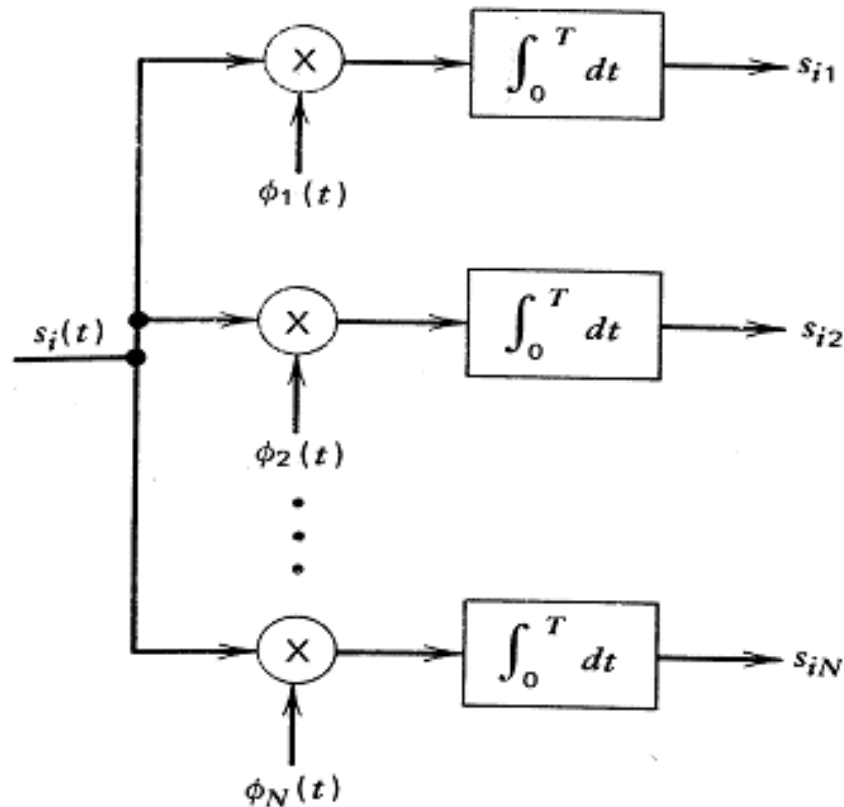
$$x(t) = \alpha.s_i(t) + n(t)$$



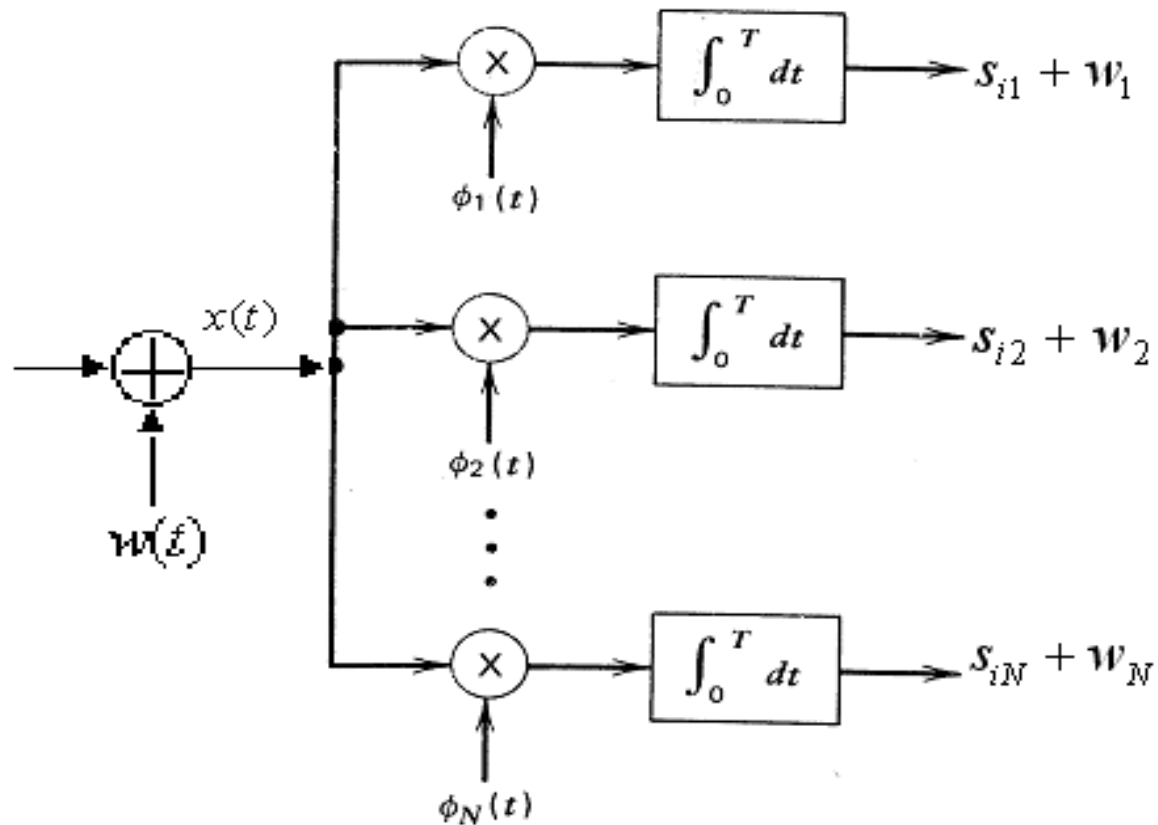
# Signal Generation



# Signal Reception



# Signal Reception in a Noisy Channel



# Signal Reception in a Noisy Channel

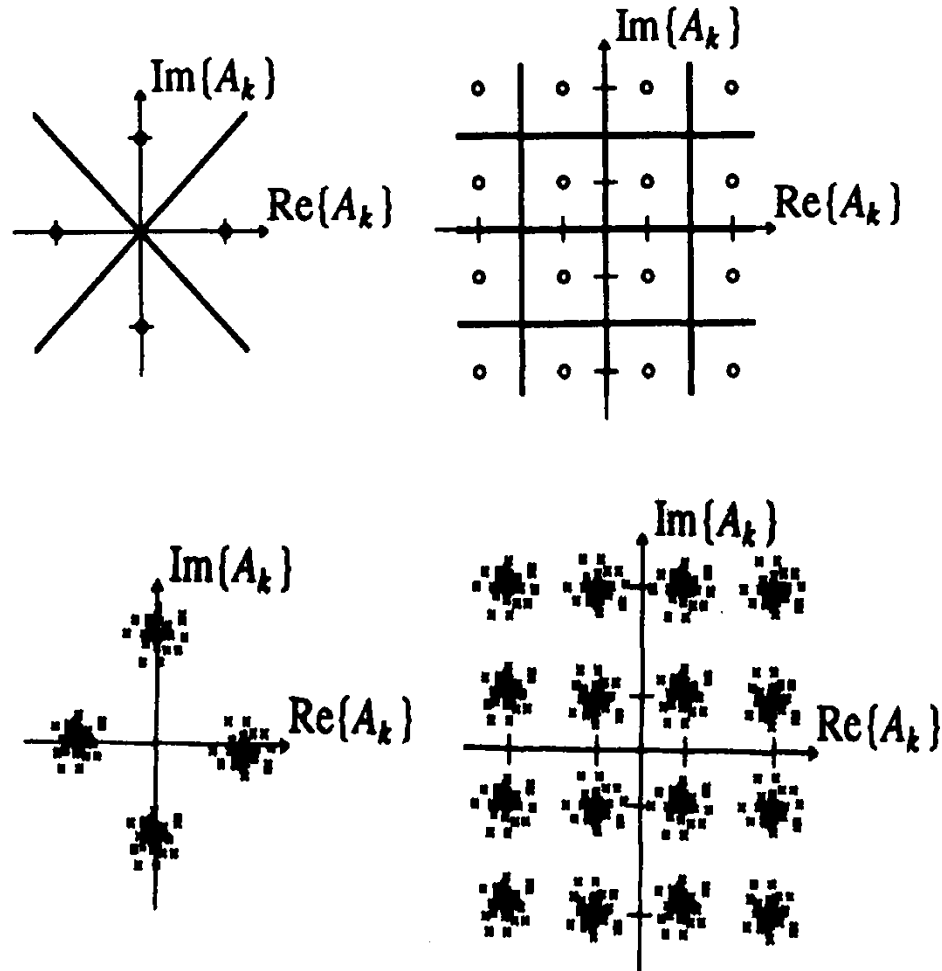
$$x(t) = s_i(t) + w(t) \quad \left\{ \begin{array}{l} 0 \leq t \leq T \\ i = 1, 2, \dots, M \end{array} \right.$$

$$x_j = \int_0^T x(t) \phi_j(t) dt = s_{ij} + w_j \quad j = 1, 2, \dots, N$$

where  $s_{ij} = \int_0^T s_i(t) \phi_j(t) dt$

$$w_j = \int_0^T w(t) \phi_j(t) dt$$

# Transmitted and Received Samples



# Exercise I

- Derive basis signals for QPSK and 8-PSK schemes
- Carry out a simulation to generate these signals using basis signals and receive the signals over noisy and noiseless channels.
- Obtain the transmit signal constellations and the receiver scatter the signals for different SNRs. Draw your inference.

# Decision rule

❖ Observation vector  $\mathbf{x}$  lies in region  $Z_j$

or if  $\sum_{j=1}^N (x_j - s_{kj})^2$  is minimum for  $k=i$

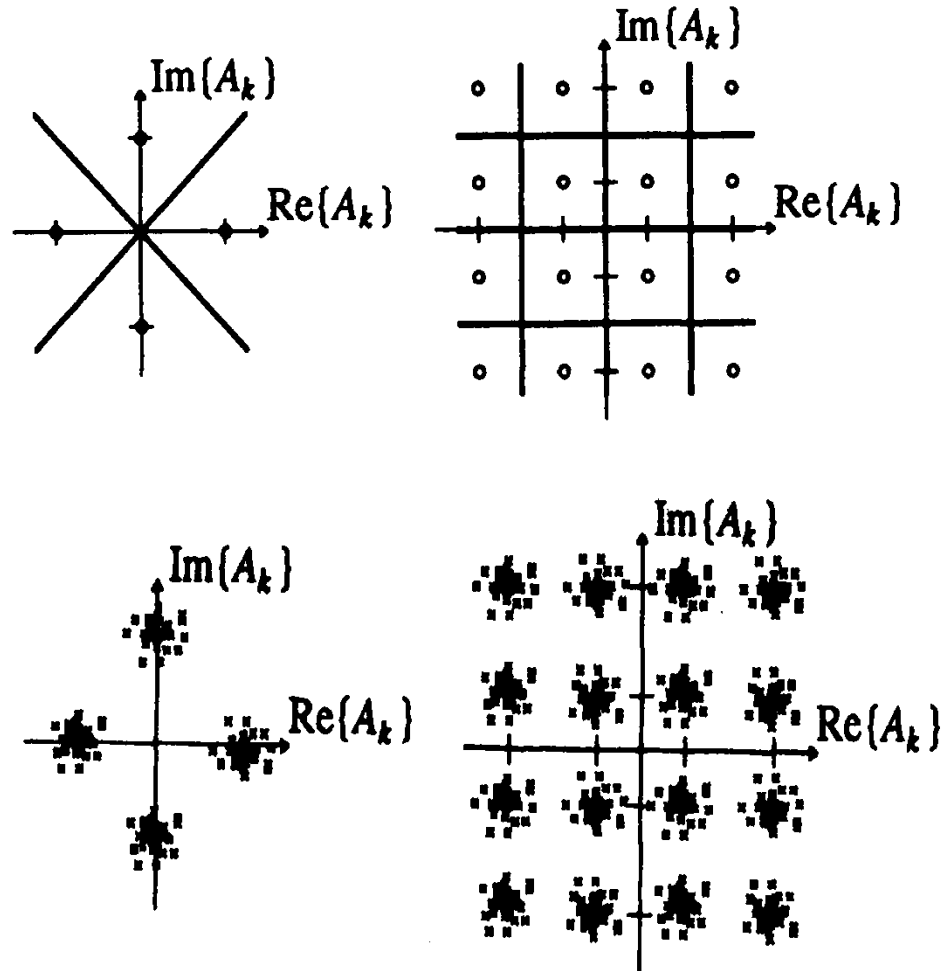
 **Squared Euclidean distance**

or if  $\|\mathbf{x} - \mathbf{s}_k\|$  is minimum for  $k=i$

---

 **Euclidean distance**

# Transmitted and Received Samples





# Decision rule

❖ Observation vector  $\mathbf{x}$  lies in region  $Z_j$

or if  $\sum_{j=1}^N (x_j - s_{kj})^2$  is minimum for  $k=i$

 **Squared Euclidean distance**

or if  $\|\mathbf{x} - \mathbf{s}_k\|$  is minimum for  $k=i$

---

 **Euclidean distance**

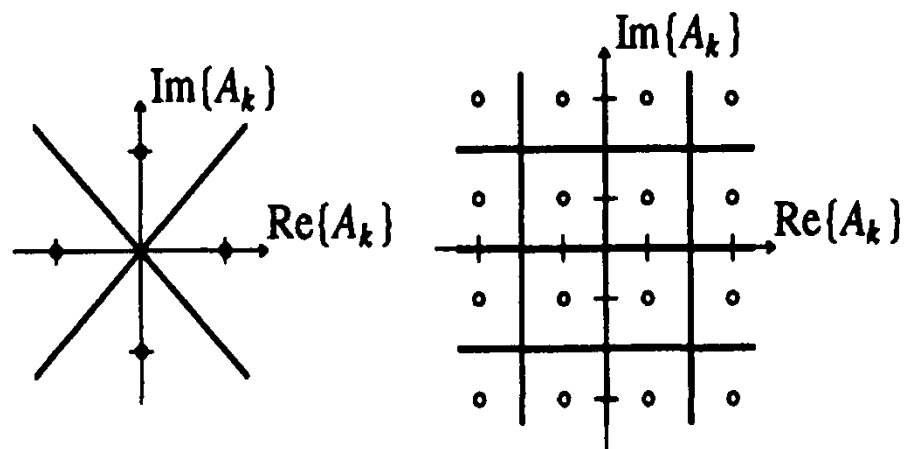
# Probability of Error

- Observation space  $Z$  is partitioned, in accordance with ML decision rule, into a set of  $M$  regions ,

$$Z_1, Z_2, \dots, Z_M$$

- Suppose  $\mathbf{m}_i$  (signal vector  $\mathbf{S}_i$ ) is transmitted
- Observation vector :  $\mathbf{x}$

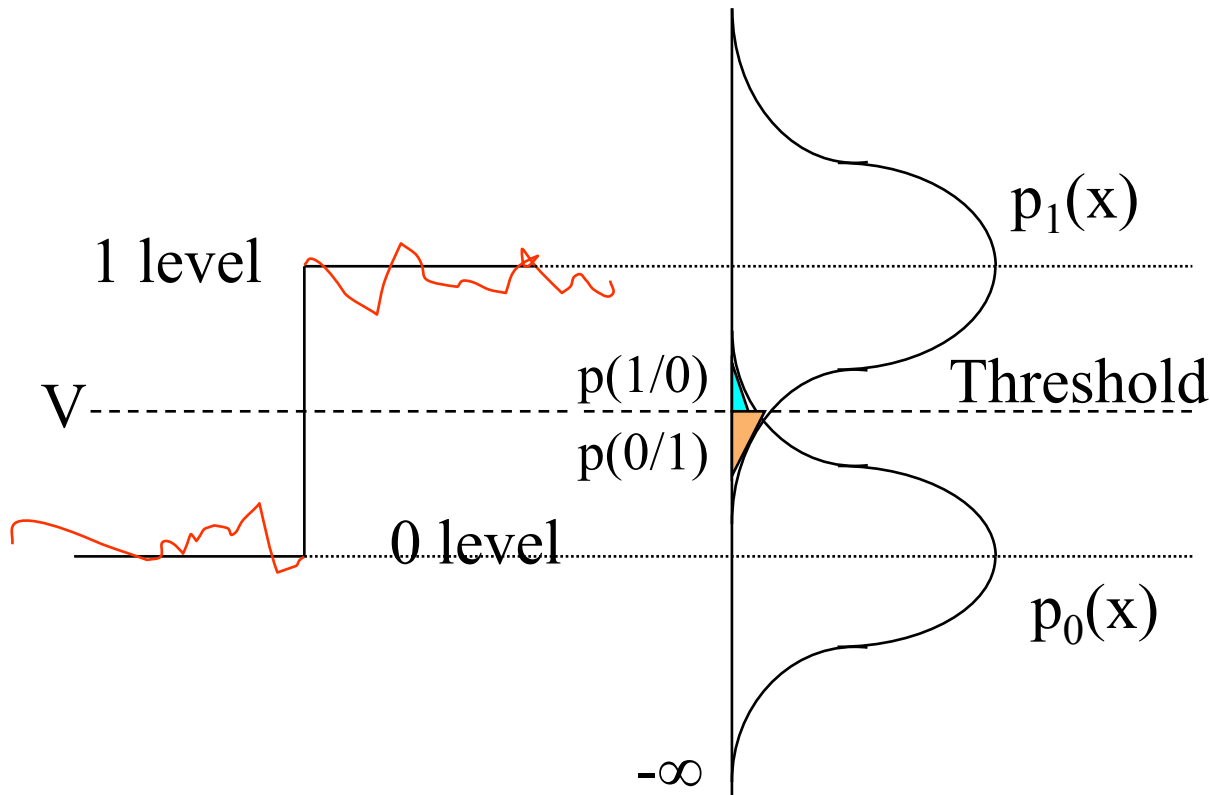
# Decision Regions



# Average probability of symbol error $P_e$

$$\begin{aligned} P_e &= \sum_{i=1}^M P(\mathbf{x} \text{ does not lie in } Z_i | m_i \text{ sent}) P(m_i \text{ sent}) \\ &= \frac{1}{M} \sum_{i=1}^M P(\mathbf{x} \text{ does not lie in } Z_i | m_i \text{ sent}) \\ &= 1 - \frac{1}{M} \sum_{i=1}^M P(\mathbf{x} \text{ lies in } Z_i | m_i \text{ sent}) \\ &= 1 - \frac{1}{M} \sum_{i=1}^M \int_{Z_i} f_{\mathbf{x}}(x | m_i) d\mathbf{x} \end{aligned}$$

# Probability of error



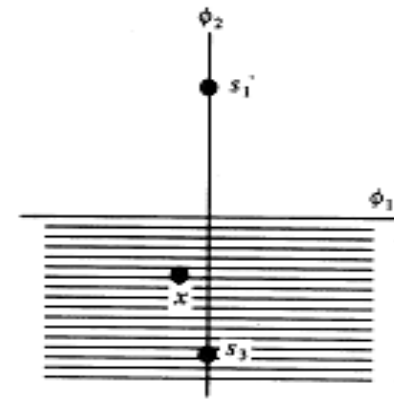
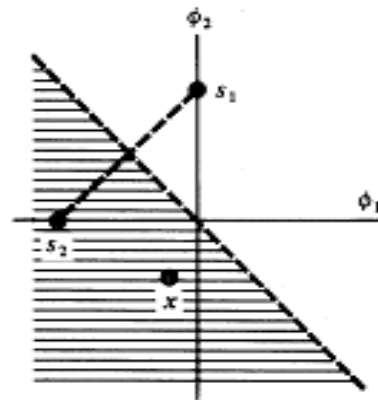
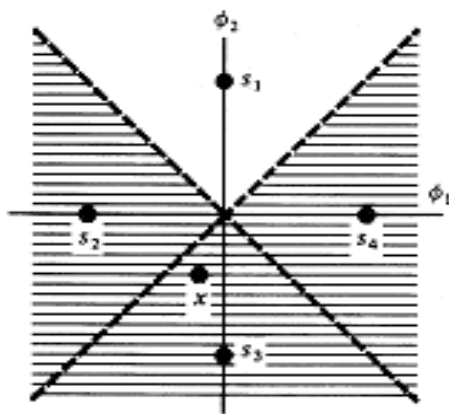
$$p(1/0) = \int_{V}^{\infty} p_0(x) dx$$

$$p(0/1) = \int_{-\infty}^{V} p_1(x) dx$$

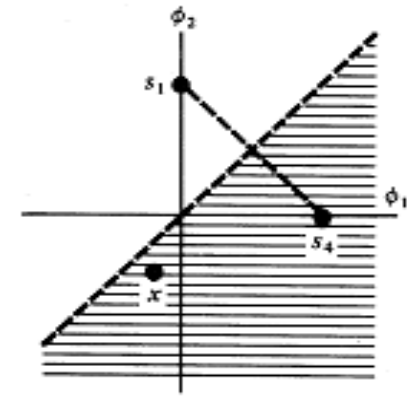
$$P_e = p(1) p(0/1) + p(0) p(1/0)$$

# Union Bound on the Probability of Error

## ■ Example of QPSK

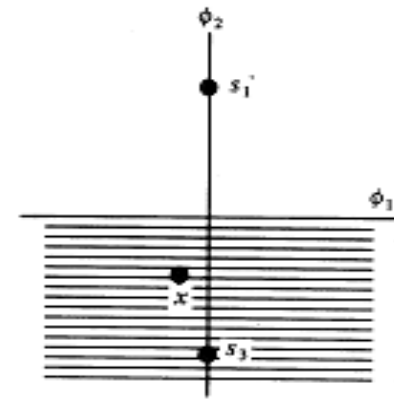
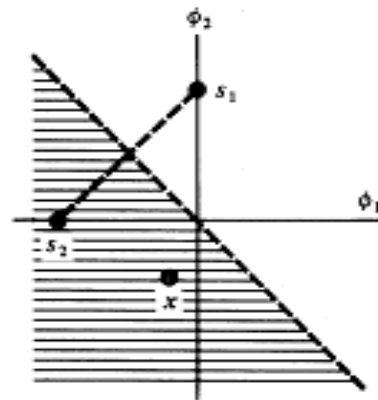
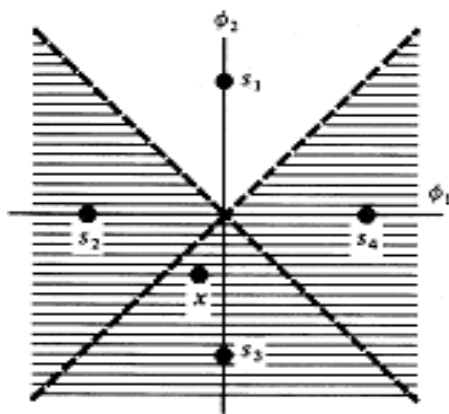


$$P_e(s_1) \leq P_2(s_1, s_2) + P_2(s_1, s_3) + P_2(s_1, s_4)$$

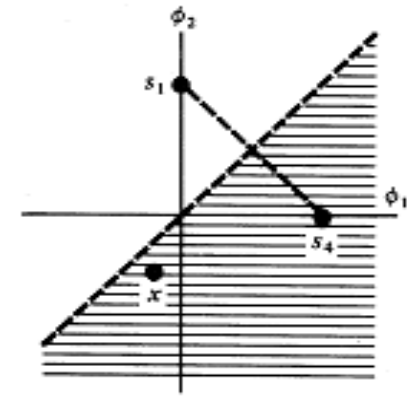


# Union Bound on the Probability of Error

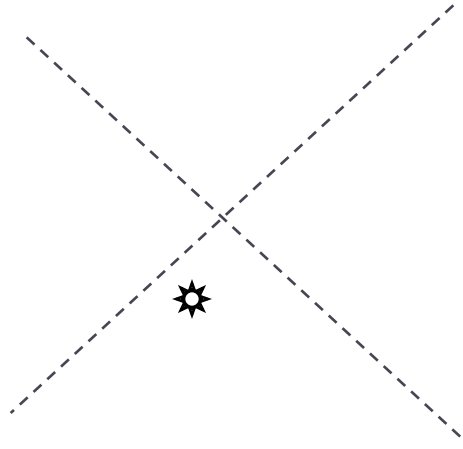
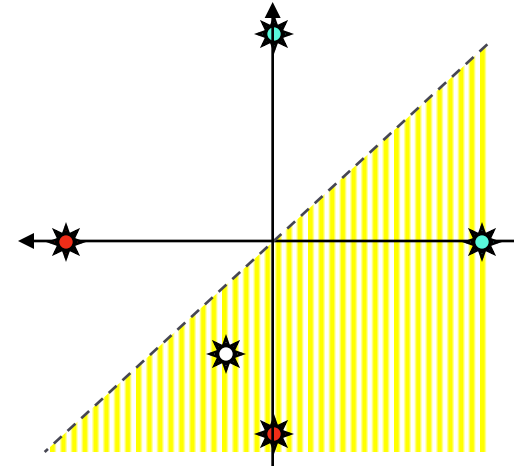
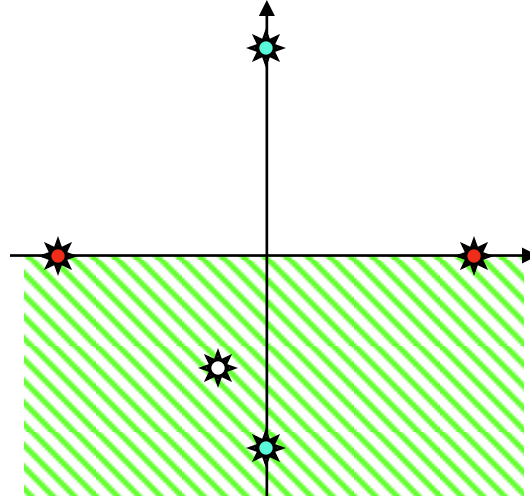
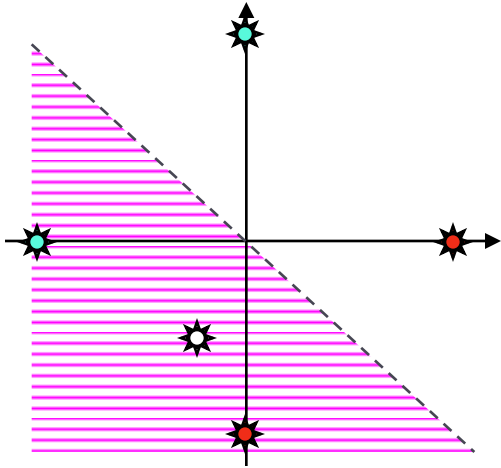
## ■ Example of QPSK



$$P_e(s_1) \leq P_2(s_1, s_2) + P_2(s_1, s_3) + P_2(s_1, s_4)$$



# Union Bound





# Union Bound on the Probability of Error

## ■ In General

$$P_e(m_i) \leq \sum_{\substack{k=1 \\ k \neq i}}^M P_2(\mathbf{s}_i, \mathbf{s}_k) \quad i = 1, 2, \dots, M$$

$$P_2(\mathbf{s}_i, \mathbf{s}_k) = P(\mathbf{x} \text{ is closer to } \mathbf{s}_k \text{ than } \mathbf{s}_i, \text{ when } \mathbf{s}_i \text{ is sent})$$

$$= \int_{d_{ik}/2}^{\infty} \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{u^2}{N_0}\right) du$$

# Union Bound on the Probability of Error

$$= \frac{1}{2} \operatorname{erfc} \left( \frac{d_{ik}}{2\sqrt{N_0}} \right)$$

$$\left( \text{where } d_{ik} = \|\mathbf{s}_i - \mathbf{s}_k\| \right)$$

$$\operatorname{erfc}(u) = \frac{2}{\sqrt{\pi}} \int_u^{\infty} \exp(-z^2) dz$$

# Union Bound on the Probability of Error

$$\therefore P_e(m_i) \leq \sum_{\substack{k=1 \\ k \neq i}}^M \frac{1}{2} \operatorname{erfc} \left( \frac{d_{ik}}{2\sqrt{N_0}} \right) \quad i = 1, 2, \dots, M$$

# Average Probability of symbol error

(equally likely signals in AWGN channel)

$$P_e = \frac{1}{M} \sum_{i=1}^M P_e(m_i) \leq \frac{1}{2M} \sum_{i=1}^M \sum_{\substack{k=1 \\ k \neq i}}^M \operatorname{erfc} \left( \frac{d_{ik}}{2\sqrt{N_0}} \right)$$

# Average Probability of symbol error

(equally likely signals in AWGN channel)

For a symmetric geometry

$$P_e = P_e(m_i) \leq \frac{1}{2} \sum_{\substack{k=1 \\ k \neq i}}^M \operatorname{erfc}\left(\frac{d_{ik}}{2\sqrt{N_0}}\right) \quad \text{for all } i$$

$$\operatorname{erfc}\left(\frac{d_{ik}}{2\sqrt{N_0}}\right) \leq \frac{1}{\sqrt{\pi}} \exp\left(-\frac{d_{ik}^2}{2N_0}\right)$$

# Hierarchy of Digital Modulation Techniques

## ➤ Coherent Systems

- ✓ Phase locked at the receiver

- ✓ Binary, M-ary

- ✓ ASK, PSK, FSK

- ✓ QPSK, MSK, QAM( Hybrid )

**Time  
Synchronism  
assumed**

# Hierarchy of Digital Modulation Techniques

## ➤ Non-coherent Systems

- ✓ No phase synchronism

- ✓ Binary, M-ary

- ✓ ASK, FSK

- ✓ DPSK : noncoherent form of binary PSK

**Time  
Synchronism  
assumed**

# Coherent and Non-coherent Detection

---

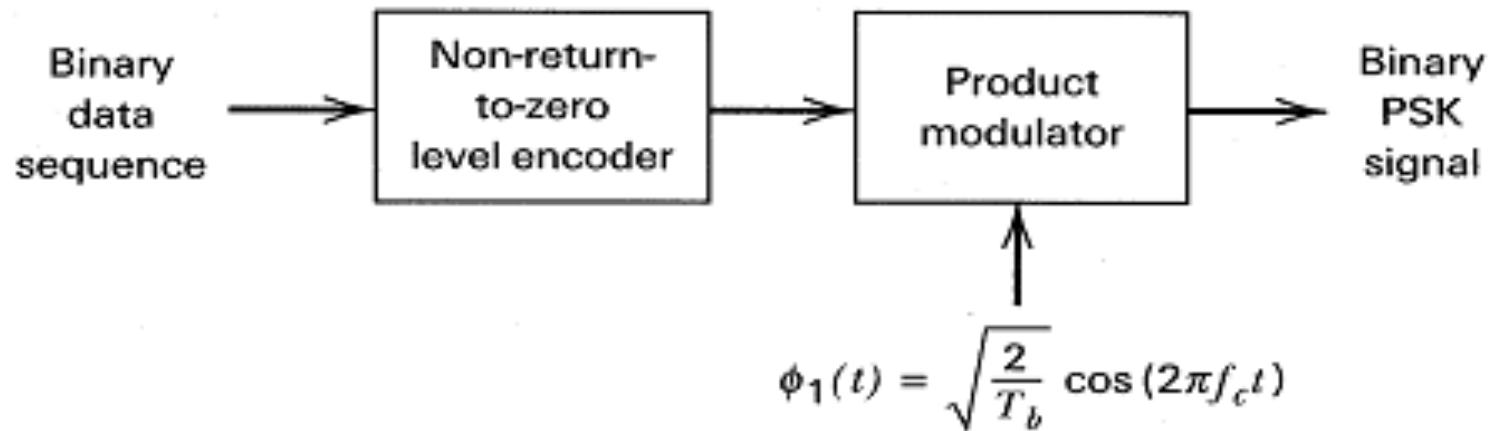
- Coherent Detection (most PSK, some FSK):
  - Exact replicas of the possible arriving signals are available at the receiver.
  - This means knowledge of the phase reference (phased-locked).
  - Detection by cross-correlating the received signal with each one of the replicas, and then making a decision based on comparisons with pre-selected thresholds.
- Non-coherent Detection (some FSK, DPSK):
  - Knowledge of the carrier wave's phase not required.
  - Less complexity.
  - Inferior error performance.



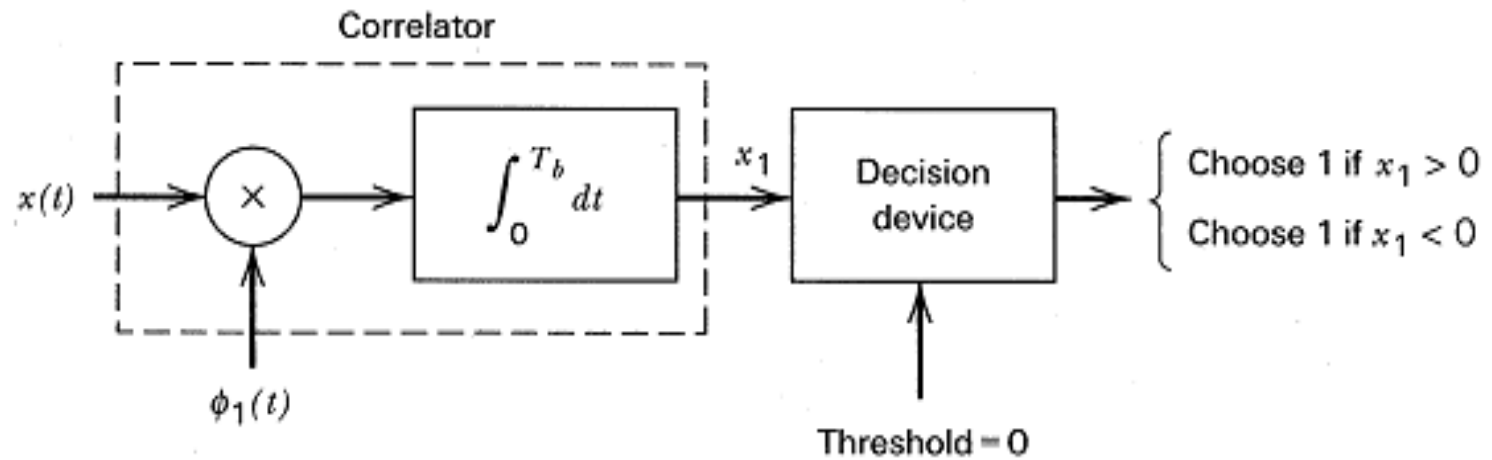
# Linear Modulation Techniques

- The envelop of the transmitted signal varies linearly with the modulating digital signal,  $m(t)$ .
- Hence usually do not have constant envelope.
- More spectrally efficient w.r.t main lobe.
- Poor power efficiency.
- Example: BPSK, QPSK, MPSK, MASK

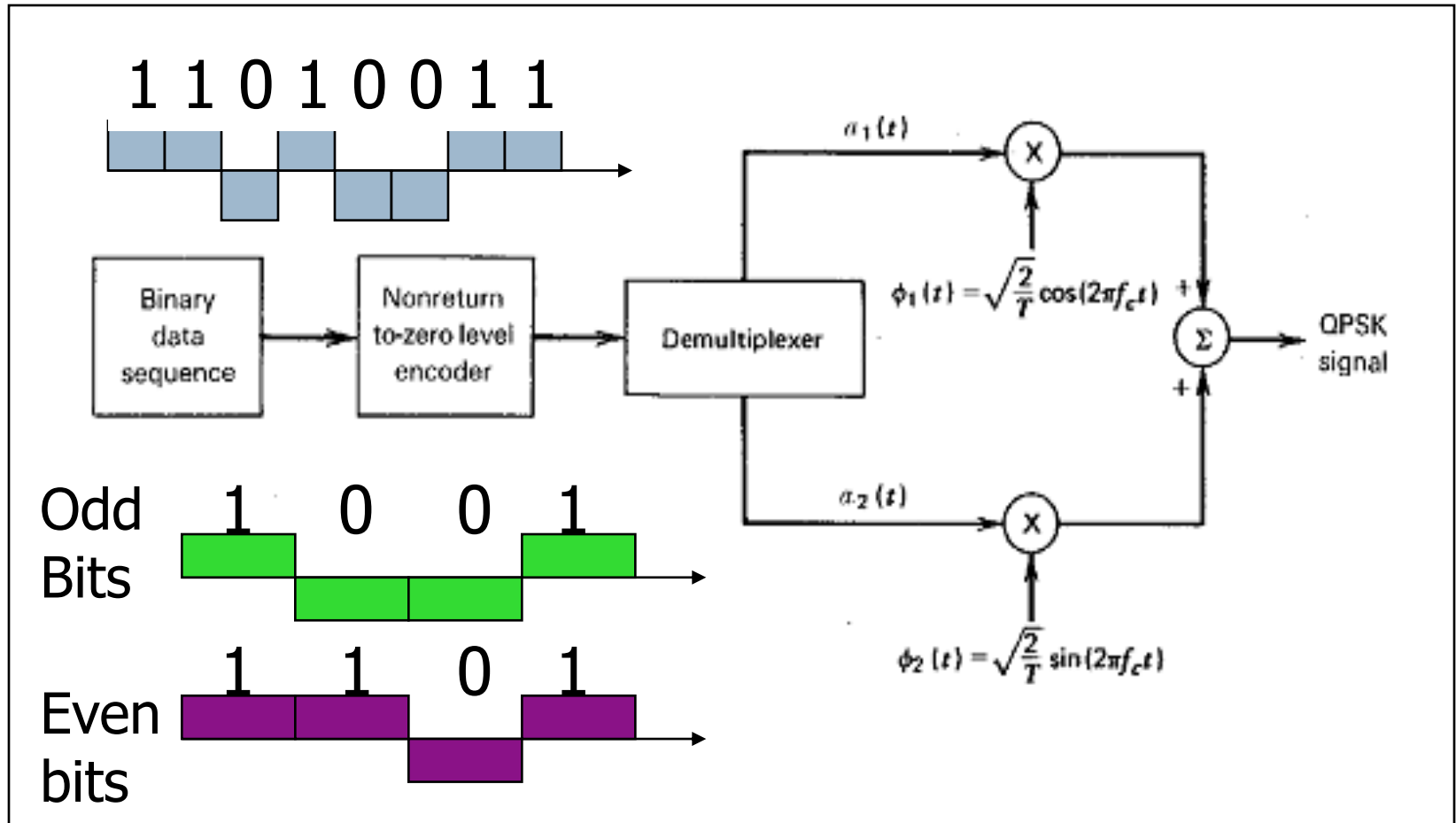
# BPSK Transmitter



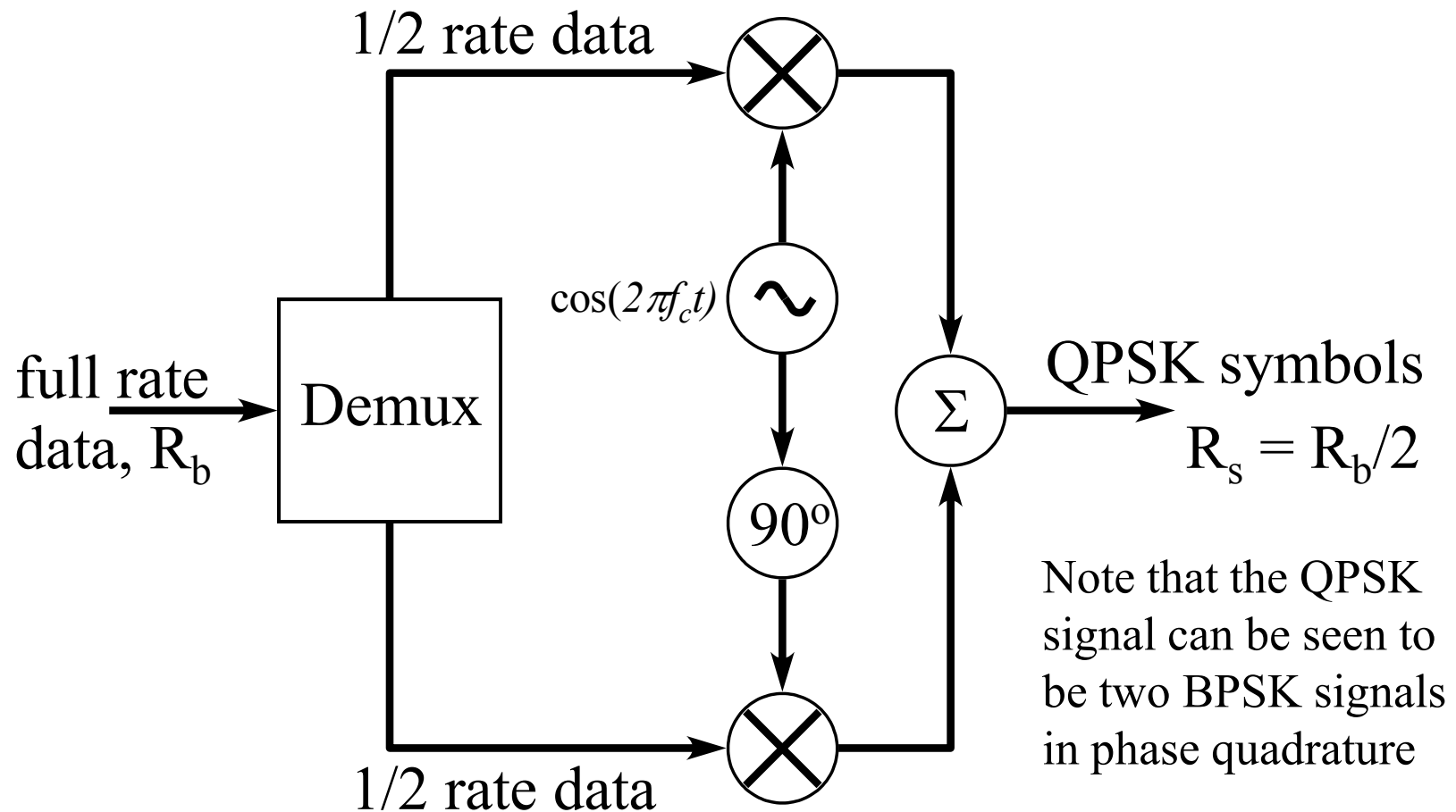
# BPSK Receiver



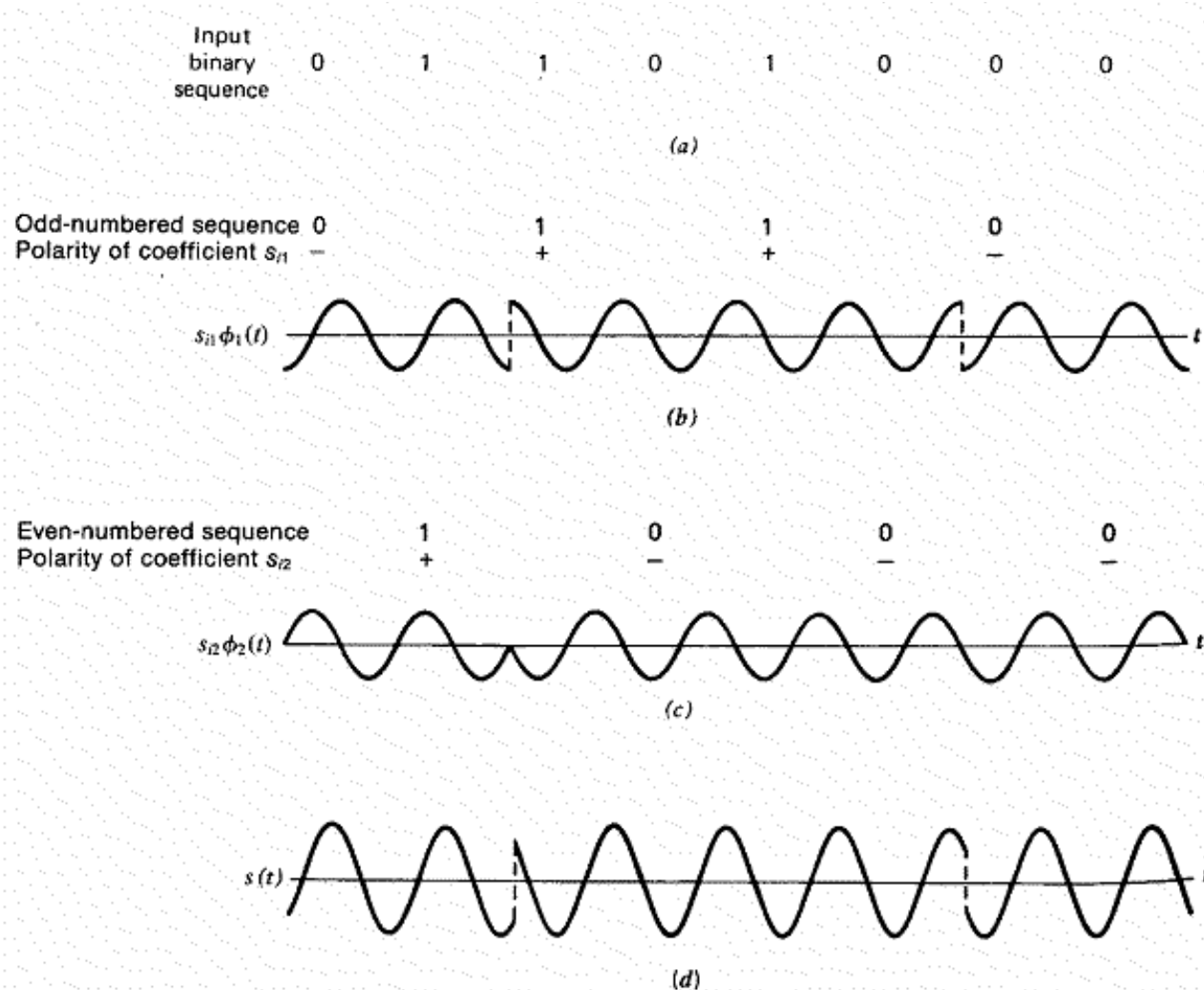
# QPSK Transmitter



# QPSK Physical Implementation

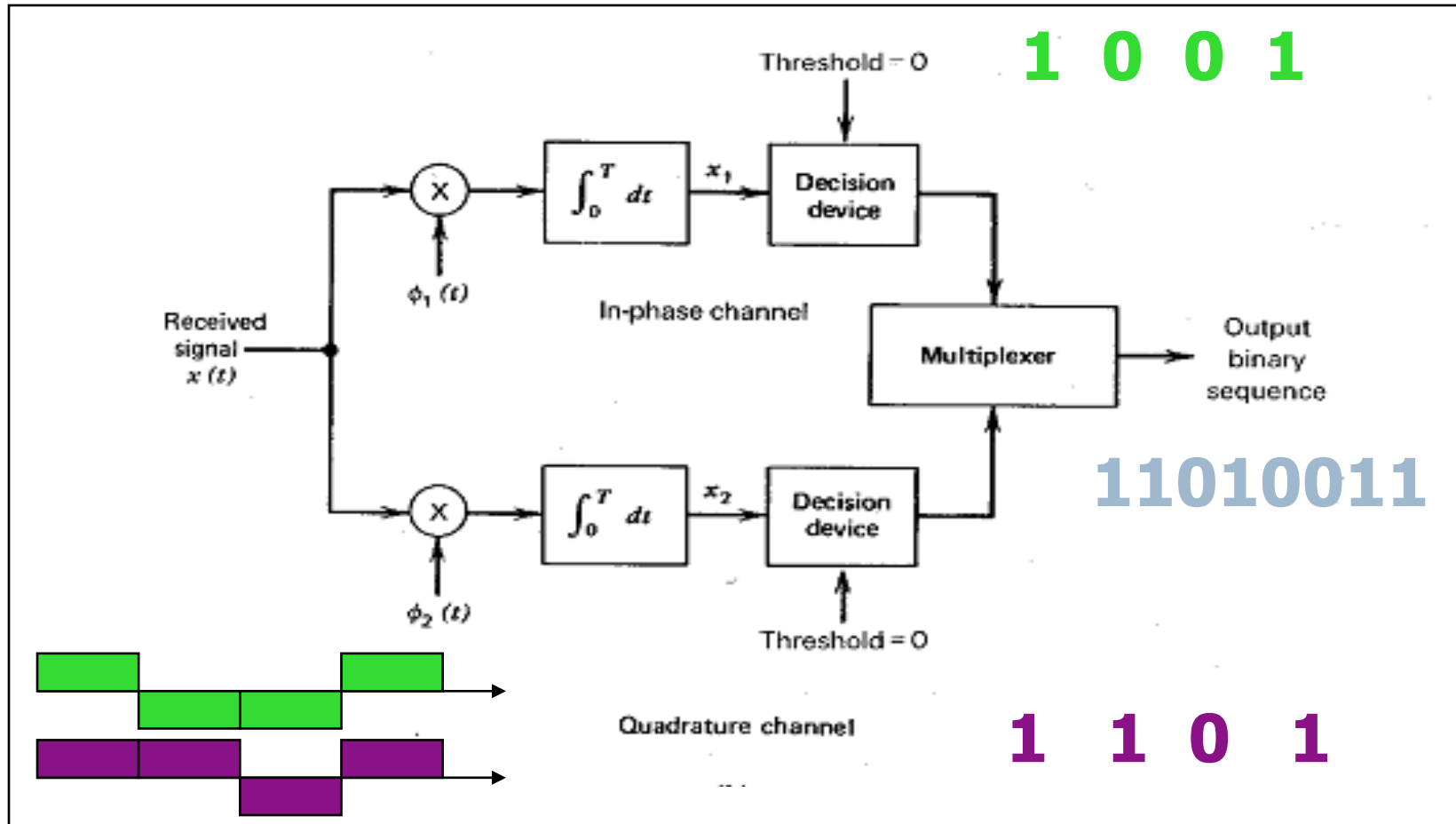


# QPSK Waveform

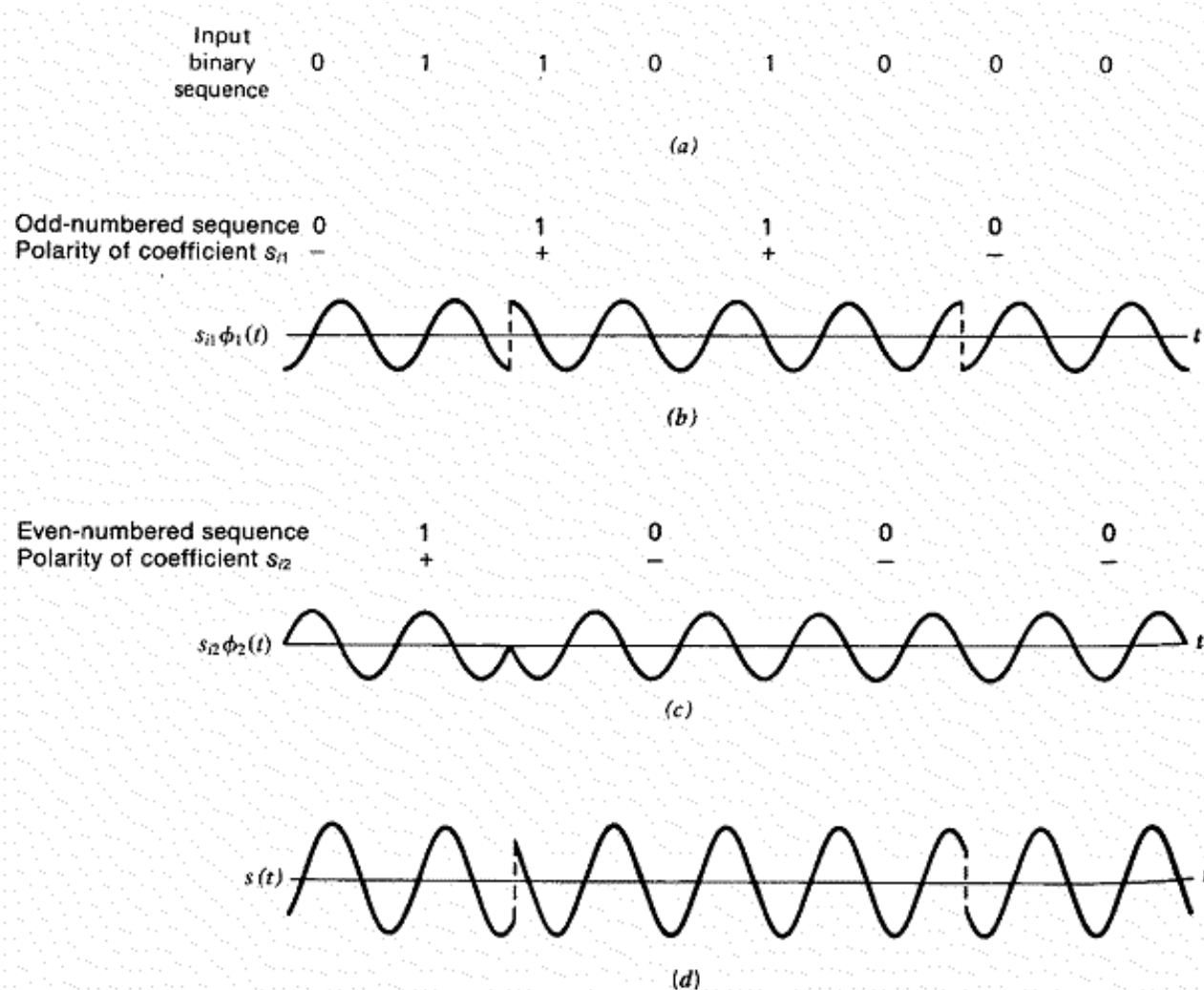


**Figure 7.7** (a) Input binary sequence. (b) Odd-numbered bits of input sequence and associated binary PSK wave. (c) Even-numbered bits of input sequence and associated binary PSK wave. (d) QPSK waveform.

# QPSK Receiver



# Motivation for OQPSK

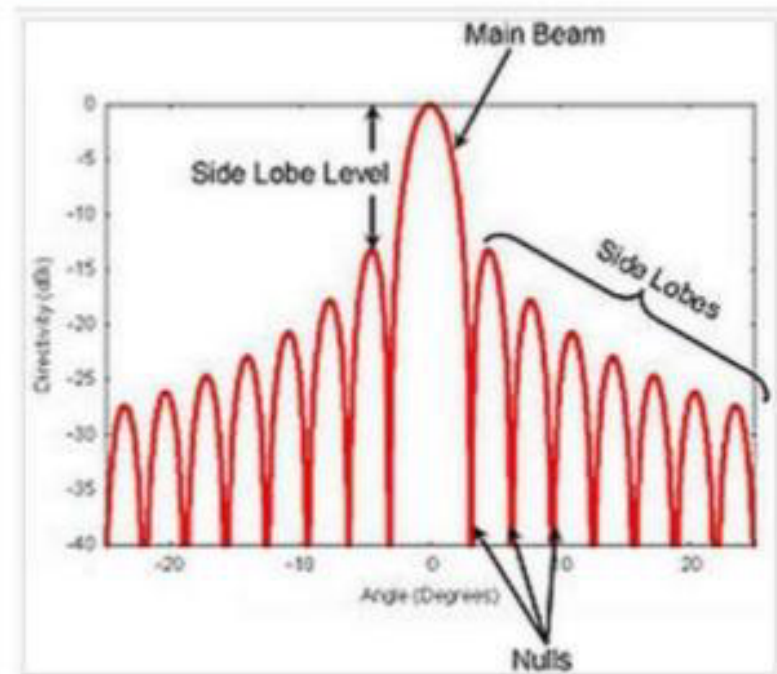
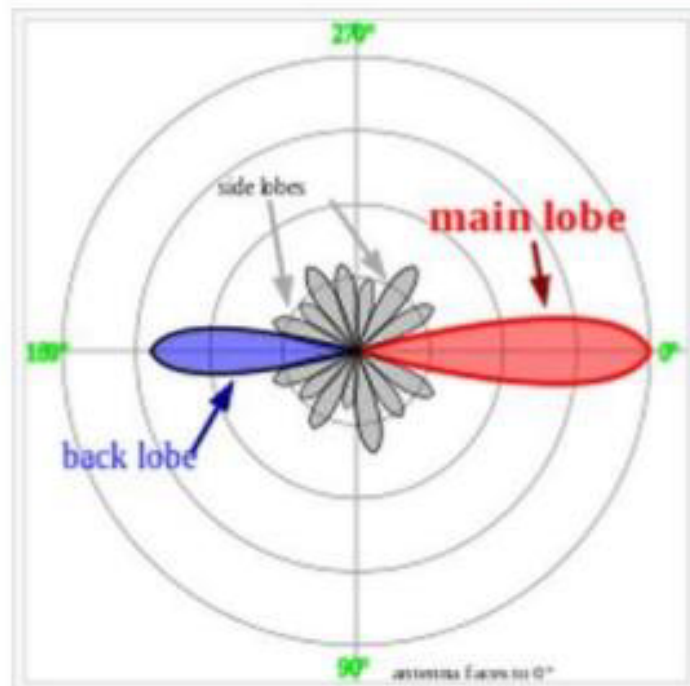


**Figure 7.7** (a) Input binary sequence. (b) Odd-numbered bits of input sequence and associated binary PSK wave. (c) Even-numbered bits of input sequence and associated binary PSK wave. (d) QPSK waveform.



## Drawback of QPSK Modulation (Application)

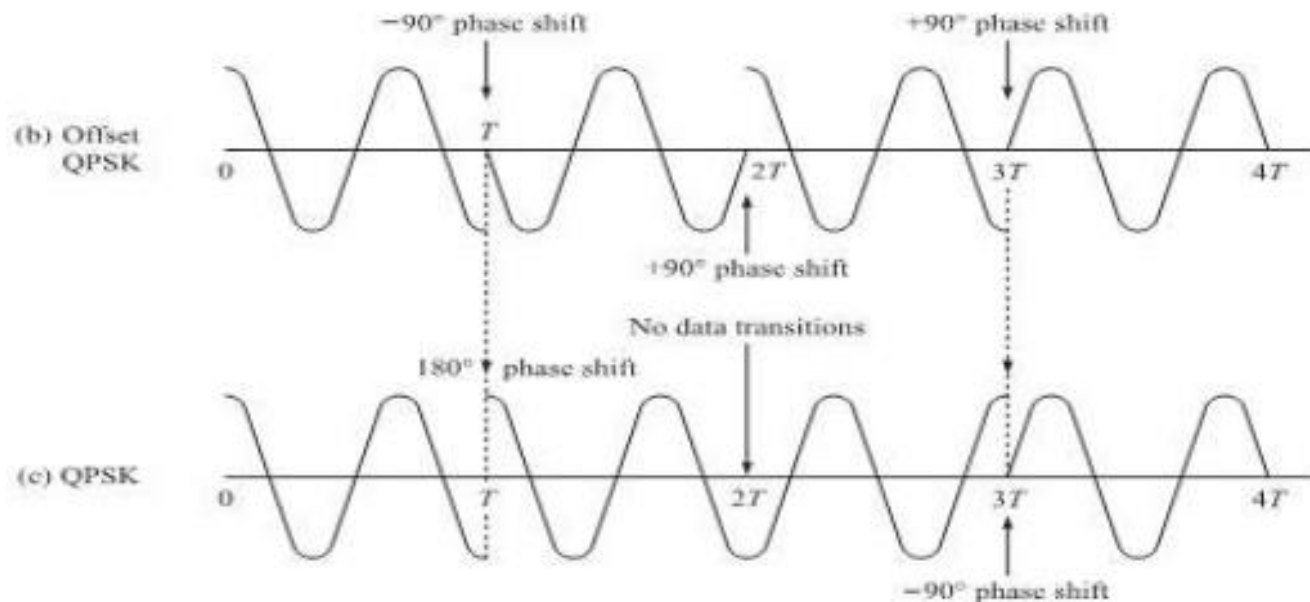
180° phase changes cause abrupt changes in the signal, resulting in large spectral **side lobes**.



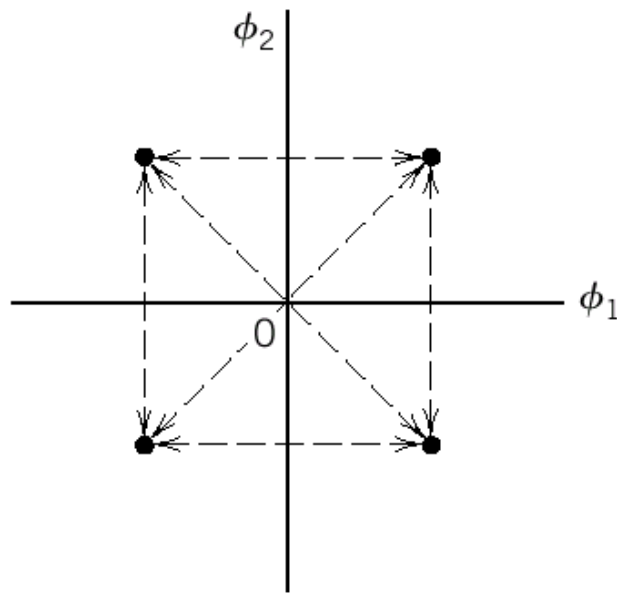
In high amplifiers and for certain satellite applications,  
**OQPSK** offers better performance

## Solution (QPSK modified to become OQPSK)

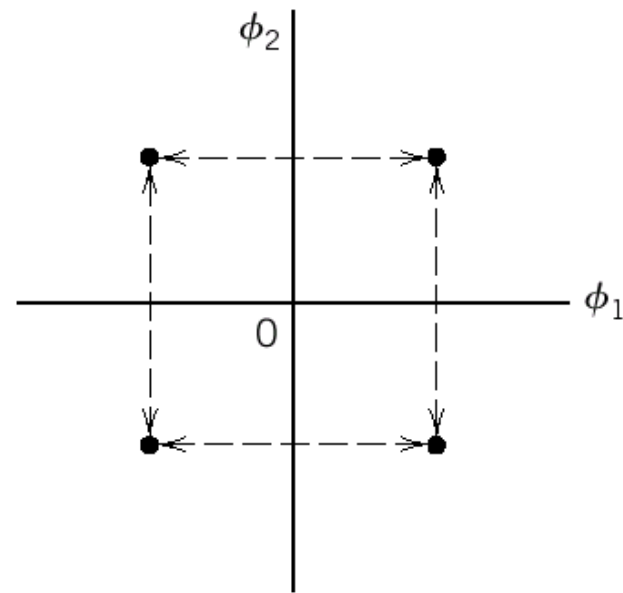
The absence of  $180^\circ$  phase jump is, however, offset by more frequent  $\pm 90^\circ$  phase shifts



Possible paths for switching between the message points in (a) QPSK and (b) offset QPSK.

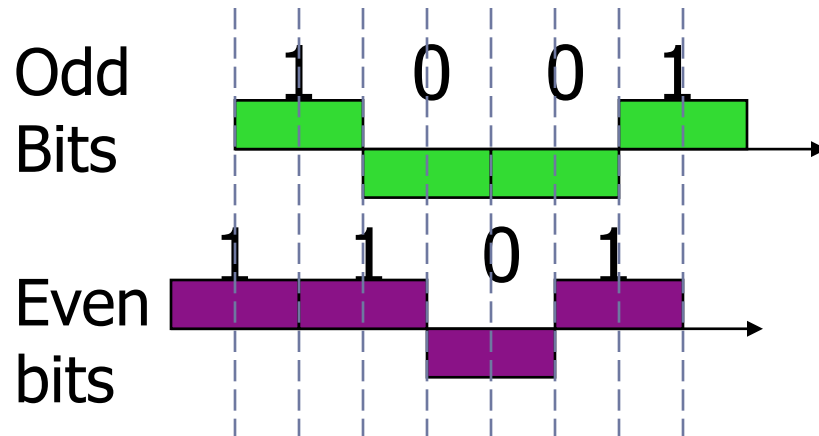
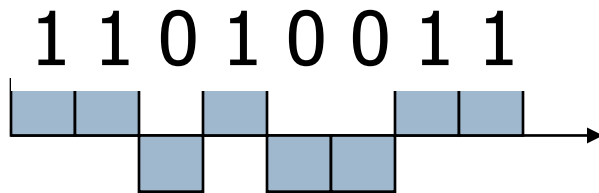
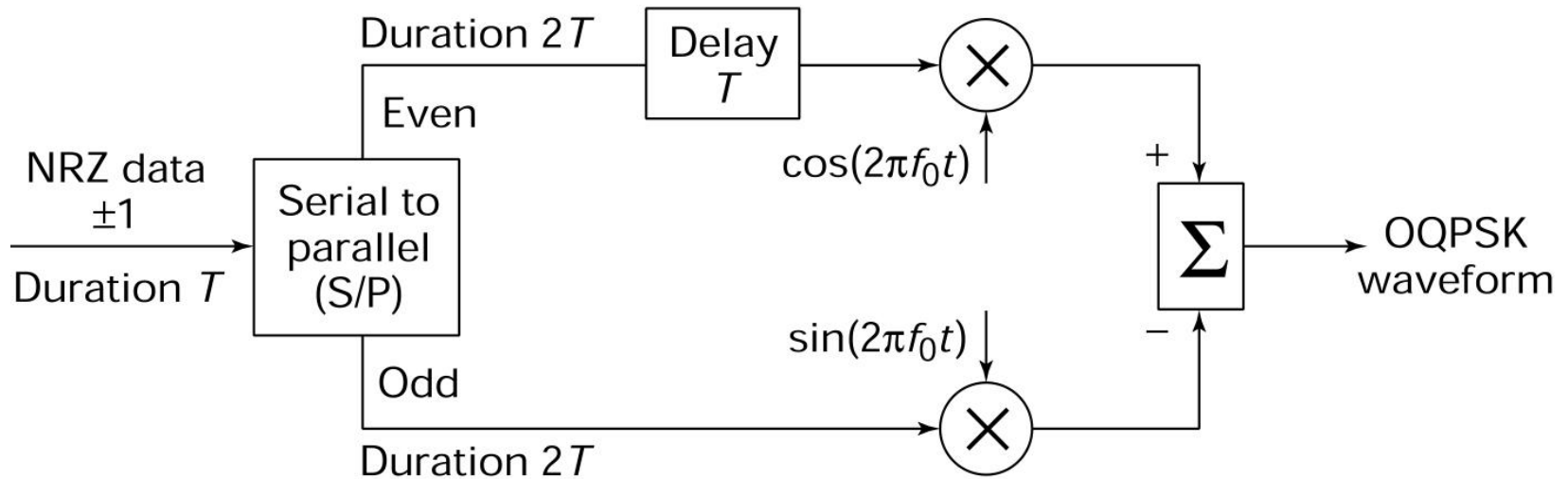


(a)



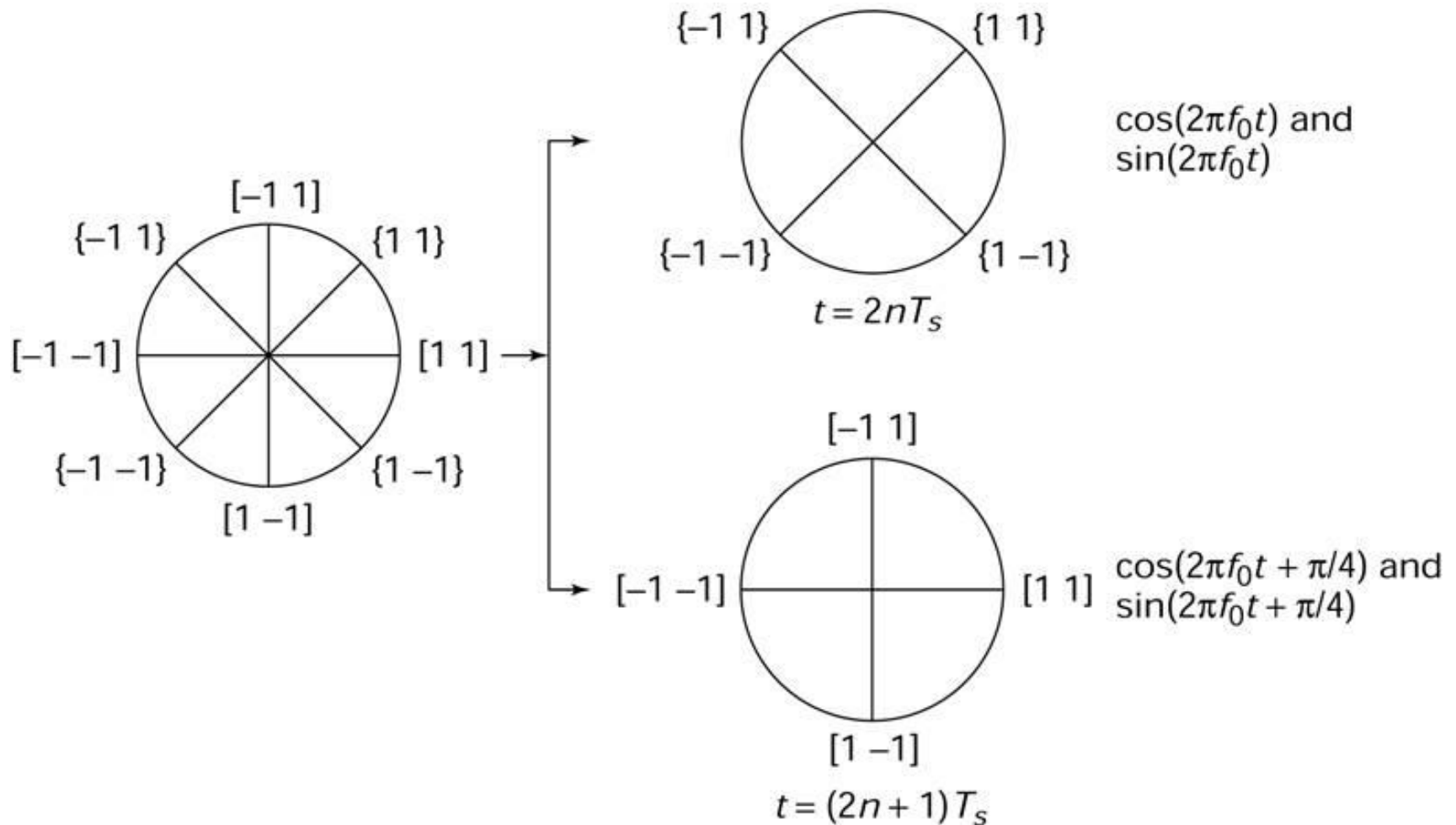
(b)

# Offset-QPSK modulator

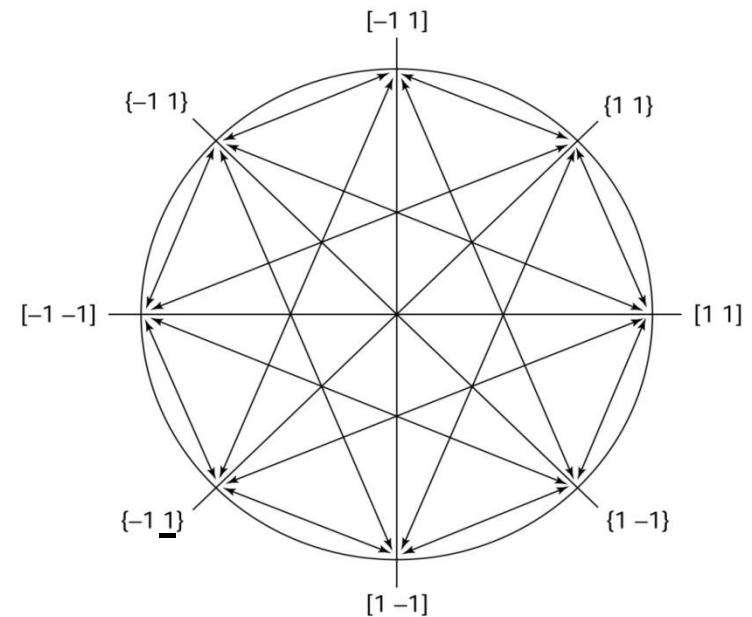
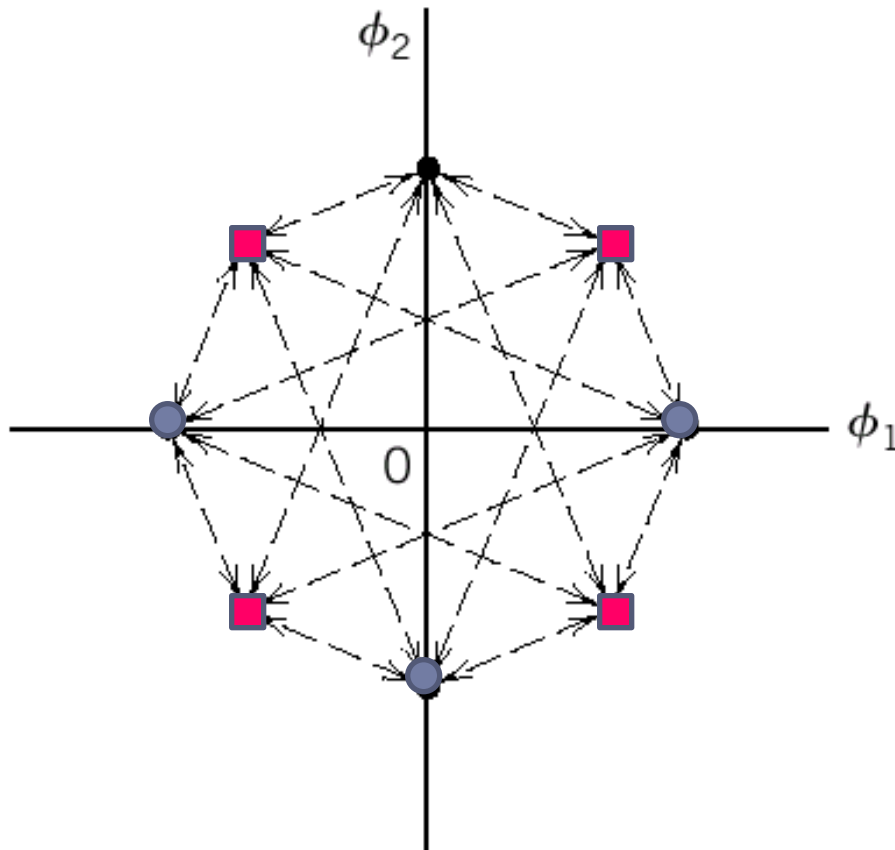


# Phase encoding for $\pi/4$ -QPSK

[ ] and { } correspond to the two respective constellations



# Eight possible phase states for the $\pi/4$ -shifted QPSK modulator



| $\frac{5\pi}{4}$ | $\frac{3\pi}{4}$ | $\frac{\pi}{4}$ |
|------------------|------------------|-----------------|
| -1               | -1               | 1               |
| I                | Q                | I               |
| -1               | 1                | 1               |
| I                | Q                | I               |
| -1               | -1               | 1               |
| I                | Q                | I               |
| 1                | -1               | 1               |
| I                | Q                | I               |

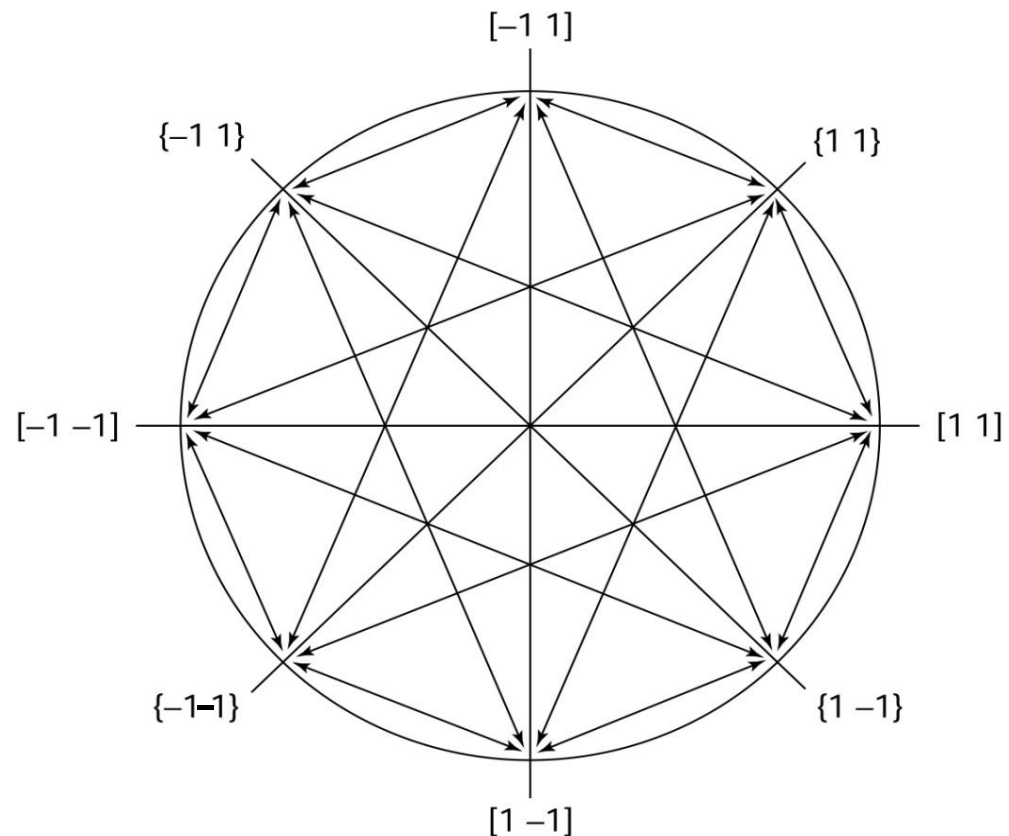
# Phase constellation associated with $\pi/4$ -QPSK

Carrier waves pattern change every alternate symbol

Phase changes in  $\pi/4$ -QPSK are limited to  $\pm 3\pi/4$  or  $\pm \pi/4$ .

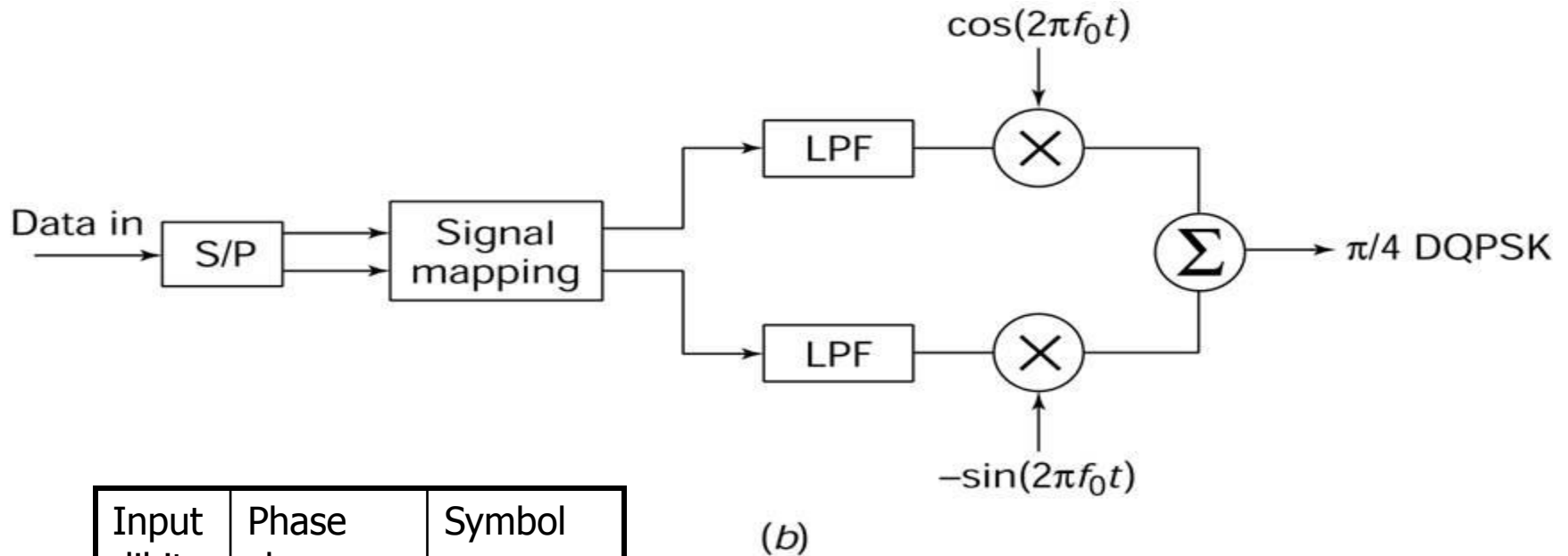
Phase changes of  $0, \pm \pi/2$ , or  $\pm \pi$  – not allowed.

Possible transitions shown by arrows.



| $\theta_k = \theta_{k-1} + \phi_k$ |          |           |          |
|------------------------------------|----------|-----------|----------|
| $\pi/4$                            | $3\pi/4$ | $-3\pi/4$ | $-\pi/4$ |
| 1 1                                | 0 1      | 0 0       | 1 0      |
| 1 1                                | -1 1     | -1 -1     | 1 -1     |

# Block diagram of the $\pi/4$ -DQPSK transmitter

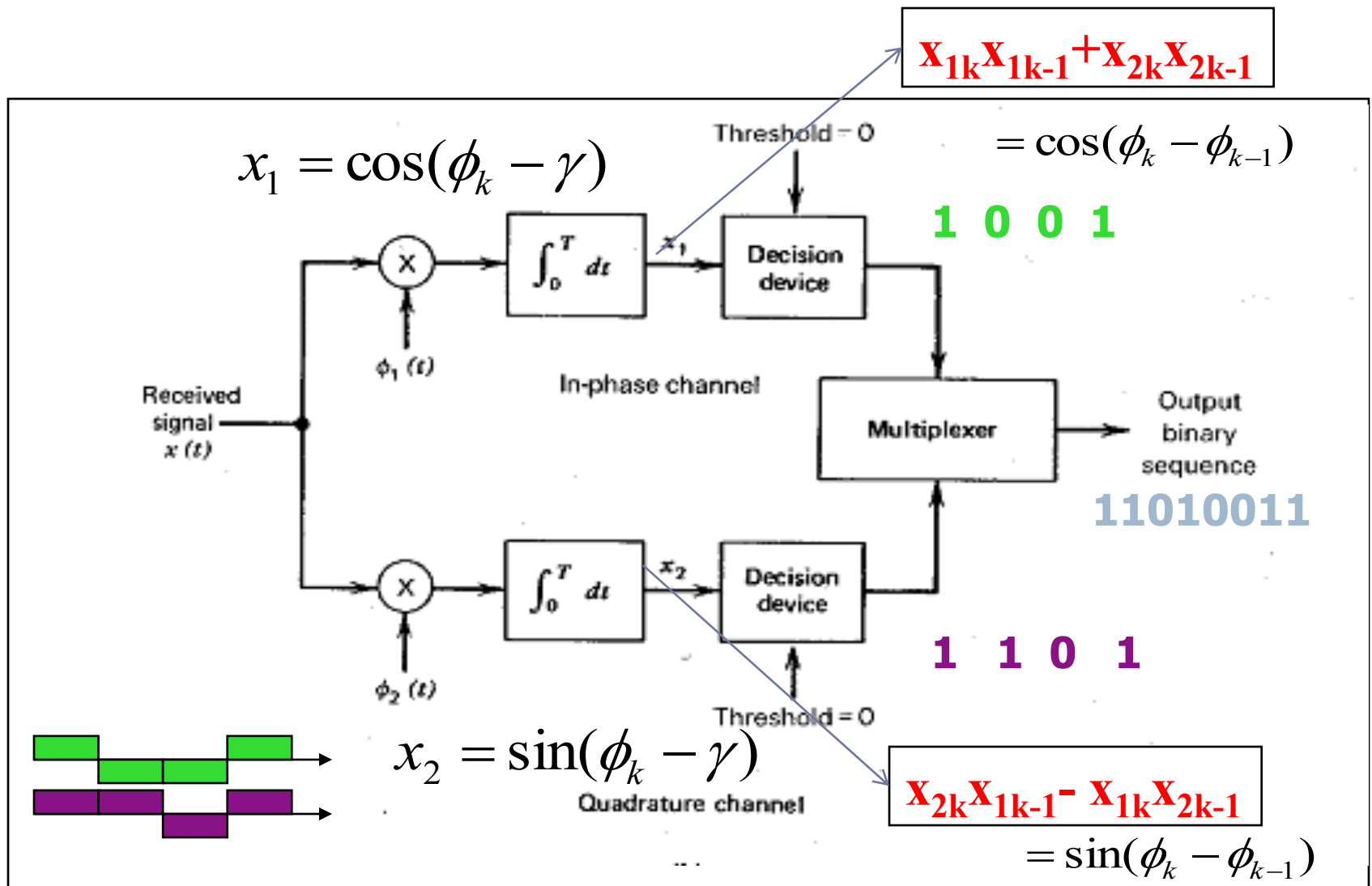


| Input dibit | Phase change | Symbol       |
|-------------|--------------|--------------|
| 11          | $\pi/4$      | $x_1 + jy_1$ |
| 01          | $3\pi/4$     | $x_2 + jy_2$ |
| 00          | $-3\pi/4$    | $x_3 + jy_3$ |
| 10          | $-\pi/4$     | $x_4 + jy_4$ |

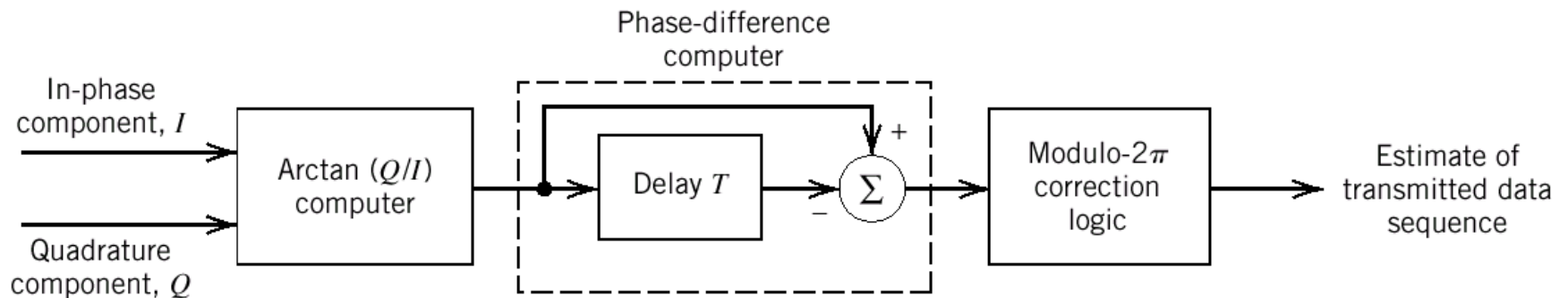
$$\theta_k = \theta_{k-1} + \phi_k$$



# $\pi/4$ – QPSK/DQPSK Receiver



# Block diagram of the $\pi/4$ -shifted DQPSK detector



# Bit Error Rate (BER) for QPSK

---

- The BER is still the probability of choosing the wrong signal state (symbol now)
- Because the signal is Gray coded (00 is next to 01 and 10 for instance but not 11) the BER for QPSK is that for BPSK:
- BER (after a lot of derivation) is given by:

$$BER_{QPSK} = \frac{1}{2} \operatorname{erfc} \left( \sqrt{\frac{E_b}{N_o}} \right) \approx \frac{0.2821}{\sqrt{\frac{E_b}{N_o}}} e^{-\frac{E_b}{N_o}}$$

Approximation  
valid for  $E_b/N_o$   
greater than  $\sim 4$  dB

Note that  $E_b$  is used here, not  $E_s$ !

# Constant Envelope Modulation

- Frequency Shift Keying
- Minimum Shift Keying
- Gaussian Minimum Shift Keying

# Frequency Shift Keying (FSK)

- Binary FSK
- Frequency of the constant amplitude carrier is changed according to the message state → high (1) or low (0)

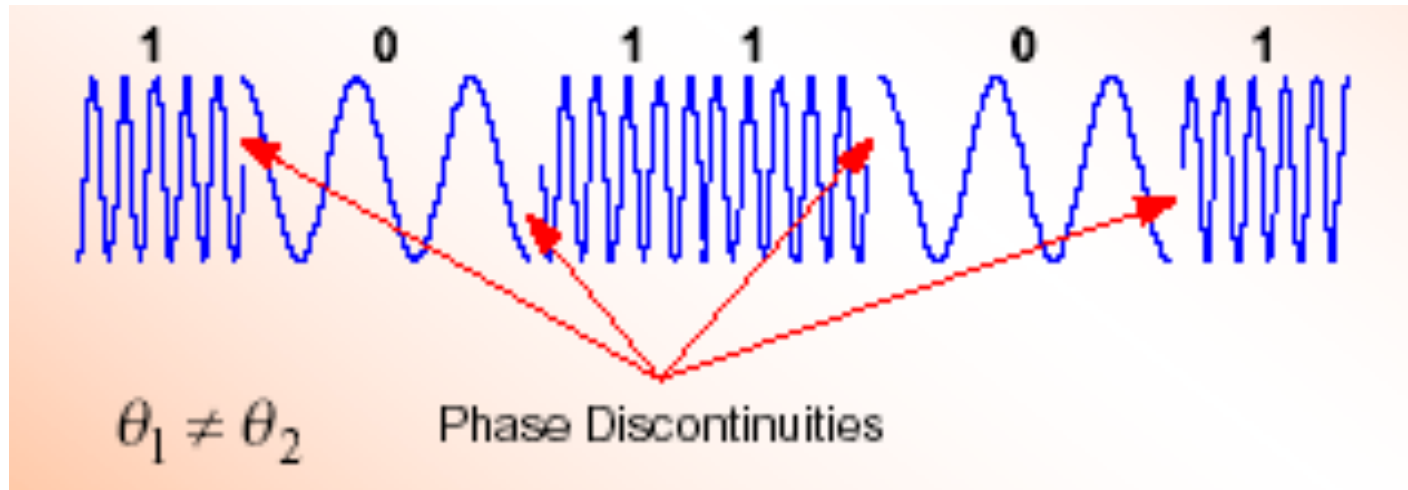
$$s_1(t) = A \cos(2\pi f_c + 2\pi\Delta f)t \quad 0 \leq t \leq T_b \text{ (bit = 1)}$$

$$s_2(t) = A \cos(2\pi f_c - 2\pi\Delta f)t \quad 0 \leq t \leq T_b \text{ (bit = 0)}$$

- Discontinuous / Continuous Phase

# Discontinuous Phase FSK

Switching between 2 independent oscillators for binary 1 & 0



- results in phase discontinuities
- causes spectral spreading & spurious transmission
- not suited for tightly designed systems

# Continuous Phase FSK

Single carrier that is frequency modulated using  $m(t)$

$$\begin{aligned} s_{BFSK}(t) &= \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta(t)) \\ &= \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi f_c t + 2\pi\Delta f \int^t m(\eta) d\eta\right) \end{aligned}$$

$$\text{where } \theta(t) = 2\pi\Delta f \int^t m(\eta) d\eta$$

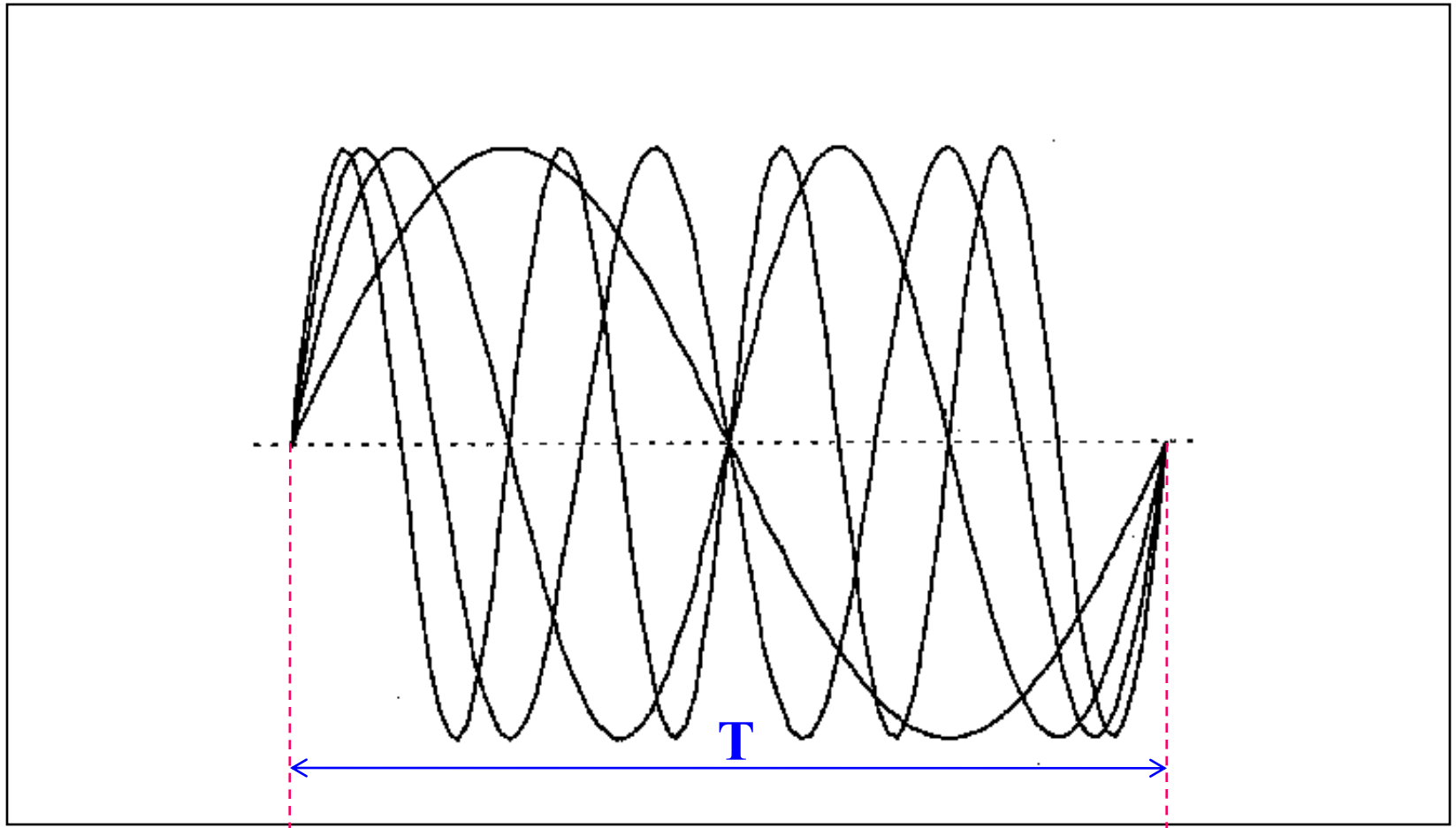
- $m(t)$  = discontinuous bit stream
- $\theta(t)$  = **continuous phase** function  
proportional to integral of  $m(t)$

# Sunde's FSK

- $f_1$  and  $f_2$  have to be orthogonal during a bit period for phase continuity and proper detection
- $f_1$  and  $f_2$  should have integer number of cycles during a bit period, say 'n' cycles of  $f_1$  and 'm' cycles of  $f_2 \rightarrow$   
$$f_1 \sim f_2 = (n \sim m)/T_b = (n \sim m) R_b = h R_b$$
- Minimum spacing if  $(n \sim m) = 1$ ,  $\Delta f = h / T_b = 1 / T_b$
- If ,  $f_c = (f_1 + f_2)/2$ , then,  $f_1 = f_c - \Delta f/2$ ;  $f_2 = f_c + \Delta f/2$ ;



# Orthogonal Signals



# Continuous Phase FSK

$$\text{for } i = 1, 2 \quad s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t) \quad 0 \leq t \leq T$$
$$= 0 \text{ otherwise}$$

$$s_1(t) = \sqrt{E_b} \sqrt{\frac{2}{T_b}} \cos(2\pi f_1 t) = \sqrt{E_b} \phi_1(t) \quad ; 0 \leq t \leq T$$

$$s_2(t) = \sqrt{E_b} \sqrt{\frac{2}{T_b}} \cos(2\pi f_2 t) = \sqrt{E_b} \phi_2(t) \quad ; 0 \leq t \leq T$$

# Continuous Phase BFSK

- Orthonormal basis functions

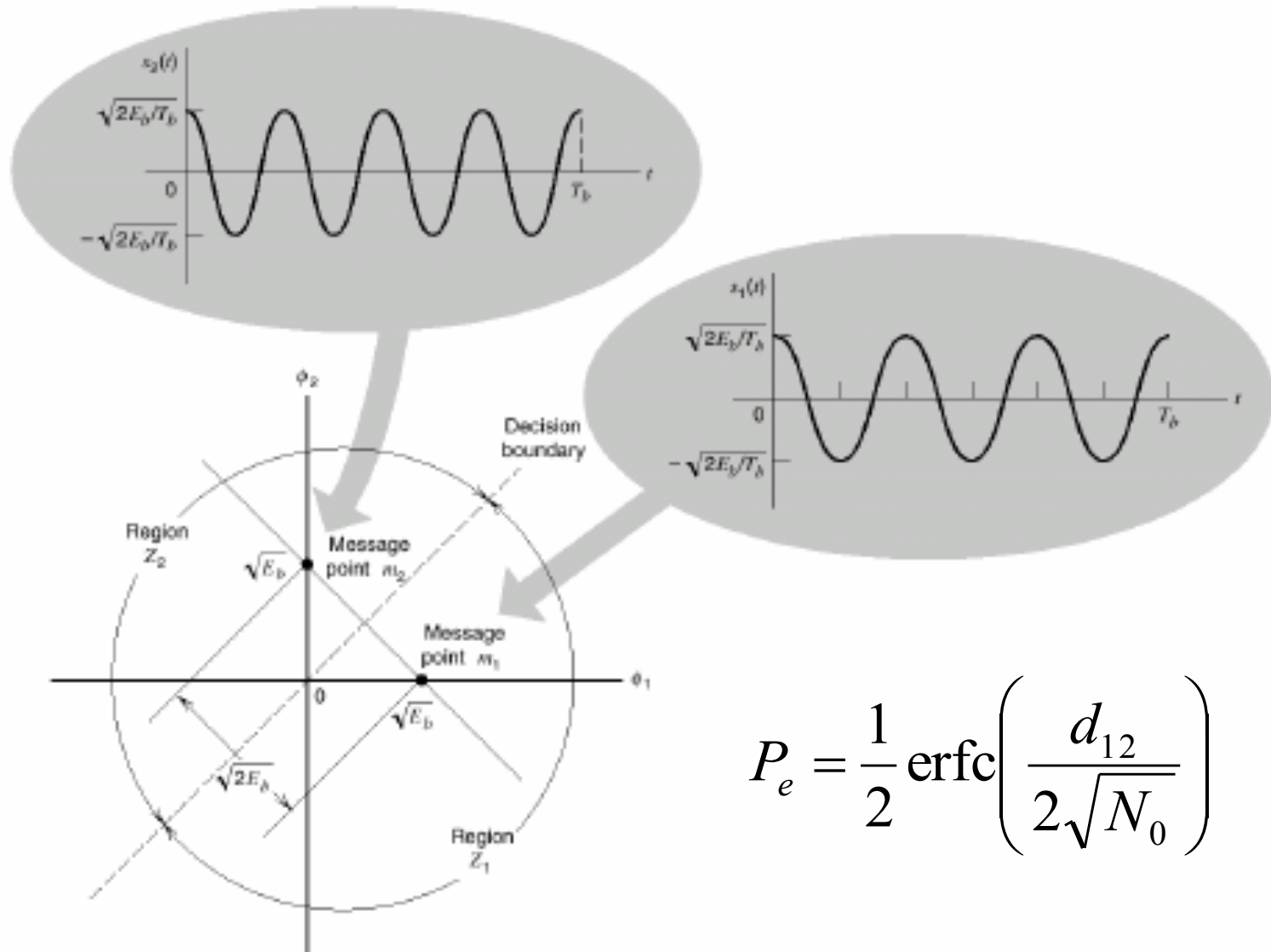
$$\phi_i(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_i t) \quad , \quad 0 \leq t \leq T_b$$

- $i = 1, 2$

$$\mathbf{s}_1 = \begin{bmatrix} \sqrt{E_b} \\ 0 \end{bmatrix}$$

$$\mathbf{s}_2 = \begin{bmatrix} 0 \\ \sqrt{E_b} \end{bmatrix}$$

# Continuous Phase BFSK



$$P_e = \frac{1}{2} \operatorname{erfc} \left( \frac{d_{12}}{2\sqrt{N_0}} \right)$$

# Probability of Error

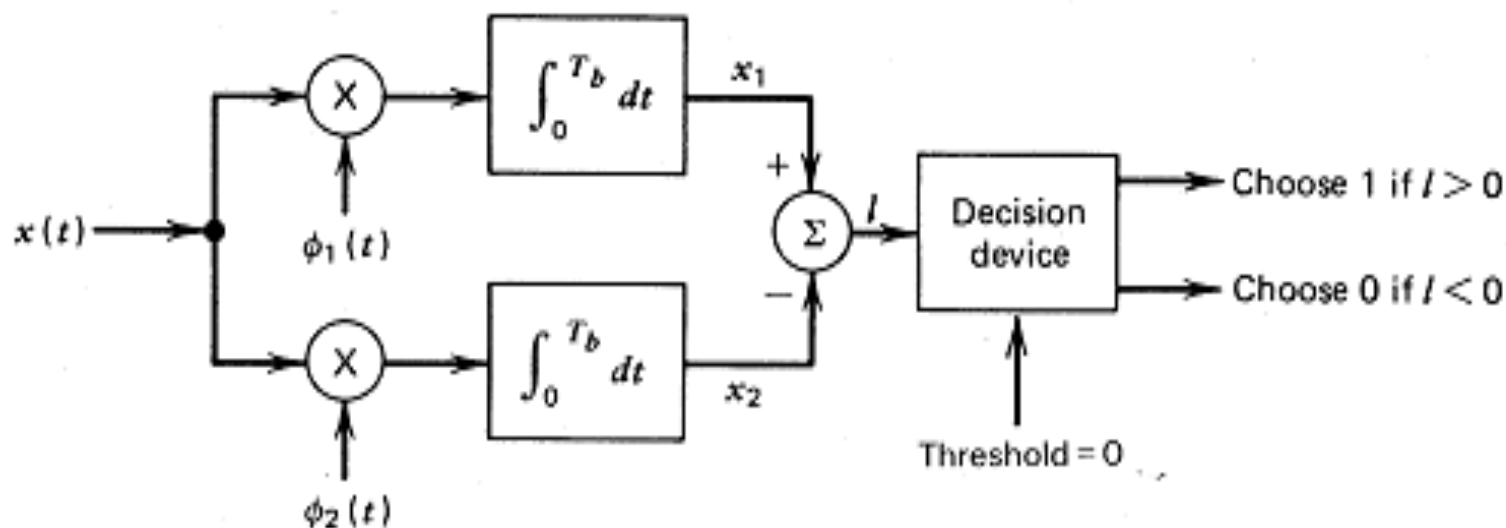
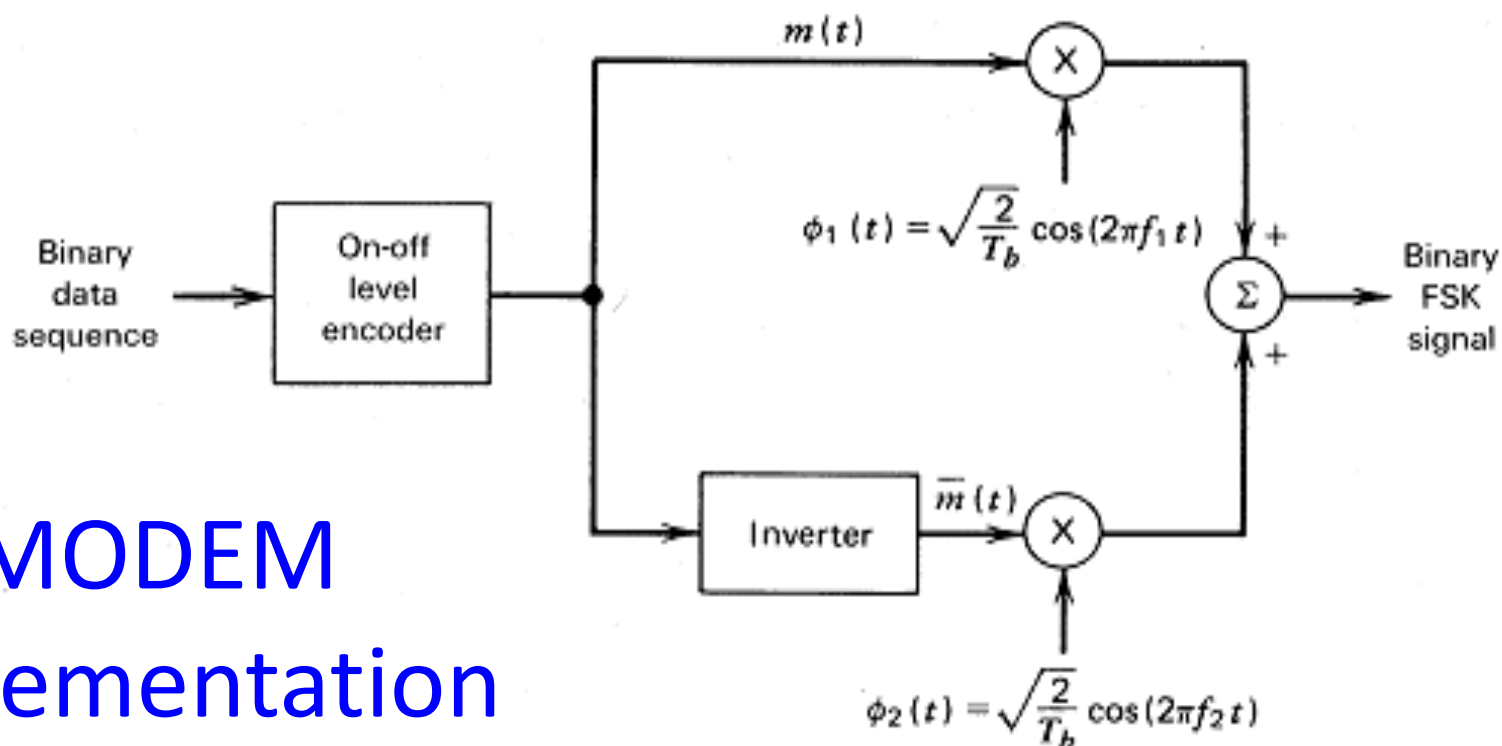
- Euclidean distance =  $d_{12} = \sqrt{2E_b}$
- Avg. prob. of symbol error = BER

$$P_e = \frac{1}{2} \operatorname{erfc} \left( \sqrt{\frac{E_b}{2N_0}} \right)$$

- Compare with BPSK ?

$$P_e = \frac{1}{2} \operatorname{erfc} \left( \sqrt{\frac{E_b}{N_0}} \right)$$

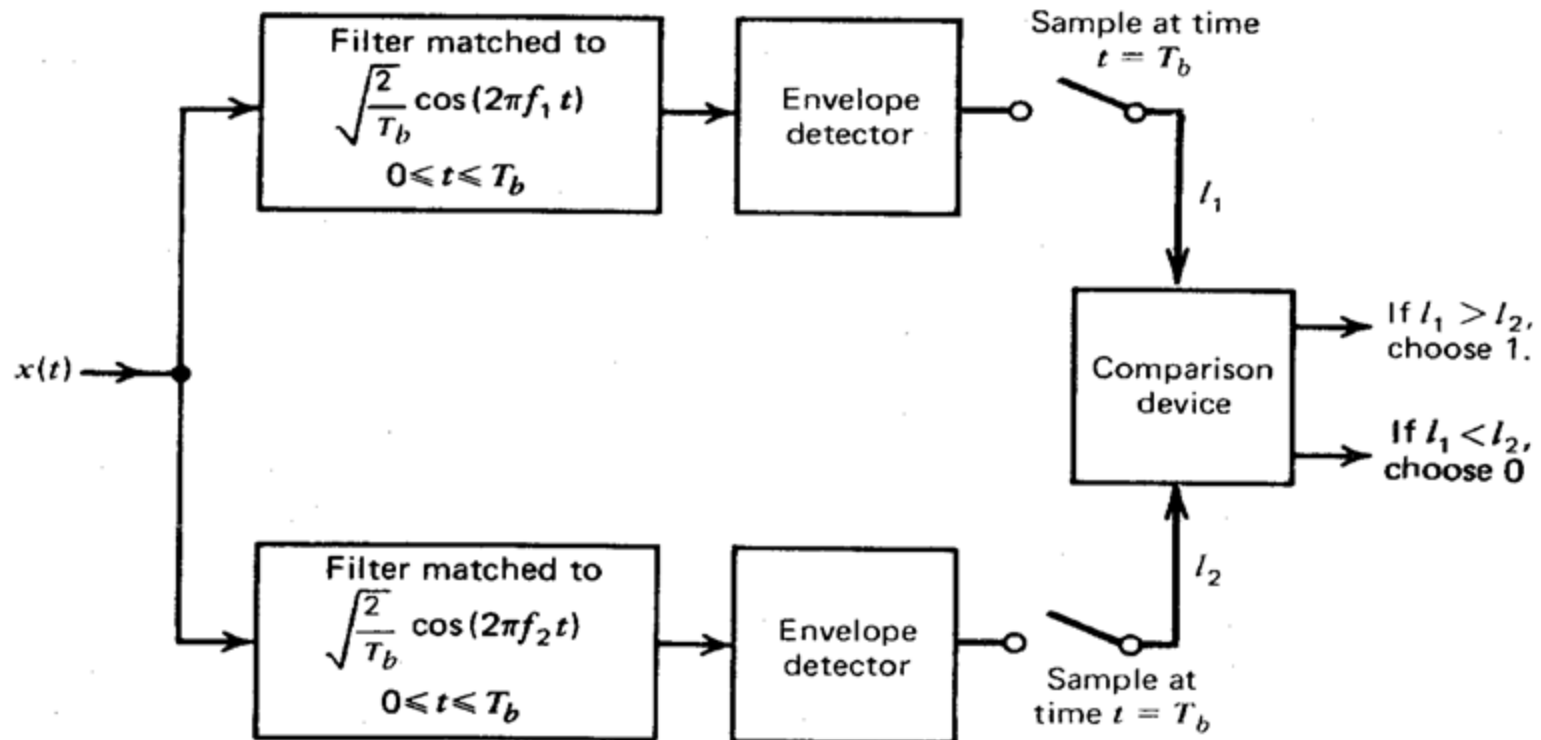
# MODEM Implementation



# Non-coherent Detection of BFSK

- Operates in noisy channel without coherent carrier reference
- Pair of matched filters followed by envelope detector
  - Upper path filter matched to  $f_H$  (binary 1)
  - Lower path filter matched to  $f_L$  (binary 0)
- Envelope detector output sampled at  $kT_b \rightarrow$  compared to threshold

# Non-coherent Detection of BFSK



$$P_{e,BFSK,NC} = \frac{1}{2} \exp\left(-\frac{E_b}{2N_0}\right)$$



# Minimum Shift Keying ( *fast* FSK)

- Type of continuous phase FSK (CPFSK)
- Spectrally efficient
- Constant envelope
- Good BER performance
- Self-synchronizing capability
- Coherent detection preferable

# Minimum Shift Keying

- Consider CPFSK  $0 \leq t \leq T_b$

$$s(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t + \theta(0)) & \text{for symbol 1} \\ \sqrt{\frac{2E_b}{T_b}} \sin(2\pi f_2 t + \theta(0)) & \text{for symbol 0} \end{cases}$$

- In the form of an angle-modulated signal

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta(t))$$

$$\theta(t) = \theta(0) \pm 2\pi \Delta f t = \theta(0) \pm 2\pi \frac{h}{2T_b} t$$

$$= \theta(0) \pm \frac{\pi h}{T_b} t, \quad 0 \leq t \leq T_b$$

**+** : symbol 1  
**−** : symbol 0

# Minimum Shift Keying

$$\left[ \begin{array}{l} f_c + \frac{h}{2T_b} = f_1 \\ f_c - \frac{h}{2T_b} = f_2 \end{array} \right] \Rightarrow \left[ \begin{array}{l} f_c = \frac{1}{2}(f_1 + f_2) \quad ; \text{carrier} \\ h = T_b(f_1 - f_2) \quad ; \text{deviation ratio} \end{array} \right]$$

**Deviation Ratio** → Difference between the two frequencies  $f_1$  and  $f_2$  normalized with respect to the bit rate  $1/T_b$ .

# MSK Phase Trellis

$$\theta(t) = \theta(0) \pm \frac{\pi h}{T_b} t \quad , \quad 0 \leq t \leq T_b$$

At  $t = T_b$

$$\theta(T_b) - \theta(0) = \begin{cases} \pi h & \text{for symbol 1} \\ -\pi h & \text{for symbol 0} \end{cases} \quad \begin{array}{l} \uparrow \text{ In phase} \\ \downarrow \text{ In phase} \end{array}$$

# Deviation Ratio h

- ❖ FSK deviation ratio / modulation index  $h = 1$   
→  $(f_1 - f_2) = 1/T_b = R_b$
- ❖ MSK deviation ratio / modulation index  $h = 1/2$   
→  $(f_1 - f_2) = 1/2T_b = R_b/2$ 
  - **Minimum frequency spacing** (bandwidth) for two FSK signals to be coherently orthogonal
  - **Minimum bandwidth that allows orthogonal detection**

# MSK Phase Trellis

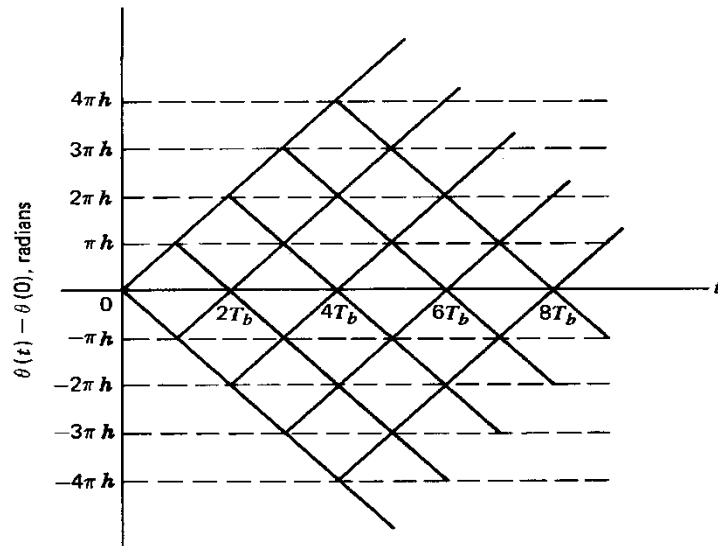


Figure 8.21 Phase tree.

**$h = 1; ?$**

Sunde's FSK  $\rightarrow \pm\pi$  change

No memory

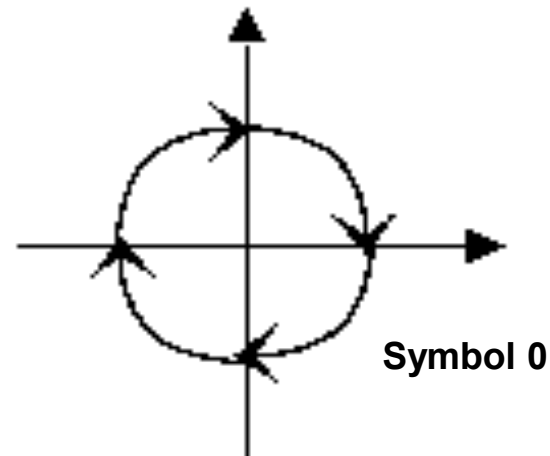
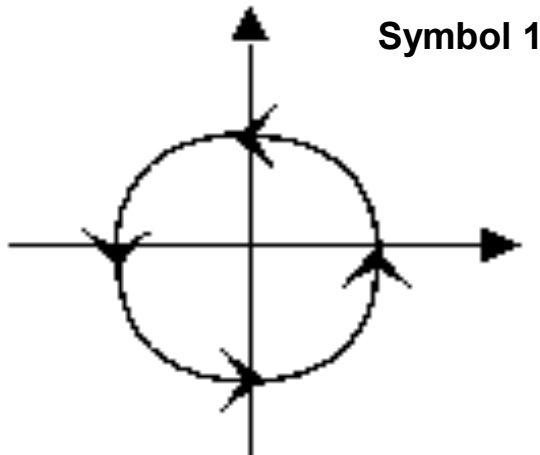
**$h = 0.5; ?$**

$t = \text{even multiple of } T$

$\rightarrow \theta(t_k) - \theta(t_{k-1}) = \pi \text{ or } 0$

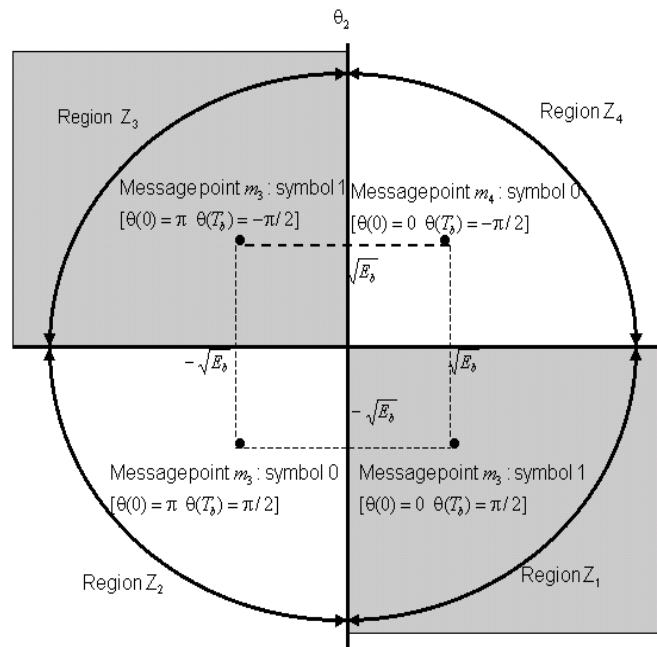
$t = \text{odd multiple of } T$

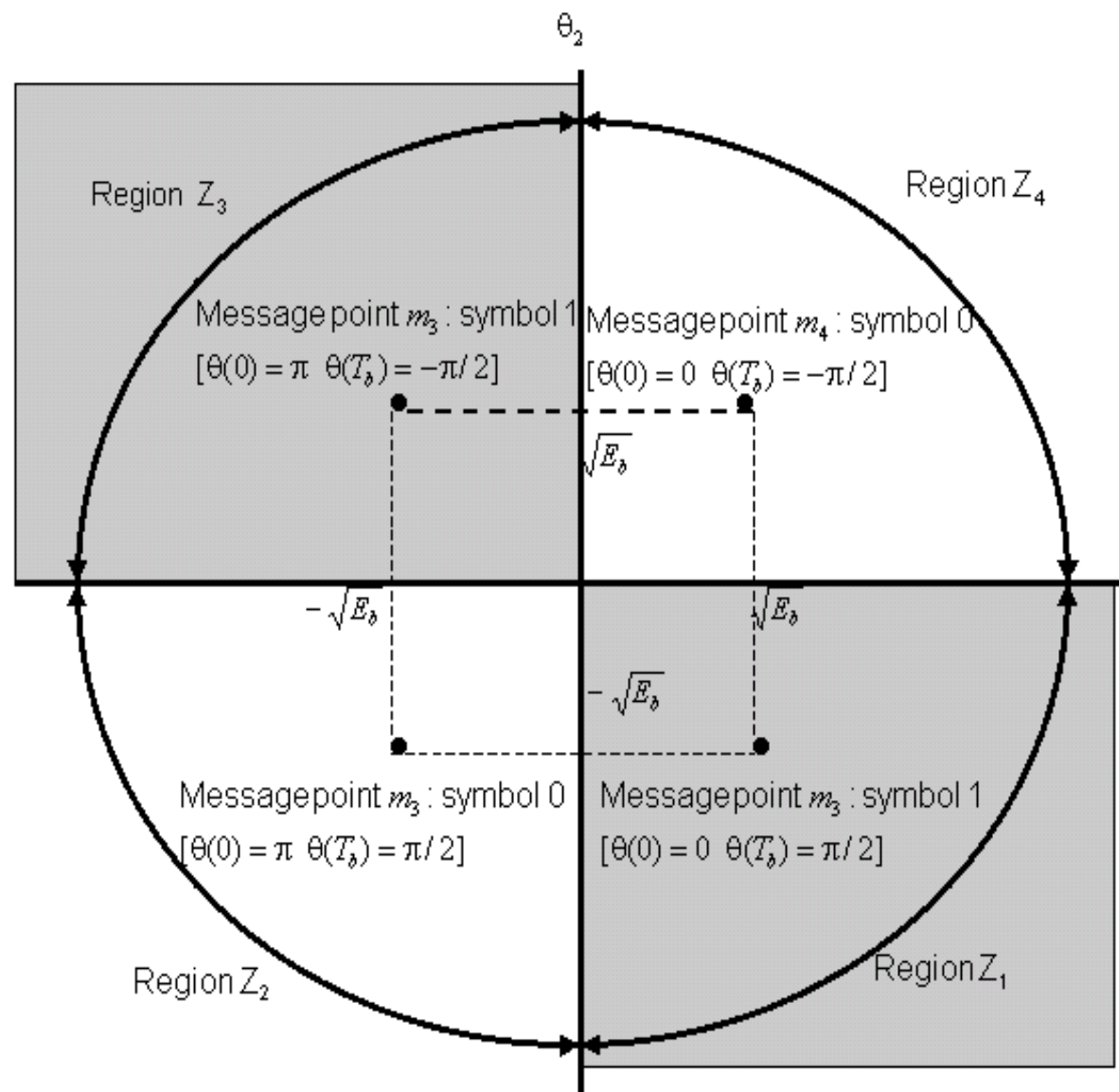
$\rightarrow \theta(t_k) - \theta(t_{k-1}) = \pm \pi/2$



# Signal-space diagram

$$\left\{ \begin{array}{lll} \theta(0) = 0 & \theta(T_b) = \frac{\pi}{2} & \rightarrow \text{transmission of symbol 1} \\ \theta(0) = \pi & \theta(T_b) = \frac{\pi}{2} & \rightarrow \text{transmission of symbol 0} \\ \theta(0) = \pi & \theta(T_b) = -\frac{\pi}{2} & \rightarrow \text{transmission of symbol 1} \\ \theta(0) = 0 & \theta(T_b) = -\frac{\pi}{2} & \rightarrow \text{transmission of symbol 0} \end{array} \right.$$





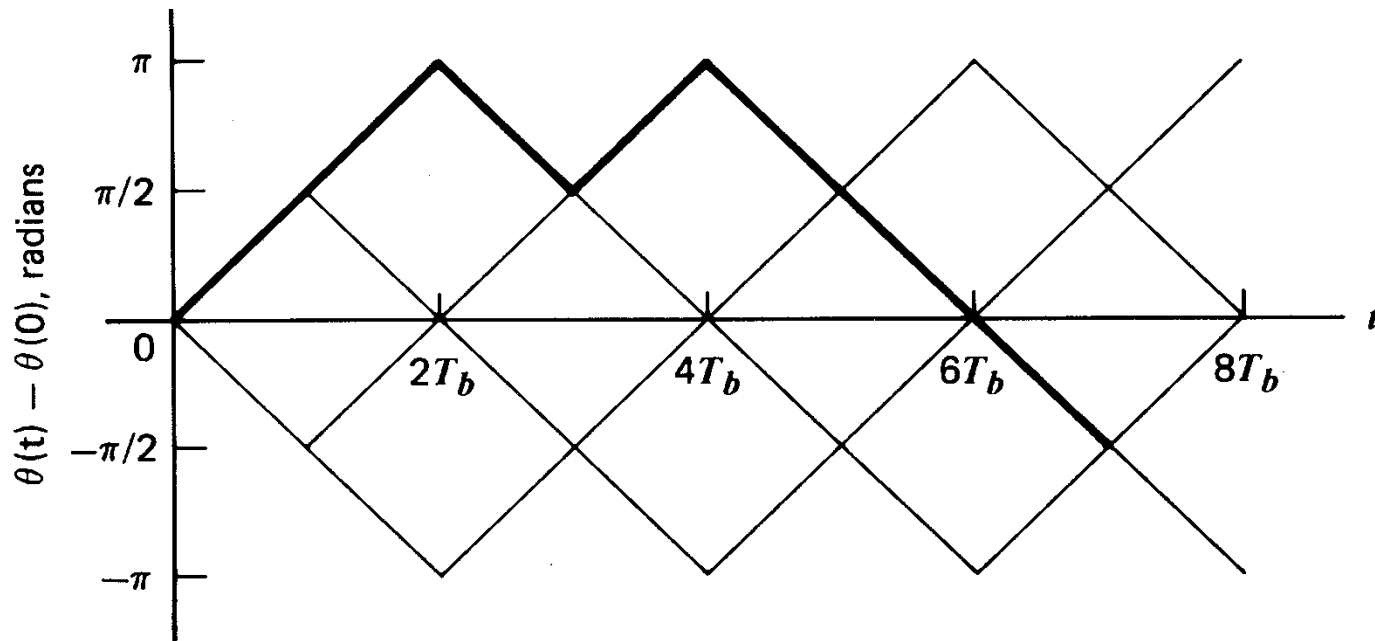


# Average Probability of Error

$$\text{BER} = \frac{1}{2} \text{erfc} \left( \sqrt{\frac{E_b}{N_0}} \right)$$

- Same as BPSK , QPSK
- Good performance  $\Leftarrow$  observation over  $2T_b$  seconds

# MSK Phase Trellis



**Figure 8.22** Phase trellis; boldfaced path represents the sequence 1101000.

# Signal-space diagram

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta(t))$$

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos[\theta(t)] \cos(2\pi f_c t) - \sqrt{\frac{2E_b}{T_b}} \sin[\theta(t)] \sin(2\pi f_c [t + T_b])$$

$$\theta(t) = \theta(0) \pm \frac{\pi}{2T_b} t \quad 0 \leq t \leq T$$

+ symbol 1

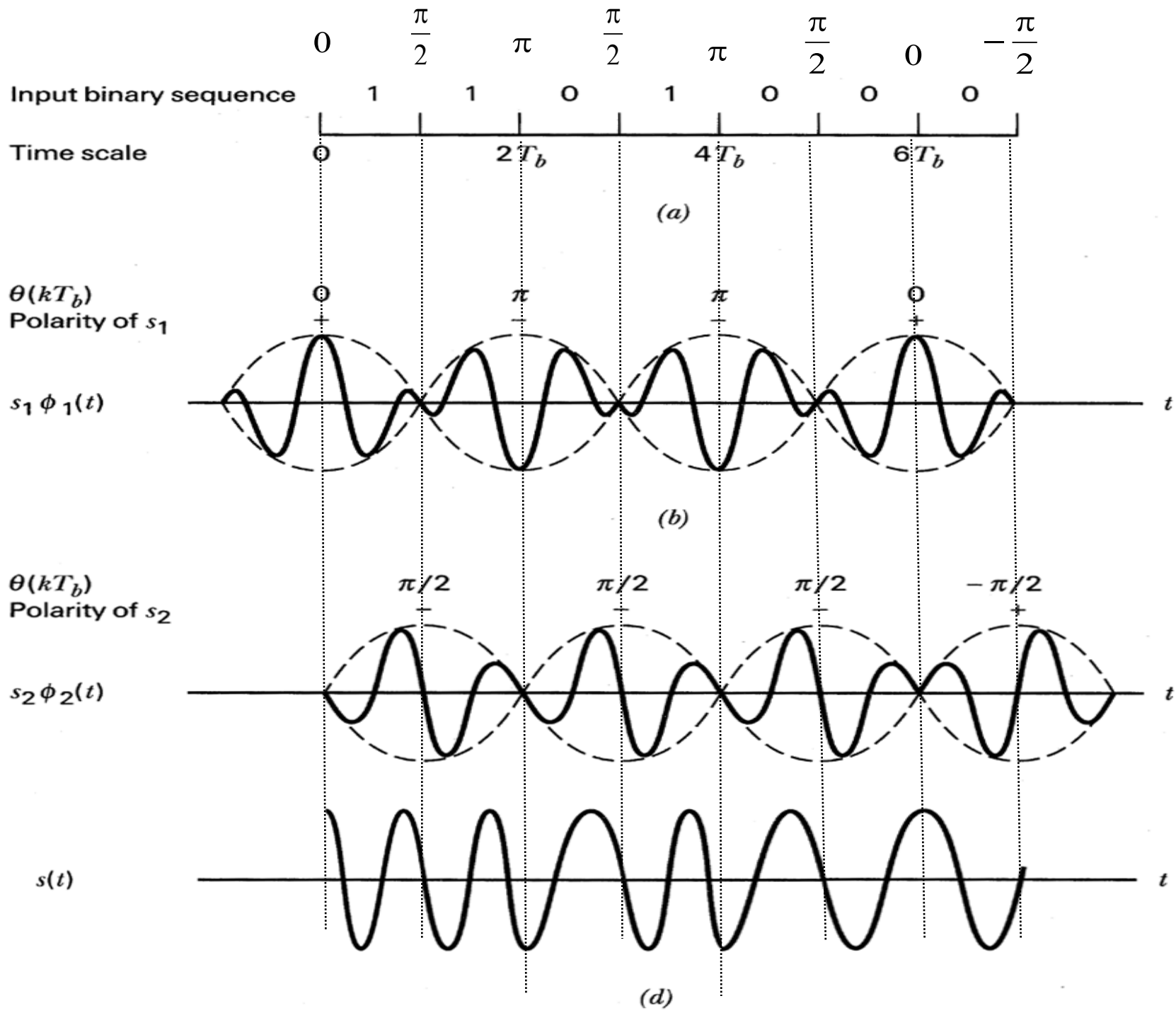
- symbol 0

# Orthonormal basis for MSK

$$\phi_1(t) = \sqrt{\frac{2}{T_b}} \cos\left(\frac{\pi}{2T_b}t\right) \cos(2\pi f_c t) \quad 0 \leq t \leq T_b$$

$$\phi_2(t) = \sqrt{\frac{2}{T_b}} \sin\left(\frac{\pi}{2T_b}t\right) \sin(2\pi f_c t) \quad 0 \leq t \leq T_b$$

$$s(t) = s_1 \phi_1(t) + s_2 \phi_2(t) \quad 0 \leq t \leq T_b$$

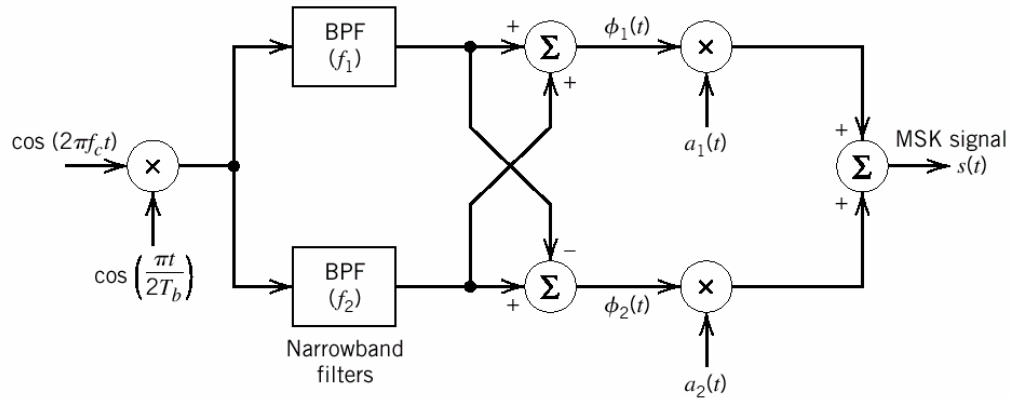


# Generation of MSK signals

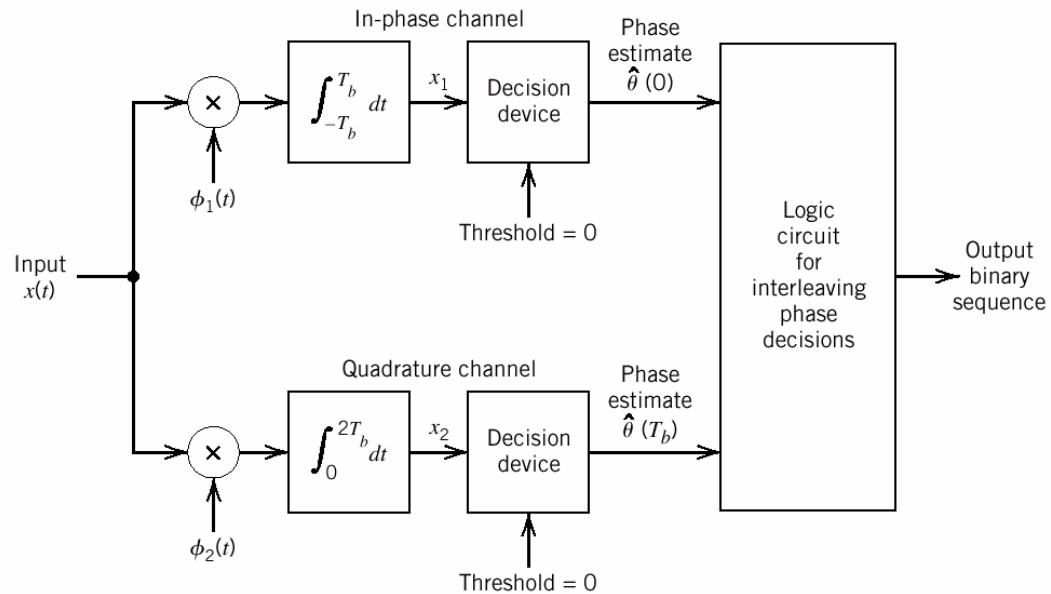
$$\begin{aligned}\phi_1(t) &= \sqrt{\frac{2}{T_b}} \cos\left(\frac{\pi}{2T_b}t\right) \cos(2\pi f_c t) \\ &= \sqrt{\frac{2}{T_b}} \left\{ \cos\left(2\pi f_c t + \frac{\pi}{2T_b}t\right) + \cos\left(2\pi f_c t - \frac{\pi}{2T_b}t\right) \right\}\end{aligned}$$

$$\begin{aligned}\phi_2(t) &= \sqrt{\frac{2}{T_b}} \sin\left(\frac{\pi}{2T_b}t\right) \sin(2\pi f_c t) \\ &= \sqrt{\frac{2}{T_b}} \left\{ \cos\left(2\pi f_c t + \frac{\pi}{2T_b}t\right) - \cos\left(2\pi f_c t - \frac{\pi}{2T_b}t\right) \right\}\end{aligned}$$

# MSK Modem



(a)



(b)

# Gaussian MSK

- **Gaussian pulse shaping to MSK**
  - **smoothens** phase trajectory of MSK signal →  
over time, **stabilizes** instantaneous frequency variations
  - results in significant additional **reduction** of  
**sidelobe levels**
- **GMSK detection** can be **coherent** (like **MSK**)  
or **noncoherent** (like **FSK**)



# Gaussian MSK

- Premodulation pulse shaping filter used to filter NRZ data
  - converts **full response** message signal into **partial response** scheme

*full response*  $\rightarrow$  baseband symbols occupy  $T_b$

*partial response*  $\rightarrow$  transmitted symbols span several  $T_b$

- pulse shaping doesn't cause pattern's averaged phase trajectory to deviate from simple MSK trajectory

# Gaussian MSK

GMSKs main advantages are:

- power efficiency - from constant envelope (non-linear amplifiers)
- excellent spectral efficiency
- pre-modulation filtering introduces ISI into transmitted signal

- if  $B_{3dB} \cdot T_b > 0.5 \rightarrow$  degradation is not severe

$B_{3dB}$  = 3dB bandwidth of Gaussian Pulse Shaping Filter

$T_b$  = bit duration = baseband symbol duration

- irreducible BER caused by partial response signaling is the cost for spectral efficiency & constant envelope

**GMSK filter can be completely defined from  $B_{3dB} \cdot T_b$**

**- customary to define GMSK by  $B_{3dB} \cdot T_b$**

# Gaussian MSK

**Impulse response** of pre-modulation Gaussian filter :

$$h_G(t) = \frac{\sqrt{\pi}}{\alpha} \exp\left(-\frac{\pi^2}{\alpha^2}\right)$$

**transfer function** of pre-modulation Gaussian Filter is given by

$$H_G(f) = \exp\left(-\alpha^2 f^2\right)$$

$\alpha$  is related to  $B_{3dB}$  by:  $\alpha = \frac{\sqrt{\ln 2}}{\sqrt{2}B} = \frac{0.5887}{B}$

# Impact of $B_{3dB} \cdot T_b$

(i) **Reducing  $B_{3dB} \cdot T_b$**  : spectrum becomes more compact

- causes **sidelobes** of GMSK to **fall off** rapidly

$B_{3dB} \cdot T_b = 0.5 \rightarrow$  2<sup>nd</sup> lobe peak is **30dB** below main lobe

**MSK**  $\rightarrow$  2<sup>nd</sup> peak lobe is **20dB** below main lobe

- MSK  $\approx$  GMSK with  $B_{3dB} \cdot T_b = \infty$

(ii) increases **irreducible error rate (IER)** due to ISI

- ISI degradation caused by pulse shaping increases
- however - **mobile channels** induce **IER** due to mobile's velocity

• if **GMSK IER** < mobile channel IER  $\rightarrow$  **no penalty** for using GMSK

# RF bandwidth

containing % power as fraction of  $R_b$

| $BT_b$    | 90%  | 99%  | 99.9% | 99.99% |
|-----------|------|------|-------|--------|
| 0.2 GMSK  | 0.52 | 0.79 | 0.99  | 1.22   |
| 0.25 GMSK | 0.57 | 0.86 | 1.09  | 1.37   |
| 0.5 GMSK  | 0.69 | 1.04 | 1.33  | 2.08   |
| MSK       | 0.78 | 1.20 | 2.76  | 6.00   |

e.g. for  $BT = 0.2 \rightarrow 99.99\%$  of the power is in the bandwidth of  $1.22R_b$

[Ish81] **BER degradation** from **ISI** caused by GMSK filtering is **minimal** at  $B_{3dB}T_b = 0.5887$

- degradation in required  $E_b/N_0 = 0.14\text{dB}$  compared to case of **no ISI**

# BER of GMSK for AWGN channel

- [Mur81] shown to perform within 1dB of optimal MSK with

$$B_{3dB}T_b = 0.25$$

- since pulse shaping causes ISI  $\rightarrow P_e$  is function of  $B_{3dB}T_b$

$$P_e = Q\left(\sqrt{\frac{2\lambda E_b}{N_0}}\right)$$

$P_e$  = bit error probability

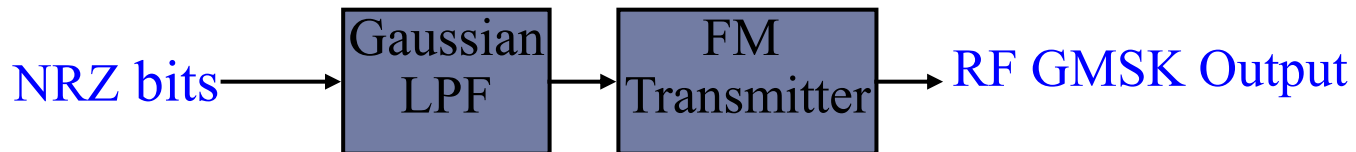
$\lambda$  is constant related to  $B_{3dB}T_b$

- $B_{3dB}T_b = 0.25 \rightarrow \lambda = 0.68$
- $B_{3dB}T_b = \infty \rightarrow \lambda = 0.85$  (MSK)

# GMSK Transmitter

(i) Pass  $m_{NRZ}(t)$  through Gaussian base band filter (see figure below)

- $m_{NRZ}(t)$  = NRZ bit stream
- output of Gaussian filter passed to FM modulator



- used in digital implementation for
  - Global System for Mobile (**GSM**)
  - US Cellular Digital Packet Data (**CDPD**)

(ii) alternate approach is to use standard I/Q modulator

# Constant Envelope Modulation

- **Constant carrier amplitude** - regardless of variations in  $m(t)$ 
  - ✓ Better **immunity** to fluctuations due to **fading** and random **noise**.
- **Improved power efficiency** - without degrading occupied spectrum
  - ✓ Use of power efficient **class C** amplifiers (non-linear)



# Constant Envelope Modulation

- **Low out of band radiation** *(-60db to -70db)*
- **Simplified receiver design**
  - ✓ **Use limiter-discriminator detection**
- **Larger occupied bandwidth**
  - ✓ **than linear modulation but smaller sidelobes**

# Combined Linear & Constant Envelope Modulation

MPSK

MFSK

MQAM

OFDM

# Mary Phase Shift Keying

Carrier phase takes 1 of  $M$  possible values – amplitude constant

$$\theta_i = 2(i-1)\pi/M, \quad i = 1, 2, \dots, M$$

**Modulated waveform:**

$$s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos\left(2\pi f_c t + \frac{2\pi}{M}(i-1)\right) \quad 0 \leq t \leq T_s, \quad i = 1, 2, \dots, M$$

$$E_s = \log_2 M \cdot E_b \quad \text{energy per symbol}$$

$$T_s = \log_2 M \cdot T_b \quad \text{symbol period}$$

written in **quadrature form** as:

Basis Signal ?

$$s_i(t) = \sqrt{\frac{2}{T_s}} \left[ \sqrt{E_s} \cos\left((i-1)\frac{2\pi}{M}\right) \cos(2\pi f_c t) - \sqrt{E_s} \sin\left((i-1)\frac{2\pi}{M}\right) \sin(2\pi f_c t) \right]$$

for  $i = 1, 2, \dots, M$

# Mary Phase Shift Keying

Orthogonal basis signals

$$\phi_1(t) = \sqrt{\frac{2}{T_s}} \cos(2\pi f_c t)$$

$$\phi_2(t) = \sqrt{\frac{2}{T_s}} \sin(2\pi f_c t)$$

defined over  $0 \leq t \leq T_s$

MPSK signal can be expressed as

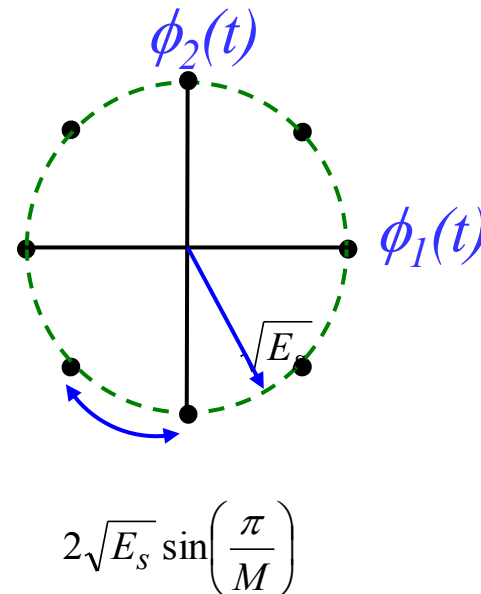
$$s_{MPSK}(t) = \left( \sqrt{E_s} \cos\left((i-1)\frac{\pi}{2}\right) \phi_1(t) - \sqrt{E_s} \sin\left((i-1)\frac{\pi}{2}\right) \phi_2(t) \right)$$
$$i = 1, 2, \dots, M$$

# Mary Phase Shift Keying

- MPSK basis has 2 signals  $\rightarrow$  2 dimensional constellation
- M-ary message points equally spaced on circle with radius  $\sqrt{E_s}$
- MPSK is constant envelope when no pulse shaping is used

**MPSK signal** can be

- coherently detected
  - $\theta = \text{Arctan}(Y/X)$
  - Minimum  $|\theta_I - \theta|$
- non-coherent detected  
with differential encoding



# Mary Phase Shift Keying

**Probability of symbol error** in AWGN channel – using distance between adjacent symbols as  $2\sqrt{E_s} \sin\left(\frac{\pi}{M}\right)$

$P_e$  = average symbol error probability in AWGN channel

$$P_e \leq 2Q\left(\sqrt{\frac{2E_b \log_2 M}{N_0}} \sin\left(\frac{\pi}{M}\right)\right)$$

When differentially encoded & non-coherently detected,  
 $P_e$  estimated for  $M \geq 4$  as:

$$P_e \approx 2Q\left[\sqrt{\frac{4E_s}{N_o}} \sin\left[\frac{\pi}{2M}\right]\right]$$

# MPSK Bandwidth Efficiency vs Power Efficiency

| $M$                     | 2    | 4    | 8    | 16   | 32   | 64   |
|-------------------------|------|------|------|------|------|------|
| $\eta_B = R_b/B_{null}$ | 0.5  | 1.0  | 1.5  | 2    | 2.5  | 3    |
| $E_b/N_0$ (dB)          | 10.5 | 10.5 | 14.0 | 18.5 | 23.4 | 28.5 |

- $\eta_B$  = bandwidth efficiency
- $R_b$  = bit rate
- $B_{null}$  = 1<sup>st</sup> null bandwidth
- $E_b/N_0$  for BER =  $10^{-6}$

*bandwidth efficiency & power efficiency assume*

- Ideal Nyquist Pulse Shaping (RC filters)
- AWGN channel without timing jitter or fading

# Mary Phase Shift Keying

## Advantages:

Bandwidth efficiency increases with  $M$

## Drawbacks:

Jitter & fading cause large increase in BER as  $M$  increases  
EMI & multipath alter instantaneous phase of signal  
– cause error at detector

Receiver design also impacts BER

Power efficiency reduces for higher  $M$

MPSK in mobile channels require **Pilot Symbols or Equalization**



# M-ary Quadrature Amplitude Modulation

- Allows amplitude & phase to vary
- General form of **M-ary QAM signal** given by

$$s_i(t) = \sqrt{\frac{2}{T_s}} \sqrt{E_{\min}} a_i \cos(2\pi f_c t) + \sqrt{\frac{2}{T_s}} \sqrt{E_{\min}} b_i \sin(2\pi f_c t)$$

$$0 \leq t \leq T_s \quad i = 1, 2, \dots, M$$

# M-ary Quadrature Amplitude Modulation

- $E_{min}$  = energy of signal with lowest amplitude
- $a_i, b_i$  = independent integers related to location of signal point
- $T_s$  = symbol period

- ✓ Energy per symbol , distance between adj. symbols isn't constant → probability of correct symbol detection is not same for all symbols
- ✓ Pilot tones used to estimate channel effects

# M-ary QAM

Assuming **rectangular pulses** - basis functions given by

$$\phi_1(t) = \sqrt{\frac{2}{T_s}} \cos(2\pi f_c t) \quad 0 \leq t \leq T_s$$

$$\phi_2(t) = \sqrt{\frac{2}{T_s}} \sin(2\pi f_c t) \quad 0 \leq t \leq T_s$$

# M-ary QAM

**QAM signal** given by:

$$s_i(t) = \sqrt{E_{\min}} a_i \phi_1(t) + \sqrt{E_{\min}} b_i \phi_2(t)$$

$$0 \leq t \leq T_s \quad i = 1, 2, \dots, M$$

**coordinates of  $i^{\text{th}}$  message point** =  $\left[ a_i \sqrt{E_{\min}}, b_i \sqrt{E_{\min}} \right]$

$(a_i, b_i)$  = element in  $L^2$  matrix, where  $L = \sqrt{M}$

# 16-ary QAM

e.g. let  $M = 16$ , then  $\{a_i, b_i\}$  given based on

$$\sqrt{E_{\min}} a_i \phi_1(t) + \sqrt{E_{\min}} b_i \phi_2(t)$$

$$\{a_i, b_i\} = \begin{bmatrix} (-3,3) & (-1,3) & (1,3) & (3,3) \\ (-3,1) & (-1,1) & (1,1) & (3,1) \\ (-3,-1) & (-1,-1) & (1,-1) & (3,-1) \\ (-3,-3) & (-1,-3) & (1,-3) & (3,-3) \end{bmatrix}$$

$$s_{11}(t) = -3\sqrt{E_{\min}} \phi_1(t) + 3\sqrt{E_{\min}} \phi_2(t) \quad 0 \leq t \leq T_s$$

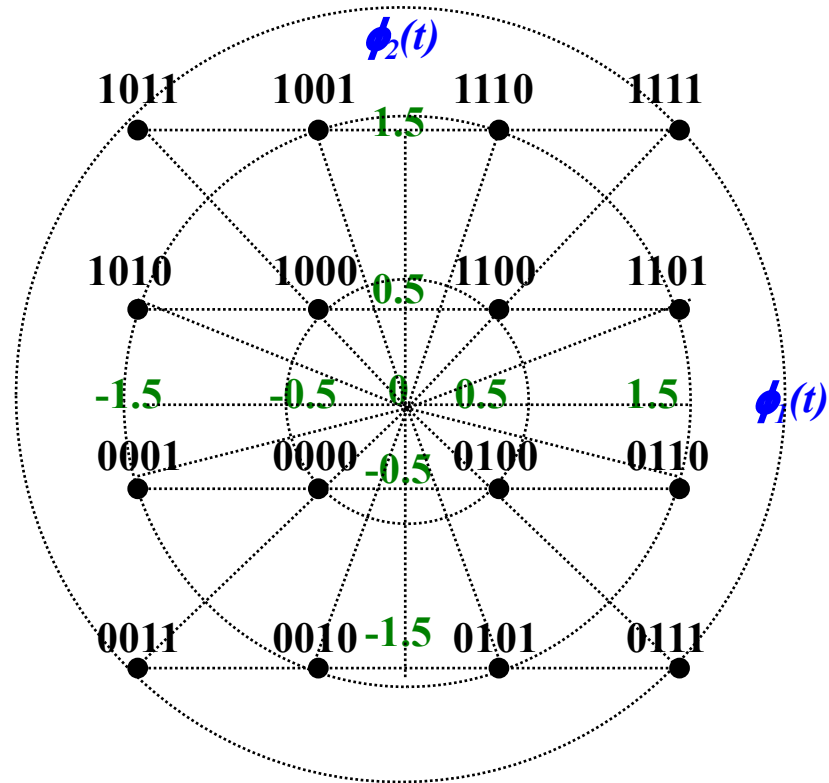
$$s_{21}(t) = -3\sqrt{E_{\min}} \phi_1(t) + \sqrt{E_{\min}} \phi_2(t) \quad 0 \leq t \leq T_s$$

# 16-ary QAM

➤ **QAM:** modulated signal is hybrid of phase & amplitude modulation

➤ Each message point corresponds to a *quadbit*

➤  $E_s$  is not constant – requires **linear channel / linear devices**



# M-ary QAM

In general, for any  $M = L^2$

$$\sqrt{E_{\min}} a_i \phi_1(t) + \sqrt{E_{\min}} b_i \phi_2(t)$$

$$\{a_i, b_i\} = \begin{bmatrix} (-L+1, L-1) & (-L+3, L-1) & \dots & (L-1, L-1) \\ (-L+1, L-3) & (-L+3, L-3) & \dots & (L-1, L-3) \\ \dots & \dots & \dots & \dots \\ (-L+1, -L+1) & (-L+3, -L+1) & \dots & (L-1, -L+1) \end{bmatrix}$$

# M-ary QAM

The average error probability,  $P_e$  for M-ary **QAM** is approximated by

$$P_e \approx 4 \left( 1 - \frac{1}{\sqrt{M}} \right) Q \left( \sqrt{\frac{2 E_{\min}}{N_0}} \right)$$

- assuming coherent detection
- AWGN channel
- no fading, timing jitter

In terms of average energy,  $E_{av}$

$$P_e \approx 4 \left( 1 - \frac{1}{\sqrt{M}} \right) Q \left( \sqrt{\frac{3 E_{av}}{(M-1) N_0}} \right)$$

**Power Spectrum & Bandwidth Efficiency of QAM = MPSK**

**Power Efficiency of QAM is better than MPSK**



# Mary QAM

## **M-ary QAM** - *Bandwidth Efficiency & Power Efficiency*

- Assume Optimum RC filters in AWGN
- Does not consider fading, jitter, - overly optimistic

| $M$                            | 4    | 16 | 64   | 256 | 1024 | 4096 |
|--------------------------------|------|----|------|-----|------|------|
| $\eta_B = R_b/B_{\text{null}}$ | 1    | 2  | 3    | 4   | 5    | 6    |
| $E_b/N_0$ (BER = $10^{-6}$ )   | 10.5 | 15 | 18.5 | 24  | 28   | 33.5 |

# Mary Frequency Shift Keying

MFSK - transmitted signals defined as

$$s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos\left(\frac{\pi}{T_s}(n_c + i)t\right) \quad 0 \leq t \leq T_s, \quad i = 1, 2, \dots, M$$

- $f_c = n_c/2T_s$
- $n_c$  = fixed integer

$$s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos\left(2\pi\left(f_c + \frac{i}{2T_s}\right)t\right)$$

$$0 \leq t \leq T_s, \quad i = 1, 2, \dots, M$$

# Mary Frequency Shift Keying

Each of  $M$  signals have

- equal **energy**
- equal **duration**
- adjacent sub carrier frequencies **separated** by  $1/2T_s$  Hz
- sub carriers are **orthogonal** to each other

# Mary Frequency Shift Keying

## **MFSK coherent detection** - optimum receiver

- receiver has bank of  $M$  correlators or matched filters
- each correlator tuned to 1 of  $M$  distinct carrier frequencies
- average probability of error,  $P_e$  (based on union bound)

$$P_e \leq (M-1)Q\left(\sqrt{\frac{E_b \log_2 M}{N_0}}\right)$$

# Mary Frequency Shift Keying

## **MFSK non-coherent detection**

- using matched filters followed by envelope detectors
- average probability of error,  $P_e$

$$P_e = \sum_{k=1}^{M-1} \binom{M-1}{k} \exp\left(\frac{-kE_s}{(k+1)N_0}\right)$$

**bound  $P_e \rightarrow$  use leading terms of binomial expansion**

$$P_e \leq \binom{M-1}{2} \exp\left(\frac{-E_s}{2N_0}\right)$$

# MFSK Performance

Impact of **increasing  $M$**  on MFSK performance

**bandwidth efficiency ( $\eta_B$ )** of MFSK decreases

- MFSK signals are bandwidth inefficient (unlike MPSK)

**power efficiency ( $\eta_P$ )** increases

- with  $M$  orthogonal signals  $\rightarrow$  signal space is not crowded
- power efficient *non-linear amplifiers* can be used without performance degradation

# MFSK Bandwidth Efficiency vs Power Efficiency

$$B_{\text{null}} = M/2T = R_b M/2\log_2 M$$

$$\eta_B = R_b/B_{\text{null}} = 2 \log_2 M/M$$

- $\eta_B$  = bandwidth efficiency
- $R_b$  = bit rate
- $B_{\text{null}}$  = 1<sup>st</sup> null bandwidth
- $E_b/N_0$  for BER =  $10^{-6}$

| $M$                            | 2    | 4    | 8    | 16  | 32     | 64     |
|--------------------------------|------|------|------|-----|--------|--------|
| $\eta_B = R_b/B_{\text{null}}$ | 1.0  | 1.0  | 0.75 | 0.5 | 0.3125 | 0.1875 |
| $E_b/N_0$ (BER = $10^{-6}$ )   | 13.5 | 10.8 | 9.3  | 8.2 | 7.5    | 6.9    |

***Bandwidth Efficiency & Power Efficiency assume***

- Ideal Nyquist Pulse Shaping (RC Filters)
- AWGN Channel Without Timing Jitter or Fading

## *Bandwidth Efficiency & Power Efficiency*

| $M$                     | 2    | 4    | 8    | 16   | 32   | 64   |
|-------------------------|------|------|------|------|------|------|
| $\eta_B = R_b/B_{null}$ | 0.5  | 1.0  | 1.5  | 2    | 2.5  | 3    |
| $E_b/N_0$ (dB)          | 10.5 | 10.5 | 14.0 | 18.5 | 23.4 | 28.5 |

### *MPSK*

| $M$                          | 4    | 16 | 64   | 256 | 1024 | 4096 |
|------------------------------|------|----|------|-----|------|------|
| $\eta_B = R_b/B_{null}$      | 1    | 2  | 3    | 4   | 5    | 6    |
| $E_b/N_0$ (BER = $10^{-6}$ ) | 10.5 | 15 | 18.5 | 24  | 28   | 33.5 |

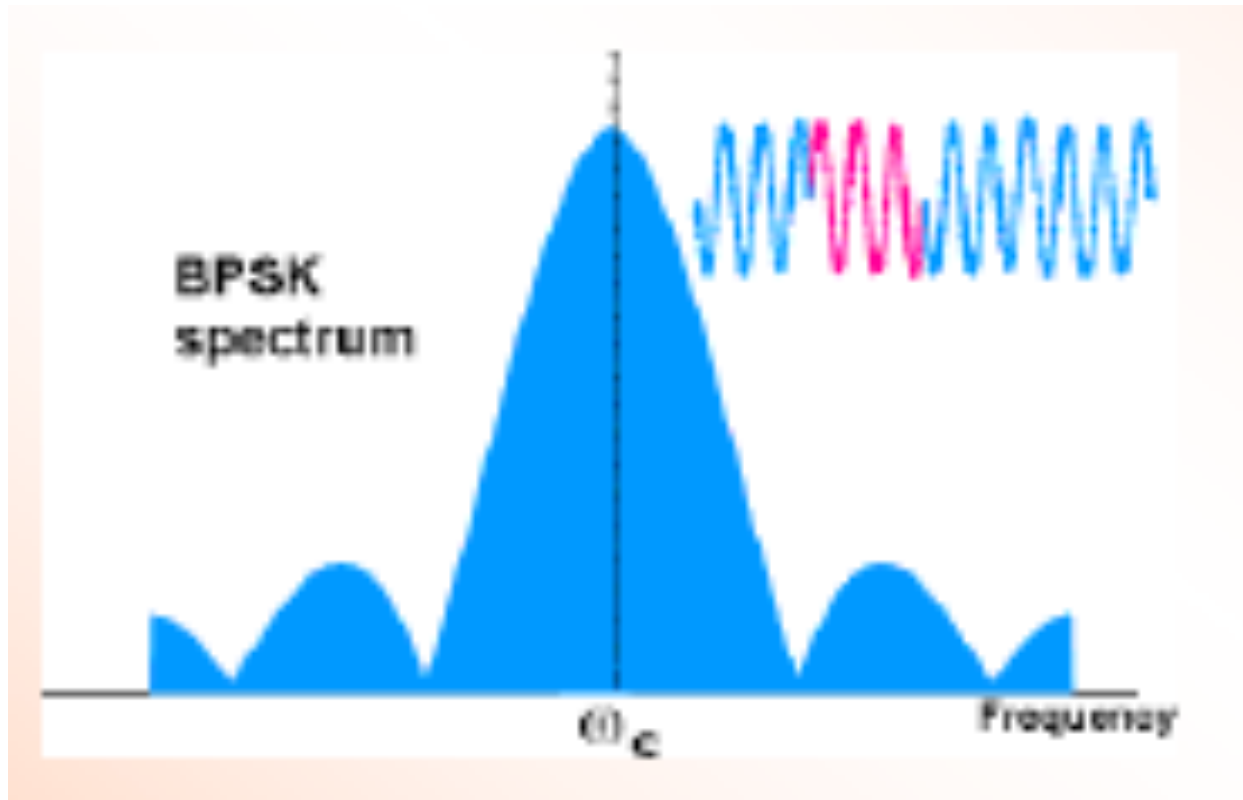
### *MQAM*

| $M$                          | 2    | 4    | 8    | 16   | 32   | 64   |
|------------------------------|------|------|------|------|------|------|
| $\eta_B = R_b/B_{null}$      | 0.4  | 0.57 | 0.55 | 0.42 | 0.29 | 0.18 |
| $E_b/N_0$ (BER = $10^{-6}$ ) | 13.5 | 10.8 | 9.3  | 8.2  | 7.5  | 6.9  |

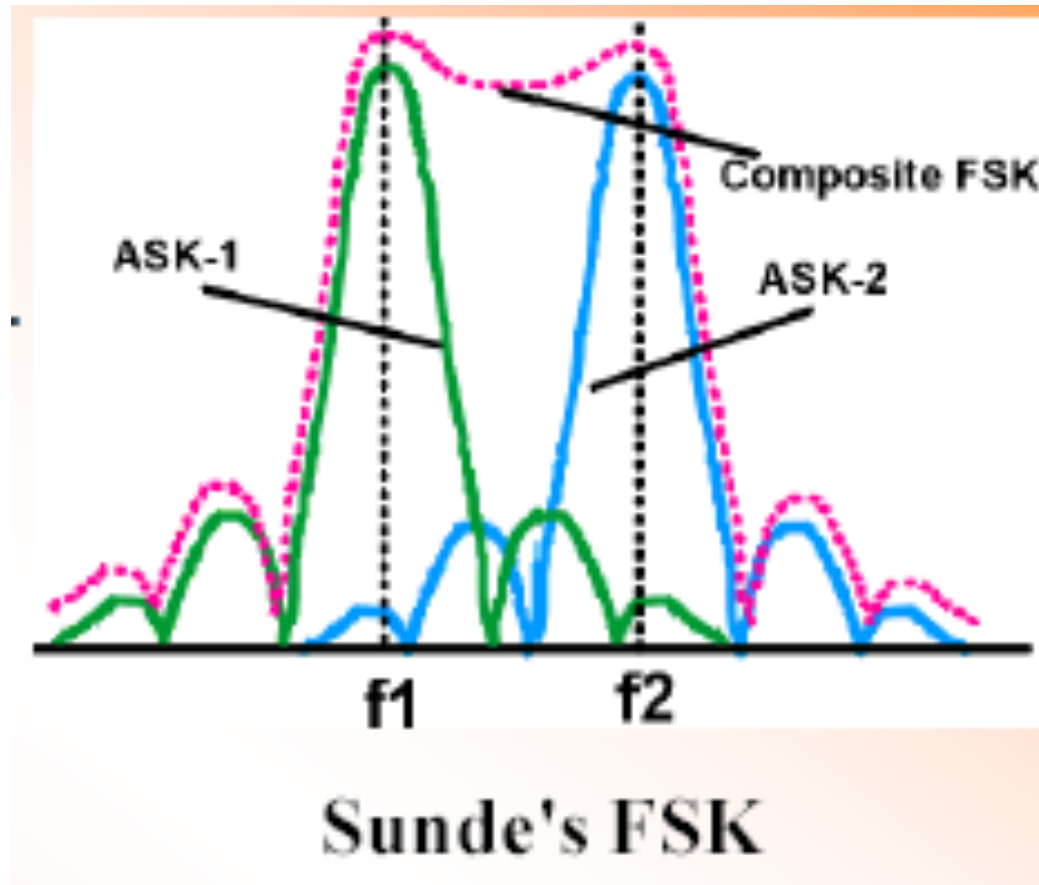
### *MFSK*



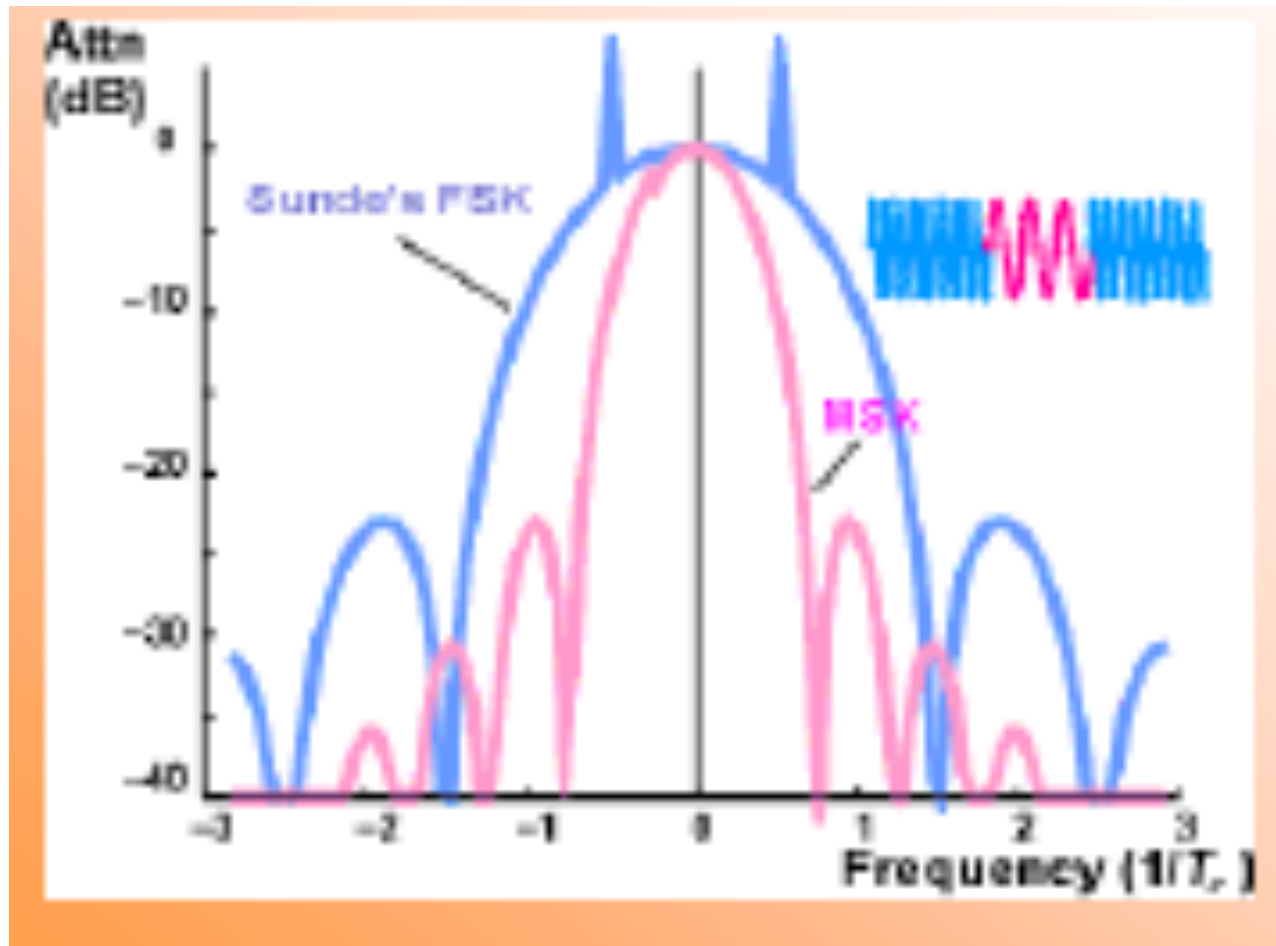
# Bandpass BPSK Spectra



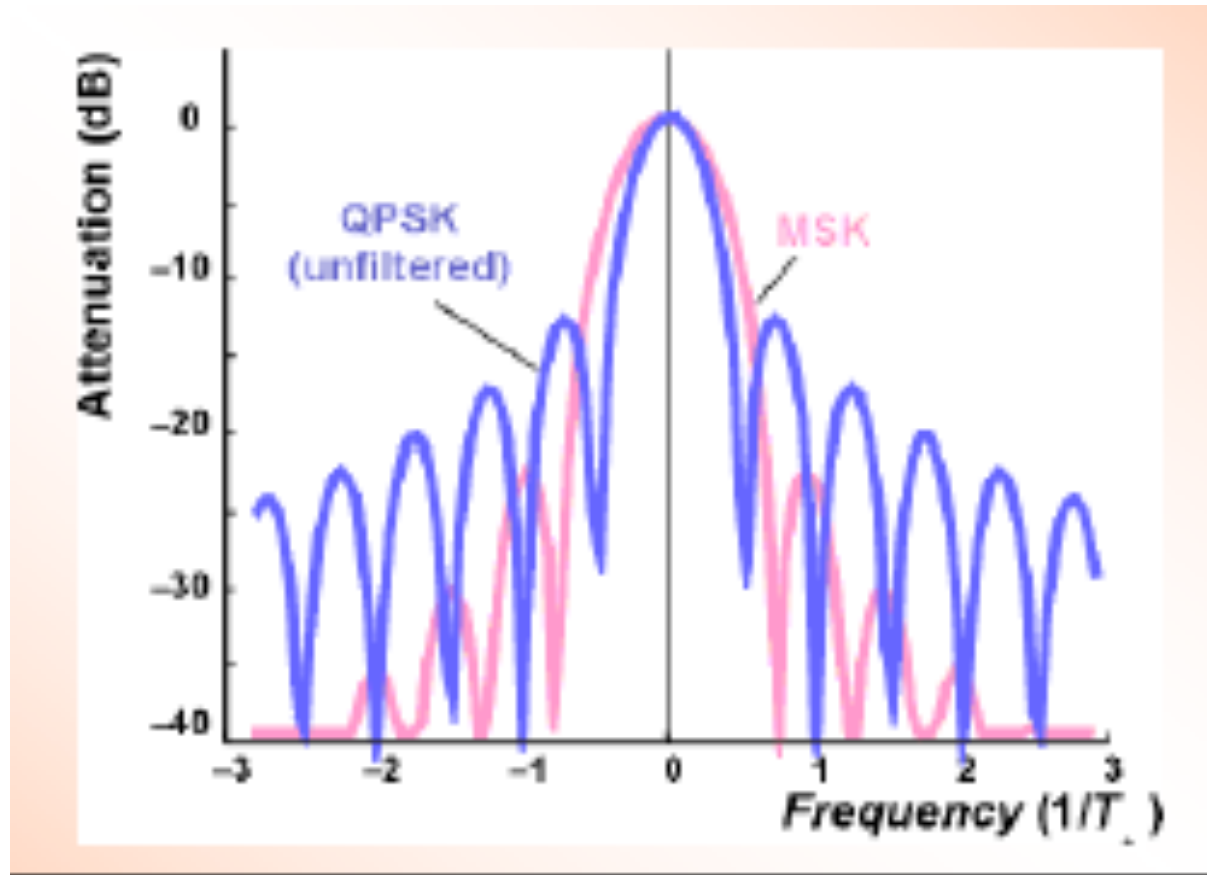
# Bandpass BFSK Spectra



# Baseband Sunde's FSK /MSK Spectra

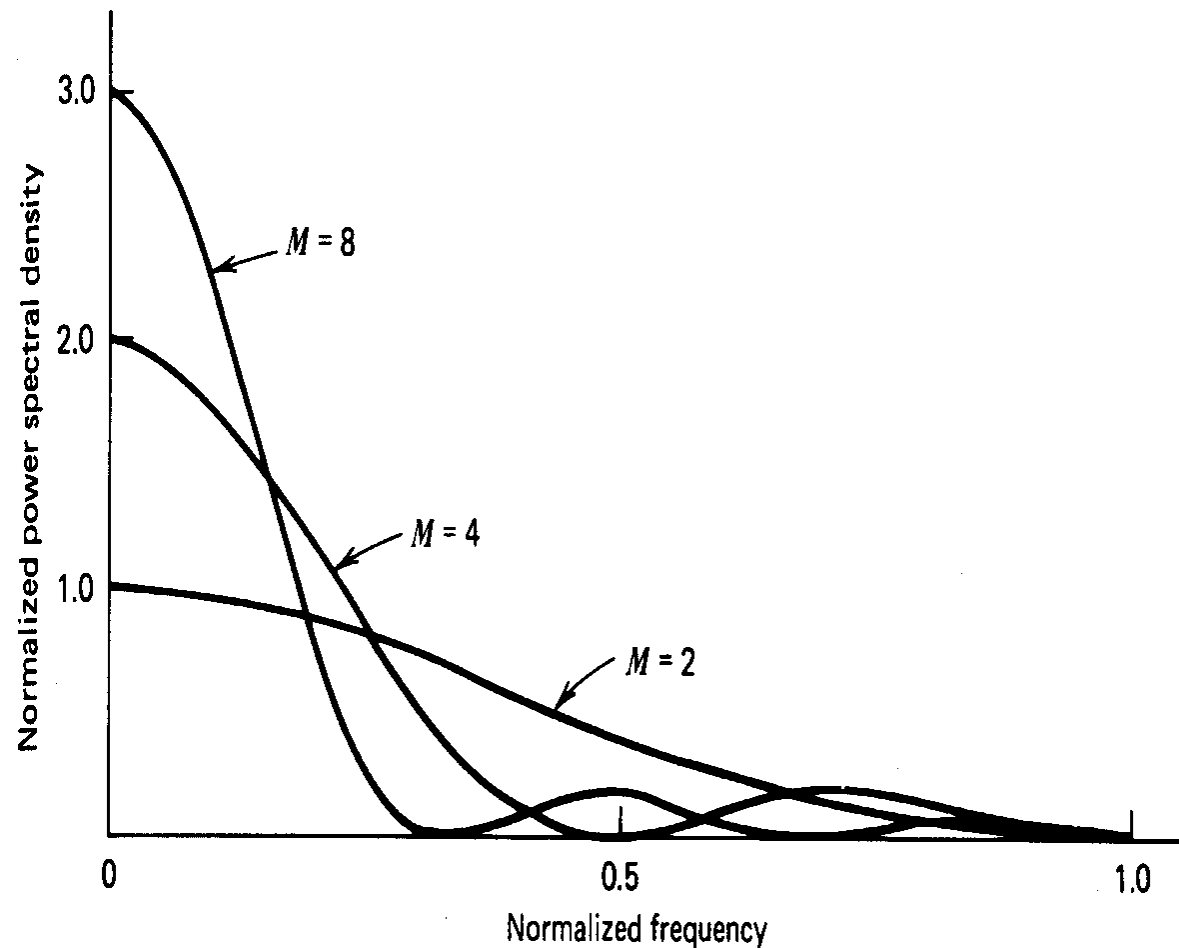


# Baseband QPSK/MSK spectra



- ✓ 99% MSK power is within bandwidth  $B = 1.2/T_b$
- ✓ 99% QPSK power is within bandwidth  $B = 8/T_b$

# Power Spectra for MPSK



**Figure 8.40** Power spectra of  $M$ -ary PSK signals for  $M = 2, 4, 8$ .

# Power Spectra for MFSK

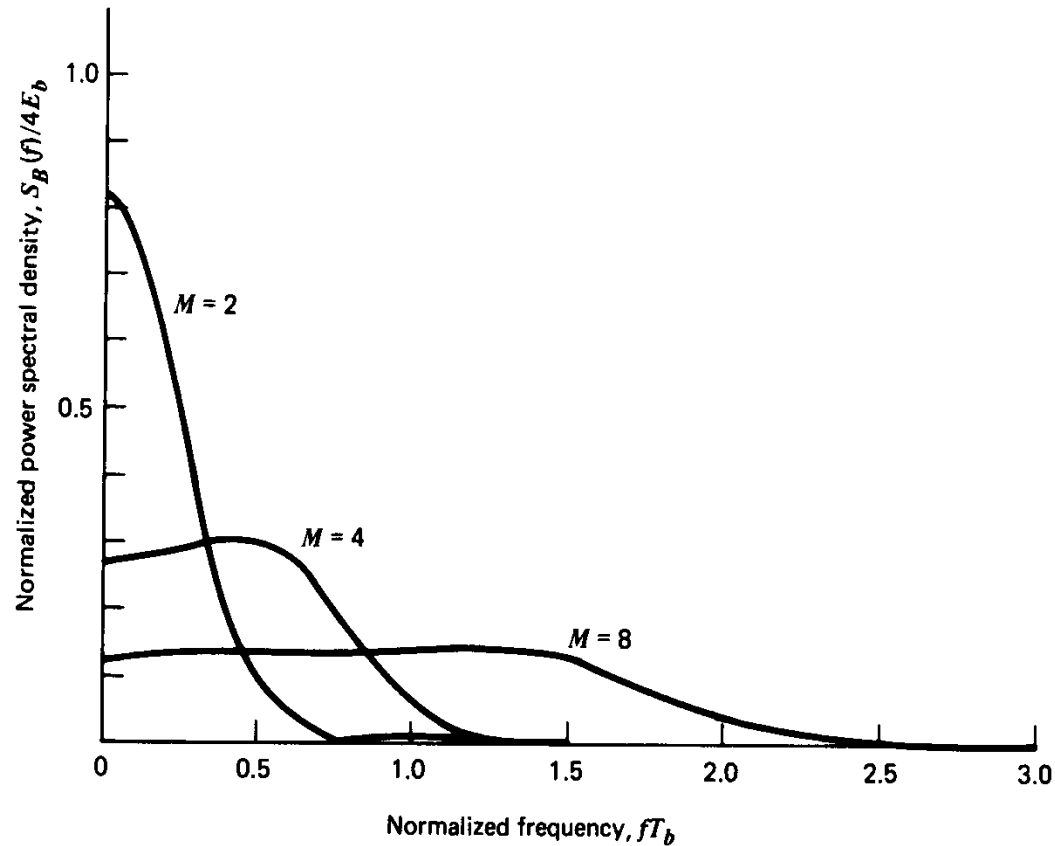


Figure 8.41 Power spectra of  $M$ -ary FSK signals for  $M = 2, 4, 8$ .